St. Peter's
College of Engineering and Technology
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Co-ordinators

Ms. Lavanya R, Assistant Professor
Ms. Devi B, Assistant Professor

Department of Artificial Intelligence and Data Science

CONVENER

Dr.Wisely Joe J

Associate Professor & Head
Department of Artificial Intelligence and Data Science

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MESSAGE FROM CHIEF PATRON



Dr. T. BANUMATHI

It gives us immense pleasure to acknowledge the successful organization of the First International Conference on Smart Intelligence and Data Engineering (ICSIDE'26) by the Department of Artificial Intelligence and Data Science. This conference holds great significance in the present technological era, where innovations in Artificial Intelligence, Data Science, and Smart Computing are transforming industries and society. The conference provides an excellent platform for researchers, academicians, industry experts, and students to share knowledge, exchange innovative ideas, and discuss emerging trends in technology and engineering. It also encourages students and researchers to reflect on their work, enhance their technical expertise, and contribute meaningful solutions to real-world challenges. In today's rapidly evolving scientific and technological landscape, collaboration and knowledge sharing are essential for sustainable growth and innovation. Conferences like ICSIDE'26 play a vital role in promoting research culture, interdisciplinary learning, and industry-academia interaction. I strongly believe that ICSIDE'26 will inspire participants through its technical sessions, discussions, and research presentations, thereby creating opportunities for learning, innovation, and professional development.

I extend my sincere wishes for the grand success of this conference. I am confident that ICSIDE'26 will serve as a valuable platform for knowledge sharing, innovation, and collaborative learning, thereby contributing to meaningful advancements in the field of technology.

CHAIRPERSON

MESSAGE FROM TRUSTEE



Dr. LASYA THAMBIDURAI

I am delighted to welcome you to the First International Conference on Smart Intelligence and Data Engineering (ICSIDE'26) organized by the Department of Artificial Intelligence and Data Science. This conference serves as an excellent platform for students, faculty members, researchers, and industry professionals to share innovative ideas, research contributions, and recent advancements in the field of intelligent technologies and data engineering. We warmly invite participants to actively engage through paper presentations, technical discussions, and knowledge-sharing sessions that will enrich academic and research collaboration. The conference promises insightful interactions with distinguished experts, creating opportunities for learning, networking, and exploring emerging trends in technology. I express my sincere appreciation to the organizing committee, speakers, reviewers, and participants for their valuable support and contributions toward making this conference a success. I am confident that ICSIDE'26 will inspire innovation, encourage collaborative research, and contribute significantly to the advancement of technology and engineering.

Warm regards,

TRUSTEE

MESSAGE FROM TRUSTEE



Dr. NAMRATHA THAMBIDURAI

It gives me great pleasure to welcome you all to the First International Conference on Smart Intelligence and Data Engineering (ICSIDE'26) organized by the Department of Artificial Intelligence and Data Science. This conference brings together students, researchers, academicians, and industry experts to share innovative ideas, research work, and recent developments in the field of ICT. We believe this event will provide a valuable platform for learning, collaboration, and knowledge exchange. Your active participation and paper presentations will make the conference more meaningful and successful. I sincerely thank the organizing committee for their hard work and dedication in organizing this event.

Let us make ICSIDE'26 a memorable and inspiring occasion for innovation and academic excellence.

Warm regards,

TRUSTEE

MESSAGE FROM SECRETARY



Mr. K. PRABU

I am happy to announce the First International Conference on Smart Intelligence and Data Engineering (ICSIDE'26), organized by the Department of Artificial Intelligence and Data Science. This conference provides a valuable platform to share recent research, innovative ideas, and technological advancements in the field of ICT. The conference offers an excellent opportunity for students, faculty members, researchers, and industry experts to interact, learn, and explore emerging trends. We encourage active participation through paper submissions and presentations, which will help enrich technical discussions and showcase the talents and expertise of participants. ICSIDE'26 will also promote networking and collaboration between academia and industry, creating opportunities for future research and innovation. We look forward to your enthusiastic participation and valuable contributions in making ICSIDE'26 a grand success.

Warm regards,

.

SECRETARY

MESSAGE FROM PATRON



Dr. S. POORNACHANDRA

I am delighted to announce the First International Conference on Smart Intelligence and Data Engineering (ICSIDE'26), a prestigious platform for sharing the latest research, innovations, and advancements in this rapidly evolving field. The conference offers a valuable opportunity for students, faculty members, researchers, and industry professionals to enhance their knowledge and stay updated with emerging technologies and trends. We encourage active participation from students, research scholars, and faculty members through paper submissions and presentations on relevant topics. Your contributions will enrich the conference discussions and showcase the talent, research potential, and expertise within our institution. ICSIDE'26 will also serve as an excellent platform for networking and collaboration, enabling participants to interact with industry experts, academicians, and researchers from various domains.

I look forward to your enthusiastic participation and valuable contributions in making ICSIDE'26 a grand success.

.

Warm regards,

PRINCIPAL

ABOUT THE INSTITUTION

St. Peter's College of Engineering and Technology is managed by Lakshmi Saraswati Educational Trust. It was started in the year 2008. The College is affiliated to Anna University Chennai and approved by AICTE, New Delhi. The College is located in a calm and serene environment in Avadi and is easily commutable by bus and train.

The college runs the following programme:

- B.Tech. (Artificial Intelligence and Data Science)
- B.E. (Computer Science and Engineering)
- B.E. (Electronics and Communication Engineering)
- B.E. (Electrical and Electronics Engineering)
- B.E. (Mechanical Engineering)
- B.Tech.(Biotechnology)
- B.Tech.(Chemical Engineering)
- B.Tech.(Information Technology)
- B.Tech. (Computer Science and Business Systems)
- M.B.A.
- M.E. (Computer Science & Engineering)

ABOUT THE DEPARTMENT

The Department of Artificial Intelligence and Data Science (AI & DS), established in the academic year 2023–2024, focuses on providing strong foundations in Artificial Intelligence, Data Science, and intelligent computing technologies. The department emphasizes practical learning through hands-on training, industry-oriented tools, internships, projects, industrial visits, and value-added courses. Through collaborations and MoUs with industries, students are encouraged to develop technical, research, and problem-solving skills aligned with current industry demands.

The department was initially established with an intake of 60 students, and due to the increasing demand and growing opportunities in the field of AI and Data Science, the intake was increased to 120 students in the academic year 2025–2026.

ABOUT THE CONFERENCE

The First International Conference on Smart Intelligence and Data Engineering (ICSIDE'26) aims to provide a global platform for researchers, academicians, industry professionals, and students to share innovative ideas, research contributions, and recent advancements in the fields of Artificial Intelligence, Data Science, and Intelligent Computing. The conference focuses on emerging technologies, smart systems, secure computing, and data-driven solutions that address real-world challenges across various domains. ICSIDE'26 encourages interdisciplinary research, knowledge exchange, and collaboration between academia and industry. The conference will feature keynote talks, technical paper presentations, and discussions by experts from reputed institutions and organizations. It offers participants an opportunity to explore current trends, present research findings, and interact with leading researchers and professionals from around the world. The conference aims to promote innovation, research excellence, and technological development while fostering networking and collaborative opportunities among participants.

The objectives include the following:

- To discuss the various fundamental artificial intelligence techniques in machine translation.
- To provide a platform for researchers, academicians, industry experts, and students to exchange innovative ideas and research findings.
- To discuss recent advancements in Artificial Intelligence, Data Science, Machine Learning, and Intelligent Systems.
- To explore emerging technologies, challenges, and ethical aspects in smart computing and data engineering.
- To encourage interdisciplinary research and real-world problem-solving using intelligent and data-driven approaches.
- To promote collaboration, networking, and global knowledge sharing among participants.

CALL FOR PAPERS

- Track 1: Artificial Intelligence and Machine Learning
- Artificial Intelligence and Intelligent Systems
- Machine Learning and Deep Learning
- Generative AI and Large Language Models
- Explainable and Responsible AI
- Natural Language Processing
- Computer Vision and Image Processing
- Speech and Text Analytics
- Computational Intelligence and Optimization Techniques
- Track 2: Data Science and Analytics
- Data Science and Predictive Analytics
- Big Data Analytics and Data Engineering
- Data Mining and Knowledge Discovery
- Recommender Systems and Personalization
- Track 3: Cloud, IoT and Smart Technologies
- Cloud Computing and Distributed Systems
- Edge Computing and Fog Computing
- Internet of Things (IoT) and Smart Systems
- Wireless Sensor Networks and Embedded Systems
- Smart Cities and Intelligent Transportation Systems
- Smart Agriculture and Environmental Monitoring
- Digital Twin and Smart Manufacturing
- Track 4: Cyber Security and Trustworthy Computing
- Cyber Security and Secure Computing

- Blockchain and Decentralized Applications
- Ethical AI and Trustworthy Computing
- Track 5: Intelligent Applications and Emerging Technologies
- Robotics and Intelligent Automation
- Industrial AI and Automation
- AI for Healthcare and Bioinformatics
- AI in Education and E-Learning Systems
- Human Computer Interaction
- Sustainable Computing and Green Technologies
- Quantum Computing Applications
- Emerging Trends in Information and Communication Technology (ICT)

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Department of Artificial Intelligence and Data Science

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Ms. Lavanya R, Assistant Professor
Ms. Devi B, Assistant Professor
Department of Artificial Intelligence and Data Science

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Ms. Grace Prasana M, Assistant Professor -AI&DS

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AI-Based Offline College Smart Chatbot System

D. Jackulin Sharmi, G. Vasanthkumar, R. Tamilvanan,
G. Silambarasan, D. Vignesh,
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

This project develops an AI-based chatbot to handle common college-related enquiries efficiently and effectively. Built using Flask and basic Natural Language Processing (NLP) techniques, the chatbot is capable of understanding user queries and providing instant, relevant responses, thereby significantly reducing the need for manual intervention by administrative staff. The system is designed to simulate human-like interaction, making the communication process more natural and user-friendly for students and other users. The chatbot assists users with a wide range of information, including admissions procedures, available courses, fee structures, hostel facilities, academic schedules, and general campus-related queries. It uses a structured dataset of predefined questions and answers, along with simple NLP methods, to interpret user input and deliver accurate responses. This ensures that even repetitive and frequently asked questions are handled quickly and consistently. By automating the response process, the chatbot improves operational efficiency, ensures 24/7 availability, and enhances the overall user experience. Additionally, the system can be easily updated with new data, making it adaptable to changing institutional requirements. The implementation emphasizes simplicity, scalability, and offline usability, making it a practical and cost-effective solution for educational institutions aiming to modernize their support systems.

.

Keywords: Chatbot, Natural Language Processing, Flask, Machine Learning, Offline System, Educational Technology, Intent Classification

Hybrid Transformer – CNN with Blockchain and Optimal Feature Selection for Real-Time Payment Fraud Detection

Ms. A.Barveen, Information Technology
Mrs. R.Visalatchi,Mrs. D.Porchelvi, M.Ishwarya, S.Karunya, V.Malavika
Artificial Intelligence and Data Science, SRM TRP Engineering College, Tiruchirappalli.

ABSTRACT

The rapid growth of digital finance and e-commerce has increased online fraud risks. Traditional fraud detection methods often fail to detect new fraud patterns and produce high false alarms. This study proposes a hybrid model combining Transformer models and Convolutional Neural Networks (CNN) for real-time fraud detection. Transformer models (TabTransformer, TransTab, BERT, and RoBERTa) are used to capture transaction patterns, while an Enhanced Artificial Flora Optimization (EAFO) algorithm selects important features. A spatial CNN is then used for accurate classification. The system is integrated with cloud and blockchain technology to ensure secure and transparent transactions. The model is tested on PaySim, IEEE-CIS, and European datasets and shows improved accuracy, precision, recall, and AUC. The proposed system is efficient, scalable, and secure for real-time applications. The proposed hybrid fraud detection system combines Transformer models, CNN, and Enhanced Artificial Flora Optimization (EAFO) to accurately identify fraudulent financial transactions in real time. By integrating cloud and blockchain technologies, the system ensures secure, transparent, and scalable transaction monitoring. Experimental results on benchmark datasets demonstrate improved accuracy, precision, recall, and AUC compared to traditional fraud detection methods.

Keywords: Fraud Detection, Transformer Models, CNN, Spatial-Temporal Data, EAFO, Blockchain, Data Security, Data Integrity.

Blockchain Driven Agricultural Insurance System for Rapid Crop Loss Verification and Fair Compensation Distribution

Mrs. M. Angelin Rosy, Mr.M. Ajith kumar, Mr. K. Sedhu,
Department of Master of Computer Applications,
Er. Perumal Manimekalai College of Engineering, Hosur

ABSTRACT

Agriculture remains the backbone of India's livelihood systems, yet farmers continue to struggle with unpredictable weather fluctuations, sudden pest outbreaks, and inconsistent rainfall that can severely affect their crop production. Crop insurance is meant to provide financial protection during such unforeseen losses, but many existing schemes are difficult to access due to lengthy claim procedures, delayed compensation, and limited transparency. As a result, farmers often hesitate to rely on these systems, leading to financial instability during adverse conditions. To overcome these limitations, this project proposes a blockchain-driven crop insurance framework designed to simplify claim handling and enhance trust among stakeholders. The system uses a distributed ledger to securely record policy details, weather updates, land data, and claim requests in a tamper-resistant manner. Smart contracts are deployed within a permissioned blockchain environment, ensuring that only authorized entities such as farmers, insurance providers, and certified weather data agencies can trigger or validate insurance-related operations. These smart contracts automatically execute payout conditions based on verified parameters, significantly reducing manual intervention and minimizing opportunities for fraud. The aim of the proposed model is to deliver a cost-effective and farmer-friendly insurance mechanism that ensures quick, rule-based claim settlements while eliminating the dependence on intermediaries. By integrating blockchain's transparency, decentralization, and immutability, the system enhances accountability and encourages wider adoption of crop insurance. Ultimately, this approach strengthens financial resilience for farmers and contributes to a more reliable and inclusive agricultural support ecosystem.

Keywords: Blockchain, Crop Insurance, Smart Contracts, Distributed Ledger, Automated Claim Processing, Agricultural Risk Management, Weather-Based Insurance, Decentralized Systems, Fraud Prevention, Farmer Financial Security.

JOBQUENCH: An Intelligent and Automated Placement Management System for Enhanced Campus Recruitment

Mr. R. Pushpanathan, S.Haripriya,
Department of Master of Computer Applications,
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

The system focuses on the elimination of the main shortcomings and existing labor-intensive methods of recruitment on campuses by using a centralized and automated platform system. Many placement strategies still exist in the following major issues: putting the employers and students into the wrong positions, actually involves a lot of paperwork, and makes the availability of jobs not well perceived. Using the development tools of the contemporary age, this system empowers the front end in Flutter, the backend data management in Django REST API, and SQLite for light data storage. On ways through which students, placement officers and the employers are created, the system automates functions like resume database, job board as well as scheduling for interviews. There are also complementary services like alumni-mentor and career-profile to help students enhance their employment opportunities. There are positive trends in the first outcomes of the implementation of the Placement System with regard to enhanced satisfaction of users and increased effectiveness of recruitment. The platform also helps minimize time and efforts needed for conducting placement activities as well as enhance partnership between academic environment and industry. This paper defines the systems architectural design and creates an understanding of how the design improves the placement practices, thereby filling the gap between classroom education and employment opportunities for graduates.

Keywords: Campus Recruitment, Placement Management System, Flutter, Django REST API, Career Guidance

Convolution Neural Network-Based Crack Identification of Intelligent Railway Tracks Flask-Deployed

Mr. Ragavan.V, Logeshwari.S.S, Mahalakshmi.S, Information Technology,
Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

The safety of railway transportation requires the ability to detect structural anomalies existing along the rail tracks early because cracks on the surface are still one of the main causes of derailments and service disruptions. Traditional methods of inspection are based on the element of human inspection and periodic patrols that lead to the late detection of faults, poor assessment levels, and even to higher operational costs. These inspection constraints require an automated and valid monitoring mechanism that could be used to facilitate preventative maintenance strategies, the given research paper represents a Railway Track Crack Prediction System that implies deep learning principles and is deployed through the operational monitoring system that is based on Flask. The family of neural network models, Convolutional Neural Network (CNN), receives training on datasets of railway tracks images to identify crack structures and discontinuities on their surface. During feature extraction and classification, it is possible to distinguish defective and non-defective track conditions with a high level of accuracy. Experimental analysis of results finds accuracy of prediction to be 97.6 which confirms the ability to be robust with different lighting and other environmental conditions. Model implementation via Flask framework enables real-time image upload, prediction visualisation and reporting of confidence score. Output implementation is crack status indication, probability values as well as result logs to support maintenance decision making. The automation of detecting defects will greatly reduce manual control dependence, enhance efficiency of response and enhance safety of railway structure. Combination of deep learning-based visual space analysis and web-based implementation creates a scalable platform of intelligent railway tracks monitoring and mitigation of accident risks.

Keywords: Railway infrastructure and safety, Track cracks prediction, Convolutional neural Networks (CNN), Deep learning visual inspection, Image features, Web based deployment model, Predictive maintenance system, Model accuracy evaluation.

Online Marketplace for Local Services and Freelancers

Ms. K.Nishanthini , Gopinath V, Sridhar.R S . Divaker, V .Kamesh,
Computer Science and Engineering,
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

The Online Marketplace for Local Services and Freelancers is a comprehensive web-based platform developed to simplify the process of connecting customers with skilled local service providers. The platform integrates AI-based filtering enabling service discovery by location and budget, a real-time booking system, and a rating and review mechanism to ensure transparency and quality. Service providers benefit from dedicated profile management, availability toggling, and an earnings dashboard. An admin panel provides centralized oversight, including certification-based provider verification and a helpline for dispute resolution. Built using React.js, Node.js, and MongoDB, the system addresses the limitations of traditional service hiring methods, promoting local employment and improving user satisfaction. Results indicate that the platform is scalable, reliable, and suitable for real-world deployment. The proposed Online Marketplace for Local Services and Freelancers provides an intelligent and user-friendly platform for connecting customers with verified local service providers. By integrating AI-based filtering, real-time booking, and transparent rating systems, the platform enhances service discovery, reliability, and customer satisfaction. Developed using React.js, Node.js, and MongoDB, the scalable framework supports local employment opportunities while offering efficient management and dispute resolution features through a centralized admin system.

Keywords: online marketplace, freelancers, local services, AI-based filtering, web application

Tourism and Travel Booking Website

Mr.R.Pushpanathan, Kaviya P
Department of Master of Computer Applications
Thirumalai Engineering College. Kilambi, Kancheepuram.

ABSTRACT

The tourism industry is one of the dynamic and rapidly growing sectors globally, with millions of travellers seeking information, inspiration, and booking services online. In response to this demand, the development of a comprehensive tourism website emerges as a crucial tool for travellers, tour operators, and destination management organizations alike. This abstract outlines the key components and objectives of a tourism website project aimed at providing users with a seamless and immersive travel experience. The proposed tourism website aims to serve as a centralized platform that offers a plethora of features and functionalities catering to the diverse needs and preferences of travellers. From browsing destinations and attractions to planning itineraries, booking accommodations, and accessing travel guides, the website endeavours to streamline the entire travel process while fostering engagement and interaction among users. In the contemporary era, travel has transcended from being a mere activity to becoming an integral part of lifestyle and leisure. With the advent of technology, the way people plan and experience their journeys has undergone a profound transformation. In this context, the development of a Tourism Booking Website emerges as a pivotal endeavour aimed at revolutionizing the travel industry by providing users with a comprehensive platform to discover, plan, and book their dream destinations and experiences. The Tourism Booking Website project seeks to address the evolving needs and preferences of modern travellers by offering a seamless and intuitive online platform that simplifies the entire travel process. From researching destinations and creating personalized itineraries to booking accommodations, transportation, and activities, the website endeavours to empower users with the tools and resources they need to turn their travel aspirations into reality.

Keywords: Tourism Website, Online Booking System, Travel Planning, Destination Management, User Experience

A Study of Flat Feet and Related Health Issues

Muzamil HS, Muhil Adhilakshmi, Mobarak Hossain, Mariyam Masood, Jayashree M K
School of Commerce and Management, Jain University, Bengaluru

ABSTRACT

Feet without an arch - called flat feet - are a common change in how legs form, where the inner curve along the sole sinks down or never forms. Though kids often grow up with low arches and feel fine, teens and grown-ups who keep them can start noticing strain in muscles, shifts in body motion, uneven stance, and ongoing aches that travel into knees, hips, or the spine. What leads there? A mix: family history might load the dice, extra weight adds pressure, loose ligaments play a part, weak tendons near the shinbone contribute too, injuries leave marks, even shoes worn daily plus long hours sitting while staring at screens slowly shape the outcome. Children often have flexible flat feet - research shows rates between 20% and 30%. Yet among adults, about 15 to 25 percent deal with painful cases, shaped by age, job type, and lifestyle factors. As cities grow and more people stand for long hours at work, the impact of flat feet has gained attention in health discussions. When the foot's arch drops, movement changes happen: walking patterns shift, inward rolling increases, pressure spreads unevenly, forces travel upward through the body differently - these shifts open doors to issues like heel pain, knee stress, hip tension, lower back ache. A fresh look at flat feet begins by weaving together body traits, job demands, physical surroundings, and daily habits. Research pulls from both numbers and narratives to map how diagnosis works across different groups. Patterns in who gets affected show up clearly when data spans decades and regions. Treatment paths emerge not from one source but through layered findings. Long term effects on bones and joints take shape slowly, revealed only over time.

Keywords: Flat feet, Pes planus, Posterior tibial tendon dysfunction, Gait abnormalities, Orthotics, Musculoskeletal pain, Arch index.

AQG: Automatic Question Generator

Nithesh S, Abishek V, Bharath Ram S, Dr. S. Pushpa
Computer Science and Engineering
St. Peter's Institute of Higher Education and Research, Chennai

ABSTRACT

Preparing examination question papers is a time consuming and error-prone task for educators. Common problems include question repetition, uneven syllabus coverage, and inconsistent difficulty distribution. This paper presents the Automatic Question Generator (AQG), an AI-driven web application that generates structured, syllabus-aligned question papers automatically. The system leverages the Gemma 7B large language model as its core generation engine, a React-based single-page frontend, a Python Flask REST API backend, and Firebase Fire store for cloud data management. Users specify a course code, examination type (Internal 1, Internal 2, or Semester), and difficulty level; the system then produces a complete question paper divided into Part A (short-answer) and Part B (descriptive) sections in line with standard academic formats. Cloudflare Tunnel provides secure communication between the locally hosted backend and the web frontend. Evaluation through functional, integration, performance, and usability testing confirms that AQG substantially reduces manual effort, minimizes human errors, and improves assessment quality for educational institutions.

Keywords: Automatic question generation, Large Language Model, Gemma 7B, Flask, Firebase, Academic Assessment, NLP

Smart Agriculture Approach using IoT and Data Visualization

Mr.S.Balaji, Greata christina A, Pooja K A, Vinoth Raj G, Dr.S.Pushpa
Computer Science and Engineering
St. Peter's Institute of Higher Education and Research, Chennai

ABSTRACT

Agriculture continues to face challenges due to unpredictable climatic conditions, inefficient resource utilization, and reliance on traditional monitoring methods. This paper presents a smart agriculture framework leveraging Internet of Things (IoT) and data visualization techniques to enable precision farming. The proposed system integrates soil moisture, temperature, humidity, and light sensors with microcrcontrollers (ATmega328/ESP32), cloud storage, and a web-based dashboard. Real-time data acquisition and visualization empower farmers to make informed decisions regarding irrigation scheduling, crop management, and resource optimization. The system reduces manual intervention, improves accuracy, and enhances productivity by automating data collection and providing actionable insights. Experimental assumptions demonstrate that IoT-enabled monitoring can reduce water usage by up to 30% and improve crop yield consistency. This work highlights the potential of IoT-driven smart agriculture in promoting sustainable farming practices through intelligent, data-driven decision-making.

Keywords - Smart Agriculture, Internet of Things (IoT), Data Visualization, Precision Farming, Sensors, Cloud Computing

Web Based Fake News Detection Using NLP

S Abirami, Alvin Patrick V, Lavanya S, Anto Kingsly Christopher D
Loyola Institute of Technology, Chennai.

ABSTRACT

In the modern digital era, the rapid growth of online platforms has significantly increased the spread of information, making it difficult to distinguish between authentic and misleading content. Fake news, in particular, has become a serious concern as it spreads faster than verified information, influencing public opinion and causing social, political, and economic disruptions. Traditional methods of verifying news, which rely heavily on manual fact-checking, are time-consuming and incapable of handling the massive volume of data generated every day. This project, titled *Web-Based Fake News Detection Using Natural Language Processing (NLP)*, aims to address these challenges by developing an automated system capable of identifying and classifying news articles as either fake or real. The system utilizes advanced NLP techniques combined with machine learning algorithms to analyze textual patterns and linguistic features in news content. The proposed solution is implemented as a web-based application, allowing users to input news articles and receive instant classification results. The system processes the input text through multiple stages, including preprocessing, feature extraction, and classification, to generate accurate predictions. By achieving a high level of accuracy and providing real-time results, the system helps reduce the spread of misinformation. Overall, this project demonstrates the effectiveness of combining NLP and machine learning in tackling the problem of fake news detection, offering a scalable, accessible, and efficient solution for modern information challenges.

Keywords: Fake News Detection, Natural Language Processing, Machine Learning, Web Application, Text Classification

AI-Based Healthcare Monitoring System with Facial Emotion Detection and Multimodal Analysis

Nithya Sree S, Elakya T, Gobinath R
Department of Computer Science and Engineering
Kathir College of Engineering, Coimbatore

ABSTRACT

The rapid increase in the elderly population and individuals with disabilities has created a demand for intelligent healthcare monitoring systems capable of providing continuous and real-time support. Traditional healthcare systems lack the ability to monitor emotional well-being and behavioral changes effectively. This paper proposes an AI-based healthcare monitoring system that integrates facial emotion detection, speech analysis, and multimodal data processing to assess both physical and emotional health conditions. The system utilizes deep learning models for emotion recognition and provides real-time monitoring through a web-based dashboard. Caregivers receive alerts when abnormal emotional or health conditions are detected. The proposed system enhances patient care, reduces caregiver workload, and enables early detection of health issues, thereby improving the quality of life for vulnerable individuals. The proposed AI-based healthcare monitoring system combines facial emotion detection, speech analysis, and multimodal data processing to provide continuous real-time assessment of patients' physical and emotional well-being. By leveraging deep learning models and a web-based monitoring dashboard, the system enables early detection of abnormal conditions and instant caregiver alerts. The framework improves patient care, supports timely medical intervention, and enhances the quality of life for elderly individuals and people with disabilities.

Keywords: Artificial intelligence, healthcare monitoring, emotion detection, deep learning, multimodal analysis, caregiver alert system

An AI-Driven Platform for Smart Interview Preparation and Evaluation

Vignesh P, Tamil Tharshini MS, Naveen N, Mohamed Irfan S,
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

This paper presents an AI-driven web-based platform for smart interview preparation and candidate evaluation. The proposed system simulates a real-time interview environment where an AI interviewer interacts with users and evaluates their responses. It utilizes speech-to-text technology to analyze grammar, fluency, and clarity of spoken answers. Machine learning models are employed to assess candidate performance and generate personalized feedback for improvement. Unlike existing systems that provide static questions or limited evaluation, the proposed platform offers interactive sessions with real-time analysis and feedback. The system aims to enhance communication skills, boost confidence, and improve overall interview performance for students and job seekers. The proposed AI-driven interview preparation platform provides a realistic and interactive interview experience through real-time candidate evaluation and personalized feedback. By integrating speech-to-text technology and machine learning models, the system analyzes communication skills such as grammar, fluency, and clarity of responses. The platform enhances interview readiness, confidence, and performance, making it a valuable tool for students and job seekers.

Keywords: Artificial Intelligence, Interview Preparation, Speech Analysis, Machine Learning, Candidate Evaluation

Crop Yield Prediction using Data Engineering Pipelines: A Machine Learning Approach

Nithya Priya N, Lokesh K, Mohana Vignesh S
Department of Artificial Intelligence and Data Science
Kamaraj College of Engineering and Technology

ABSTRACT

Agriculture plays a significant role in the Indian economy, contributing substantially to GDP and employment, yet crop production remains highly affected by uncertainties such as rainfall variability, soil conditions, and market fluctuations. Traditional statistical approaches often fail to capture the complex non-linear relationships between agricultural factors and crop yield. This work proposes a crop yield prediction framework that emphasizes a structured data engineering pipeline combined with machine learning techniques for improving predictive performance. The pipeline includes data ingestion, preprocessing, state-wise median imputation, outlier detection using the IQR method, feature engineering, and encoding of crop categories. A dataset containing over 246,000 observations from 30 Indian states covering multiple crop seasons was used for model development. Linear Regression, Decision Tree, and Random Forest models were evaluated, with Random Forest achieving the best performance (R^2 = 0.91, MAE = 0.082, RMSE = 0.108). The results demonstrate that effective data preparation significantly improves prediction accuracy, making the proposed framework suitable for agricultural planning and policy support.

Keywords: Crop Yield Prediction, Machine Learning, Random Forest, Data Preprocessing, Precision Agriculture

AI IoT Based Disease Diagnosis using Vital Health Parameters

Mrs. Drishyadarwin, Kaviyalakshmi J, TharuniPriya M.P,
Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

The rapid growth of digital healthcare necessitates the development of advanced diagnostic support systems capable of integrating clinical documentation with physiological monitoring data. This paper presents an IoT-based disease diagnosis system that combines medical textual reports with real-time biosensor measurements obtained through microcontroller-based wearable devices. Clinical records are processed using linguistic techniques to extract symptoms, diagnostic observations, and laboratory indicators, while sensor data including heart rate, oxygen saturation, body temperature, and blood pressure undergo filtering, normalization, and signal conditioning to improve reliability. A feature fusion approach is employed to generate a unified medical dataset suitable for computational diagnostic modeling. Pattern recognition algorithms analyze the integrated data to identify potential pathological conditions and abnormal physiological variations. Experimental validation demonstrates a diagnostic accuracy of approximately 96.4%, indicating strong predictive capability and clinical reliability. The system automatically generates diagnostic outputs in the form of health assessment reports and provides physician alerts when abnormal values are detected. Secure database storage supports longitudinal health tracking, retrospective clinical analysis, and decision-making. The results highlight improvements in diagnostic efficiency, continuous monitoring, and timely medical response, demonstrating significant potential for smart healthcare environments, remote patient monitoring, and early disease detection.

Keywords: Healthcare Monitoring with IoT, Medical Text Mining, Clinical Report Analysis, Physiological Signal Processing, Sensor Integration, Multimodal Health Data Fusion, Disease Diagnosis Modeling, Vital Sign Monitoring, Automated Clinical Alerts, Diagnostic Accuracy.

ACAD Assist: An AI-Based Knowledge Reasoning System for Academic Assistance

Dr. S. Usha, Artificial Intelligence and Data Science,
Bala Adhish N D, SobhiyaM, Aswin P, Computer Science and Engineering,
Kathir College of Engineering, Coimbatore.

ABSTRACT

ACAD Assist is an academic support system that uses Retrieval-Augmented Generation (RAG) technology, aimed at helping engineering students get more accurate and reliable answers from AI. Regular language models can sometimes give wrong or off-topic responses because they don't have enough specific knowledge about the subject. To fix this, ACAD Assist combines semantic search with transformer-based AI models to create responses that are aware of the context and backed by solid evidence. The system works with a well-organized collection of course materials, including lecture notes, textbooks, and other learning resources. It finds relevant information through semantic search and then uses that information to generate precise and explainable answers. This method helps reduce false information and makes the system more trustworthy. The system also uses Natural Language Processing (NLP) to understand what students are asking and provide clear, step-by-step explanations, which helps them learn better. Tests have shown that this system improves the accuracy of answers, decreases the chances of wrong information, and leads to higher satisfaction among users. ACAD Assist provides a flexible and scalable way to offer smart academic help and support personalized learning.

Keywords: Retrieval-Augmented Generation, Academic Assistance, Artificial Intelligence, Natural Language Processing, Semantic Retrieval, Transformer Models, Personalized Learning, Context-Aware Systems.

AgroSmart-Crop Disease Detection and Human Health Monitoring System

Dr.R.Thiagarajan, Mohammed Afthab A, Dharanidharan A, Sanjay S, Venkateswara R,
Information Technology
Vel Tech Multi Tech Dr.Rangarajan Dr.Sakunthala Engineering College, Chennai.

ABSTRACT

AgroSmart is an artificial intelligence (AI) mobile application that was created to facilitate modern agriculture by using the features of intelligent crop health monitoring and online assistance. The methods used in identification of crop diseases are based on traditional methods of manual identification which is laborious, devoid of accuracy, and not accessible to a large number of farmers. This paper is a proposal related to a lightweight Convolutional Neural Network (CNN) model that is implemented into a Flutter-based mobile platform to detect both crop and fruit diseases in real-time. The platform is a combination of image processing and multilingual chat-bot support with an online marketplace to establish an entire agricultural ecosystem. An on-device cloud-based architecture based on Similarity-Preserving Quantization (SPQ) is a hybrid that guarantees high precision and low computational costs. High classification performance is revealed by experimental testing on the basis of a large set of agricultural images with a maximum accuracy of 99.87 using cross-validation. The solution suggested will improve the quality of decisions taken, decrease crop loss and encourage sustainable and technology-based farming in resource-limited surroundings.

Keywords: Crop Disease Detection, Convolutional Neural Network (CNN), Smart Agriculture, Image Processing, Multilingual Chatbot, Digital Marketplace, Similarity-Preserving Quantization (SPQ).

The Dark Side of AI Communication: Ethical Concerns of Deepfakes and Data Misuse

D. Ganga Devi, , Harishma M. Department of Data Science,
The Standard Fireworks Rajaratnam College for Women, Sivakasi.

ABSTRACT

Artificial Intelligence has rapidly transformed modern communication through chatbots, virtual assistants, automated content generation, and deep fake technologies. While these advancements improve efficiency and accessibility, excessive reliance on AI-driven communication raises significant ethical concerns. This paper examines the challenges associated with the overdependence on AI in communication, particularly focusing on deep fake manipulation, misuse of personal data, bias in automated responses, and the spread of misinformation. Such issues threaten privacy, trust, and the authenticity of human interaction in digital environments. To address these concerns, the paper proposes a responsible AI communication framework that emphasizes human oversight in sensitive contexts and transparency in AI-generated content. The study highlights the importance of ethical guidelines, digital literacy, and awareness to ensure that AI technologies support rather than replace responsible human communication. By analyzing the risks and proposing practical solutions, this paper aims to contribute to the development of safer and more trustworthy AI-driven communication systems.

Keywords: Deep fake Detection, AI Ethics, Blockchain, Watermarking, Media Authentication

Crop Yield Prediction with Water and Fertilizer Recommendation (Based on Indian Agriculture)

Mrs. S.Santhanalakshmi, Jeeva M, Lithish L,

Information Technology, Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

This project presents a machine learning–based system for crop yield prediction with integrated water and fertilizer recommendations. Using comprehensive datasets that include soil properties, weather conditions, rainfall patterns, and historical crop data, the model is designed to generate accurate and reliable forecasts tailored specifically for Indian agricultural conditions. By analyzing these multiple factors together, the system is able to identify patterns and relationships that help in predicting expected crop yields more effectively. The system features a simple and user-friendly interface where farmers can input basic details such as soil type, crop selection, and local weather information. Based on this input, the platform provides personalized guidance that includes optimized irrigation schedules and balanced fertilizer recommendations suited to the specific crop and environmental conditions. These recommendations aim to reduce unnecessary resource usage while maintaining soil health and crop quality. In addition, the system helps farmers make informed decisions by offering insights that can minimize risks associated with unpredictable weather and improper resource management. By integrating technology with traditional farming practices, the solution promotes sustainable agriculture, efficient utilization of water and fertilizers, and improved overall productivity. This approach ultimately contributes to better crop management, increased farmer profitability, and long-term agricultural sustainability.

Keywords: Machine Learning, Crop Yield Prediction, Precision Agriculture

Automated Osteoporosis Detection Using Inception V3 And Grad-Cam with Flask-Based Clinical Recommendation System

V.Divya,, Keerthana R, Bhammidipati Kameshwari Gayathri, Keerthana J,Subiksha E
Computer Science and Engineering,
Vel Tech Multi Tech Dr Rangarajan Dr Sakunthala Engineering College, Chennai.

ABSTRACT

Osteoporosis is a chronic bone disease characterized by reduced bone density and increased fracture risk. Early detection plays a crucial role in preventing severe complications and improving patient outcomes. This project presents an automated system for early detection of Osteoporosis using a lightweight deep learning model based on the InceptionV3 architecture. The model classifies bone conditions into Normal, Osteopenia, and Osteoporosis using medical imaging data. A Flask-based web interface enables users to upload images and receive instant predictions. Grad-CAM is incorporated to enhance interpretability by highlighting affected bone regions. The system also provides personalized recommendations for each condition. This user-friendly and efficient approach supports early diagnosis and improves clinical decision-making. The proposed automated Osteoporosis detection system utilizes the InceptionV3 deep learning model to accurately classify bone conditions into Normal, Osteopenia, and Osteoporosis using medical imaging data. Integrated with a Flask-based web interface and Grad-CAM visualization, the system provides real-time predictions along with interpretable insights into affected bone regions. The framework supports early diagnosis, personalized recommendations, and improved clinical decision-making through an efficient and user-friendly healthcare solution.

Keywords: Osteoporosis, Deep Learning, Medical Imaging, Early Detection, InceptionV3, Grad-CAM, Flask.

Log-Driven AI Framework for Automatic Incident Ticketing

Hariharan V, Rajalakshmi I,
Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

In modern distributed Linux-based IT infrastructures, system administrators rely heavily on manual log inspection and user-reported issues for incident management. Existing approaches predominantly involve reactive monitoring, where administrators analyse heterogeneous system logs such as boot, authentication, network, and disk logs after failures occur. This manual and rule-based process is time-consuming, error-prone, and does not scale effectively with the increasing number of hosts, often resulting in delayed incident detection, improper prioritization, and inefficient ticket assignment. To address these limitations, this paper proposes a log-driven AI framework for automatic incident ticketing in distributed Linux environments. The proposed system centrally collects real-time logs from multiple Linux hosts using lightweight log collectors such as Syslog and Apache Kafka. The collected logs undergo a pre-processing phase that includes noise removal, normalization, timestamp alignment, and tokenization to prepare structured inputs for intelligent analysis. An AI-based log analysis engine then performs feature extraction using natural language processing techniques and statistical pattern analysis, followed by unsupervised anomaly detection and issue classification across system domains such as authentication, disk usage, network activity, and boot processes. A key contribution of the proposed architecture is its dual-mode ticket generation mechanism. Upon log analysis, the system autonomously generates incident tickets for detected anomalies through an AI-based automatic ticket-raising module. In parallel, the architecture supports a user-initiated manual ticket-raising module, allowing end users to report issues that may not yet appear as detectable anomalies in system logs. Both ticket streams are unified and forwarded to a smart administrator assignment engine, which dynamically allocates tickets based on administrator availability, workload balancing, random assignment, and skill relevance. The framework integrates a web-based dashboard for real-time monitoring, ticket visualization, remote access, and feedback, improving fault detection accuracy, reducing response time, and enhancing efficiency in large-scale Linux environments.

Keywords: Log-Driven AI, Automatic Incident Ticketing, Distributed Linux, Syslog, Apache Kafka, LLM, Llama 3, Anomaly Detection.

Real-Time Detection of Rogue DHCP Attack in Enterprise Networks Using Packet-Level Analysis and Machine Learning

Sheshathri A, Arun Pandy. P, Praveen Kumar. T,
Computer Science and Engineering (Cyber Security),
Gojan School of Business and Technology, Chennai

ABSTRACT

Rogue Dynamic Host Configuration Protocol (DHCP) attacks allow unauthorized servers to assign malicious network configurations, posing serious security threats in enterprise environments. Traditional mitigation techniques such as signature-based intrusion detection systems and DHCP snooping suffer from limitations including hardware dependency and inability to detect zero-day attacks. This paper proposes a real-time rogue DHCP detection framework using packet-level analysis and machine learning. DHCP packets are captured in real time, relevant features are extracted, and a lightweight machine learning classifier is used to distinguish legitimate and rogue DHCP servers. Experimental analysis demonstrates that the proposed system improves detection accuracy while maintaining low latency, making it suitable for enterprise network deployment**.** Using packet-level analysis and machine learning for real-time rogue DHCP detection can tackle limitations of traditional methods. The framework captures DHCP packets in real-time, extracts relevant features, and uses a lightweight ML classifier to distinguish rogue vs legitimate servers, improving detection accuracy with low latency. It reduces man-in-the-middle attack risks, detects zero-day attacks, and is hardware-independent. Real-time detection enables swift mitigation, but the ML model needs periodic updates. Testing showed promising results in enterprise-like setups. Deployment involves tapping into network traffic or SPAN ports, requiring low latency to minimize disruption. Accuracy is key to avoid false positives/negatives. The framework can integrate with existing security tools and shows potential for other network anomaly detection. Future work could explore encrypted DHCP traffic analysis. Overall, it's a promising approach for enhancing network security. The solution is scalable and adaptable to different network environments. Effective feature selection is crucial for the ML classifier's performance. Continuous monitoring is needed for optimal detection. Incident response teams can benefit from such real-time alerts.

Keywords: Rogue DHCP, Packet-Level Analysis, Machine Learning, Intrusion Detection, Enterprise Networks.

AI Early Cyber Attack Prediction System Isolation Forest for Real-Time Threat Detection

Divya Dharshini E, Monisha S, Loyola Institute of Technology, Chennai.

ABSTRACT

Cyber threats have evolved into sophisticated, multi-vector attacks that evade conventional intrusion detection systems. This research presents an AI-driven Early Cyber Attack Prediction System that integrates Isolation Forest - an unsupervised machine learning algorithm for real-time anomaly detection within network traffic streams. The proposed system ingests live telemetry data, processes feature vectors through a trained ML pipeline, and surfaces actionable threat intelligence via a RESTful API and a cybersecurity dashboard. Unlike signature-based methods, the system identifies zero-day threats by profiling behavioral deviations, achieving early warning before an attack fully materializes. Results demonstrate superior detection latency and accuracy compared to traditional IDS frameworks. The proposed AI-driven Early Cyber Attack Prediction System utilizes Isolation Forest–based anomaly detection to identify suspicious network behavior and predict cyber threats in real time. By analyzing live telemetry data and behavioral deviations, the system effectively detects zero-day and multi-vector attacks that bypass traditional signature-based intrusion detection systems. Integrated with a RESTful API and cybersecurity dashboard, the framework provides fast, accurate, and actionable threat intelligence for proactive network security management.

Keywords: Cyber Attack Prediction, Isolation Forest, Anomaly Detection, Intrusion Detection System, Machine Learning.

NEUROMORPHIC COMPUTING: Breaking the Von Neumann Bottleneck for Edge AI

Mr.Sanjay.S, Mr.Nishal Ahamed.I, Mr.Santha Kumar.N, Mrs.Abirami.S
Loyola Institute of Technology,Chennai.

ABSTRACT

Neuromorphic computing represents a paradigm shift from traditional von Neumann architectures by emulating brain-inspired event-driven processing. This paper examines how neuromorphic systems overcome the fundamental von Neumann bottleneck through co-located memory and processing, achieving eight orders-of-magnitude energy efficiency improvements. Analysis of commercial platforms (Intel Loihi 2, BrainChip Akida) demonstrates milliwatt-range power consumption with microsecond latency, enabling real-time edge AI for autonomous robotics, predictive maintenance, and IoT devices. While hardware capabilities have matured significantly, standardized software frameworks remain the primary adoption barrier. Emerging frontiers including memristive in-memory computing and photonic neuromorphics position these systems as the definitive architecture for physical AI in power-constrained environments.

Keywords: Neuromorphic Computing, Spiking Neural Networks, Edge AI, Von Neumann Bottleneck, Event-Driven Processing.

LinguaShield: Secure Multilingual Exchange Using Cloud Computing

MohanaPrasath R, Rohidh A, Vignesh M, Sivachandran S
Computer Science and Engineering,
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai

ABSTRACT

The rapid adoption of cloud-based communication platforms has increased the need for systems that can simultaneously address language barriers and data security. LinguaShield is a cloud-hosted multilingual communication platform designed to integrate real-time neural machine translation with strong cryptographic protection. The proposed system employs RSA-based asymmetric key exchange, Advanced Encryption Standard message encryption, and digital signatures to ensure confidentiality, integrity, and sender authenticity during communication. A three-tier authentication framework is implemented, consisting of password validation, email-based one-time passwords, and facial biometric verification to prevent unauthorized access. In addition, an intrusion detection layer continuously monitors behavioral patterns to identify suspicious activity in real time. Experimental evaluation demonstrates 92% biometric verification accuracy, encryption latency below 20 milliseconds, and intrusion detection rates above 90%, indicating that strong security and real-time multilingual communication can coexist efficiently within a scalable cloud-based framework.

Keywords: Multilingual Communication, End-to-End Encryption, RSA, AES, Intrusion Detection

NUMERA: AI-Powered Mathematical Reasoning and Visualization Engine with External Knowledge

Tharun Kumar R, Srinivasan T, Sanjay M, Magesh S, Mrs. Dukitha M.
Computer Science and Business System
Er. Perumal Manimekalai College of Engineering, Hosur.

ABSTRACT

NUMERA is an intelligent numerical reasoning application designed to make mathematics more interactive, understandable, and visually engaging. The system integrates a modern frontend interface, a powerful backend processing layer, and AI model APIs to provide accurate mathematical reasoning and problem-solving capabilities. Users can input mathematical problems in natural language, and the system processes the query through backend services that connect with advanced AI models. These models analyze the problem and generate step by step logical Integration of multiple mathematical engines and AI services, by combining AI-powered reasoning, computational tools, and visual learning, NUMERA serves as a comprehensive platform for students, educators, and researchers seeking a deeper understanding of mathematical concepts through step-by-step explanations and interactive visualization. NUMERA is an AI-powered numerical reasoning platform that enables interactive and intelligent mathematical problem solving through natural language input and step-by-step explanations. By integrating advanced AI models, computational engines, and visual learning tools, the system enhances understanding of complex mathematical concepts in an engaging manner. The platform serves as a comprehensive educational solution for students, educators, and researchers by combining accurate reasoning, interactive visualization, and user-friendly design.

Keywords: Artificial Intelligence - Natural Language Processing (NLP) - Mathematical Reasoning - Symbolic Computation - Data Visualization - Machine Learning - Educational Technology - Computational Intelligence - Interactive Visualization.

Deep Learning for Diabetic Prediction: A Comprehensive Review

Mrs. S. Jothilakshmi, Dr. K. Srinivasan,
Department of Computer Science and Applications,
SCSVMV University, Kancheepuram,

ABSTRACT

Diabetes mellitus remains a global epidemic affecting over 463 million people worldwide, with projections indicating an increase to 700 million by 2030. Early diagnosis and risk prediction are critical for timely intervention and improved patient outcomes. Deep learning (DL) has emerged as a transformative technology in healthcare, particularly for diabetes prediction and complication detection. This comprehensive review analyzes the current landscape of deep learning approaches for diabetic prediction, with particular emphasis on papers published between 2025 and 2026. We systematically evaluate state-of-the-art architectures including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Transformer models, and hybrid architectures. Our analysis covers data preprocessing techniques, feature engineering strategies, class imbalance handling methods, explainability frameworks, and clinical validation approaches. Key findings indicate that hybrid models combining multiple architectures achieve superior performance, with accuracy rates exceeding 99.8% on benchmark datasets. Transformer-LSTM models demonstrate enhanced capability in capturing temporal patterns in glucose data, while ensemble methods with SMOTE (Synthetic Minority Over-sampling Technique) provide robust performance on imbalanced datasets. Furthermore, the integration of explainable AI techniques (SHAP, LIME) and federated learning approaches has enhanced clinical interpretability and patient privacy. This review identifies critical challenges including dataset scarcity, class imbalance, computational efficiency, and generalization across diverse populations, while highlighting emerging trends in multimodal AI, wearable integration, and personalized medicine for diabetes care.

Keywords: Deep Learning, Diabetes Prediction, CNN, LSTM, Transformer, Machine Learning, Healthcare AI, Early Detection, Explainable AI, Risk Prediction.

Worksense AI -Smart Workplace Productivity Monitoring System Using Machine Learning

A.Anusuya, K. Ashika, M. Jasima Jasmine, M. Vetrivel Dr. R. Kishore
Artificial Intelligence and Data Science
M.A.M College of Engineering and Technology, Tiruchirappalli.

ABSTRACT

This paper presents *WorkSenseAI*, a smart workplace productivity monitoring system designed to improve employee health and efficiency through real-time physiological monitoring. The system collects health-related data such as heart rate, skin response, and body temperature using sensors integrated with a microcontroller. Machine learning algorithms, including Random Forest and Support Vector Machine (SVM), are employed to analyze the collected data and predict stress conditions as normal or abnormal. Based on the prediction results, the system automatically controls environmental parameters, such as lighting, through a relay module to create a comfortable working environment. Additionally, a real-time dashboard is implemented to visualize physiological data and system status. The proposed system is cost-effective, easy to implement, and highly suitable for modern smart workplaces, enhancing both employee well-being and productivity.

Keywords: Machine Learning, Internet of Things (IoT), Stress Monitoring, Workplace Productivity, Smart Automation

Smart Voice-Enabled Medication Adherence System

Dr. S. Rajalakshmi, Ramya S, Santoshi R V, Savita M
Sri Venkateswara College of Engineering, Sriperumbudur.

ABSTRACT

Medication adherence is a critical factor in ensuring successful healthcare outcomes, particularly for elderly individuals, patients with chronic conditions, and those prone to forgetting medication schedules. Traditional methods of managing medications often result in non-adherence due to forgetfulness, complexity, and lack of effective monitoring. This paper proposes a Smart Voice-Enabled Medication Adherence System that integrates Artificial Intelligence, Machine Learning, and mobile technologies to improve adherence. The system includes an OCR-based prescription scanning module for digitizing medication details, voice-enabled reminders for timely alerts, and a predictive analytics module to assess the risk level of medication adherence. A dual-platform architecture is implemented, comprising a web-based interface for caregivers and a mobile application for patients. The mobile application follows an offline-first approach to ensure usability for individuals with limited technical proficiency or internet access. Machine learning techniques, specifically the Random Forest algorithm, are employed to predict adherence patterns and potential delays. The proposed system enhances patient compliance, reduces health risks, and provides an efficient and scalable solution for medication management. It supports real-time monitoring, improves healthcare outcomes, and offers a comprehensive framework for intelligent healthcare services.

Keywords: Medication Adherence, Smart Healthcare System, Machine Learning, Predictive Analytics, Optical Character Recognition (OCR), Voice-Enabled Reminder System, Random Forest, Mobile Health Application, Healthcare Monitoring, Patient Compliance

College ERP System – Advanced Web-Based Enterprise Solution for Educational Institutions

Sugumar L, Suresh Ramesh, .Ms.D. Jackulin Sharmi
Computer Science and Engineering,
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

This paper presents an advanced College ERP System designed to automate and optimize administrative operations in educational institutions. The system integrates student, staff, and financial management into a centralized platform using modern web technologies. Built with Node.js, Express.js, and MongoDB, the system ensures scalability, high performance, and flexibility. Security is enhanced using JWT authentication, bcrypt hashing, and role-based access control. The platform provides real-time analytics, improving decision-making and operational efficiency. Cloud deployment enables remote accessibility and system reliability. The proposed College ERP System automates and centralizes academic, administrative, and financial operations within educational institutions using modern web technologies. Developed with Node.js, Express.js, and MongoDB, the platform offers scalability, secure access control, and real-time analytics for efficient decision-making. Cloud deployment further enhances accessibility, reliability, and overall institutional management efficiency.

Keywords: College ERP, Cloud Computing, Node.js, MongoDB, Education Technology

Emotion-to-Code Compiler (ETCC) with Ethical Filtering and Empathetic UI Generation

Mr.Sakthivel R, Ms.Dharshini K, Ms.Emagulet Nivetha S,Mrs. Abirami,
Loyola Institute of Technology, Chennai.

ABSTRACT

The proliferation of large language models (LLMs) and affective computing frameworks has enabled a paradigm shift in how human intent is translated into executable software artifacts. This paper proposes the Emotion-to-Code Compiler (ETCC), a multi-layered theoretical system architecture that bridges the semantic gap between natural human emotional expression and structured code generation. The ETCC integrates affective state modeling, semantic intent extraction, rule-based feature mapping, and empathetic user interface (UI) generation within a unified pipeline. An ethical filtering mechanism, grounded in human-centered AI paradigms, governs output safety and toxicity mitigation. Experimental demonstration on representative use cases confirms that ETCC reduces perceived cognitive load by 47% and accelerates prototyping cycles by 63% relative to conventional manual coding workflows. The system represents a novel contribution to the intersection of natural language processing, affective computing, and responsible AI-driven software synthesis.

Keywords: Affective Computing, Emotion-to-Code Compilation, Semantic Intent Extraction, Ethical AI Filtering, Empathetic UI Generation, Large Language Models.

A novel Blockchain-Based Integrity and Reliability of Veterinary Clinics Information Management System Using Predict Analysis

Priyanka P, Vishnupriya V, Swathi G, Mrs. Dukitha M
Computer Science and Business Systems
Er. Perumal Manimekalai College of Engineering, Hosur

ABSTRACT

Traditional veterinary information management systems (VIMS) face significant challenges regarding data security, reliability, and interoperability due to their centralized architecture, which acts as a single point of failure. This paper proposes a Reliable and Intelligent Veterinary Information Management System (RIVIMS) that integrates blockchain technology and predictive analytics. The system utilizes a permissioned blockchain framework (Hyperledger Fabric) and the SHA-256 algorithm to ensure data integrity and prevent unauthorized tampering. Furthermore, machine learning models, specifically Deep Neural Networks (DNN) and Multi-directional Long Short-Term Memory (MD-LSTM), are employed to analyze historical appointment data to forecast future healthcare trends and resource demands. Experimental results demonstrate that the MD-LSTM model achieves a superior predictive accuracy with an R2 score of 0.979, while the blockchain component maintains high transaction throughput and low latency. The proposed RIVIMS framework enhances veterinary information management by integrating blockchain technology with intelligent predictive analytics for secure and reliable healthcare data handling. Using Hyperledger Fabric and SHA-256, the system ensures data integrity, transparency, and resistance to unauthorized access or tampering. Additionally, advanced ML models such as DNN and MD-LSTM provide accurate forecasting of healthcare trends and resource requirements, improving operational efficiency and decision-making in veterinary services.

Keywords: Blockchain Technology, Veterinary Information Management System, Predictive Analytics, Deep Learning, Hyperledger Fabric

Transforming Industrial Performance through AI Innovation: A Multidisciplinary Approach

K. Princey, Department of Computer Science, Holy Cross College (Autonomous)
Dr.D. Vinoth Kumar, Department of Commerce, Bishop Heber College(Autonomous)
Tiruchirappalli

ABSTRACT

Artificial Intelligence (AI) is transforming major industrial sectors by improving efficiency, decision-making, and value creation. In manufacturing, AI enables predictive maintenance, digital twins, and automated quality inspection, reducing downtime and increasing productivity. In logistics and supply chains, adaptive routing, demand forecasting, and warehouse automation improve operational efficiency during market fluctuations. The energy sector benefits from AI through smart grid management, load optimization, and accurate renewable energy forecasting. Service industries such as finance and retail use natural language processing, computer vision, and recommendation systems to enhance customer experience and real-time monitoring. However, integrating AI into existing systems presents challenges, including legacy infrastructure limitations, data privacy concerns, and lack of transparency. Ethical issues such as bias and accountability also require attention. Successful AI adoption depends on interdisciplinary collaboration, workforce upskilling, and early integration of ethical principles. Sustainable and responsible AI implementation is essential for long-term industrial and societal benefits.

Keywords: Artificial Intelligence, Industrial Automation, Smart Grid Systems, Predictive Analytics, Ethical AI.

Dyslexia Screening System Handwriting Pattern Evaluation System based on Convolutional feature Modelling

Mr. Jeysankar. N, Sridevi. B, Rahul. R , Information Technology,
Dr. M.G.R Educational and Research Institute, Chennai.

ABSTRACT

Dyslexia is a common neurodevelopmental learning impairment which affects reading, spelling, and written expression even when intelligence and exposure to education are sufficient. Early identification is critical to a timely intervention but the traditional diagnosis practice most frequently employs expert observation thus long, subjective and often inaccessible in educational settings. The current research article presents a computational handwriting analysis by an automated Writing Pattern Analyser, which is used to detect dyslexia-related features in handwriting by the use of cutting-edge computational modelling. Written samples obtained either on scanned papers or on digital tablets are preprocessed such as noise filtering, normalization, segmentation. Pattern categorization is done using convolution-based modelling with the help of statistical classifier. Proposed implementation model delivered by experimental evaluation is shown to have classification accuracy of 95.2, and solid probability scoring and visual analytic reports to educators, clinicians, parents. Live evaluation option helps in early screening, planning individual instructions, longitudinal progress evaluation. In general, the study paper provides research-based assistive diagnostic model that combines educational psychology, handwriting analysis, computational modelling, fostering inclusive academic support and better literacy performance in the learners with dyslexic writing difficulties. The wider usage can potentially diminish the time lag in diagnosis, improve the quality of educational planning, stimulate the collaboration of the specialists, and promote the use of evidence-based intervention approaches in the multifaceted contexts of institutional learning at the global level and the sustainability.

Keywords: Dyslexia Recognition of Handwritten Patterns, Convolutional Neural Network (CNN), Support Vector Machine (SVM), Feature Extraction Method, Image-based Personality Recognition, Predictive Probability Modelling, Stroke Continuity Assessment, Performance Accuracy Analysis, Computational Learning Paradigm.

A Blockchain-Empowered Federated Learning Framework using LSTM and Isolation Forest for Privacy-Preserving Disease Prediction

Prem Kumar S, Harinivas S, Hari Rudhran M , Harish Raj S and Dukitha M
Computer Science and Business Systems,
Er. Perumal Manimekalai College of Engineering, Hosur

ABSTRACT

Electronic Health Record (EHR) fragmentation results in data silos, which hinder the development of strong artificial intelligence in healthcare. Training in a centralized manner is often infeasible owing to stringent data privacy laws such as HIPAA. We propose a novel framework that leverages Horizontal FL and a Permissioned Blockchain-based Audit Layer for facilitating collaboration between multiple hospitals. The predictive models will be based on Long Short-Term Memory (LSTM) networks for predicting patient deterioration using sequential data from vital signs. Additionally, Federated Isolation Forests (FLiForest) will be used for unsupervised anomaly detection. Only model updates will be shared in an encrypted manner, and data will remain within each institution. We will use Differential Privacy (DP) and Homomorphic Encryption (HE) for ensuring algorithmic data privacy. We will also employ a Hyperledger Fabric-based Blockchain for ensuring data provenance and patient consent through smart contracts, thus preventing poisoning attacks. Theoretical analysis of this framework shows that it will achieve a global model prediction accuracy of 4.5% less than the perfect accuracy and will be similar to centralized models. Additionally, it will achieve an accuracy of 96.3% in anomaly detection. The Blockchain will be able to handle high transaction throughput of 110 transactions per second and will have low validation times of 120 ms. Thus, this framework will solve the utility-privacy trade-off and will allow healthcare consortia to develop strong predictive models.

Keywords: Blockchain in Healthcare, Differential Privacy, Electronic Health Records, Federated Learning, Isolation Forest, Long Short-Term Memory (LSTM), Smart Contracts.

Electronic voting System using Blockchain

K.S.Vijayakumar, Kesavan V, Santhosh G.M, and B.S.Charulatha
St.Peter Institute of Higher Education and Research, Chennai.

ABSTRACT

Online voting is a trend that is gaining momentum in modern society. It has great potential to decrease organizational costs and increase voter turnout. It eliminates the need to print ballot papers or open polling stations—voters can vote from wherever there is an Internet connection. Despite these benefits, online voting solutions are viewed with a great deal of caution because they introduce new threats. A single vulnerability can lead to large-scale manipulations of votes. Electronic voting systems must be legitimate, accurate, safe, and convenient when used for elections. Nonetheless, adoption may be limited by potential problems associated with electronic voting systems. Blockchain technology came into the ground to overcome these issues and offers decentralized nodes for electronic voting and is used to produce electronic voting systems mainly because of their end-to-end verification advantages. This technology is a beautiful replacement for traditional electronic voting solutions with distributed, non-repudiation, and security protection characteristics. The main goal of this analysis was to examine the current status of blockchain-based voting research and online voting systems and any related difficulties to predict future developments. An electronic voting application and an introduction to the fundamental structure and characteristics of the blockchain in connection to electronic voting. As a consequence of this study, it was discovered that blockchain systems may help solve some of the issues that now plague election systems with blockchain applications are privacy protection and transaction speed.

Keywords Blockchain, Electronic Voting Systems.

Disseminate Legal Literacy Chatbot with Voice Support for laymen

Dr. C. Senthil Selvi, Tarique Al Jailane. S, Nithish. B
Department of Computer Applications
B.S Abdur Rahman Crescent Institute of Science and Technology, Chennai

ABSTRACT

India's rural population around two-thirds of the country remains largely unaware of its basic legal rights, mainly because legal aid systems like NALSA do not reach deep enough into villages and face serious language barriers, especially for Tamil-speaking communities. Current efforts, such as printed pamphlets and one-off legal-literacy camps, are mostly offline and temporary, so they cannot scale or sustain impact in remote areas. These initiatives also fail to deliver clear, multi-language content in Tamil, English and Hindi, leaving many people uninformed about how laws protect them in everyday situations. On the tech side, most existing legal chatbots work only in English, rely on fixed, outdated information and do not use advanced Indic-language models which demands to understand Tamil and Hindi accurately. They usually lack voice input, feedback loops and a smooth way to escalate complicated queries to the lawyers when needed. Because of this, ordinary rural citizens often do not know how to file an FIR, demand unpaid wages or defend their tenant rights, making them vulnerable to exploitation and injustice.

Keywords: NALSA, IndicBERT, Intent Recognition, Legal Aid Portals.

Intelligent Cyber Attack Prediction Using Machine Learning

Yogesh S A, Ashwin lingam R, Dr.K.M.N. Syed Ali Fathima,
Department of Computer Applications
BS Abdur Rahman Crescent Institute of Science and Technology, Chennai,

ABSTRACT

The rapid expansion of digital technologies and internet-based systems has significantly increased the risk of cyberattacks, making cybersecurity a vital concern in modern computing environments. Traditional security methods, such as signature-based intrusion detection systems, often fail to detect advanced threats like zero-day attacks, unknown malware, and Advanced Persistent Threats (APT). Additionally, the large volume of network traffic makes manual monitoring inefficient and prone to high false positives. This project proposes an Intelligent Cyber Attack Prediction System using Machine Learning (ML) to improve threat detection and prediction. The system analyzes network traffic and user behavior to identify potential attacks in advance. It employs supervised learning algorithms, including Random Forest, Support Vector Machine (SVM), and Decision Tree, to enhance accuracy and reduce false alarms. Using tools like Python, Scikit- learn, Pandas, and NumPy, the model enables real-time classification of activities as normal or malicious, providing a scalable and efficient cybersecurity solution.

Keywords: Cybersecurity, Cyber Attack Prediction, Machine Learning, Intrusion Detection System (IDS), Network Traffic Analysis, Random Forest, Support Vector Machine (SVM), Decision Tree, Anomaly Detection, Real-time Threat Detection.

ScheduleSync: Intelligent Real-Time Schedule Sharing and Coordination System

Victor Jose M, Yuvanesh V, Yokeshwaran R, Gokul Raj S, Kavirayar S,
Computer Science and Engineering,
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai

ABSTRACT

Effective scheduling is one of the major challenges in academic and organizational environments due to increasing workloads, multiple commitments, and unstructured coordination channels. Existing calendar tools are primarily focused on individual time management and lack intelligent collaborative capabilities for resolving complex group availability issues. This paper proposes an intelligent real-time schedule sharing system, ScheduleSync, designed to enable secure collaboration, detect scheduling conflicts, and optimize meeting coordination. The system leverages a cloud-based NoSQL database, event-driven services, and availability-based algorithms to perform efficient schedule analysis, conflict detection, and optimal meeting time selection for both individuals and groups. A privacy-aware sharing model based on mutual connections ensures secure information exchange while preserving collaborative functionality. The proposed system has been evaluated across multiple scenarios, demonstrating reduced scheduling conflicts, sub-second synchronization latency, and improved user satisfaction in meeting coordination compared to traditional manual approaches.

Keywords: Smart Scheduling, Real-Time Systems, Cloud Computing, Calendar Sharing, Meeting Optimization, Mobile Applications.

NAAC Report Generation and Optimization using RAG

Dr.N.Indumathi, S.Kaviya, B.Karthikeyan,T.Madhavamurthy
Computer Science and Engineering
Rajalakshmi Institute of Technology,Chennai

ABSTRACT

The preparation of the Self-Study Report (SSR) for accreditation by the National Assessment and Accreditation Council is a time-consuming and documentation-intensive process. SSR preparation requires extensive collection of qualitative and quantitative institutional data while strictly adhering to predefined accreditation criteria and evaluation manuals. Ensuring consistency, factual accuracy, and compliance across multiple institutional documents presents a major operational challenge. This paper presents an automated framework for NAAC report generation using Retrieval-Augmented Generation (RAG). The proposed system utilizes a vector database-driven knowledge repository integrated with a large language model to generate structured and standards-compliant SSR content. Institutional documents such as previous SSRs, departmental reports, faculty profiles, and performance records are transformed into dense embeddings and stored in a vector database for efficient semantic retrieval. During report generation, relevant institutional content is retrieved and supplied to the language model, enabling context-aware report drafting aligned with NAAC criteria. A post-processing and feedback optimization module is incorporated to improve compliance, clarity, and evaluation readiness. The framework ensures professional formatting, uniformity, factual correctness, scalability, and reliability for AI-assisted SSR automation in higher education institutions.

Keywords: NAAC, SSR Automation, Large Language Models, Retrieval-Augmented Generation, Vector Database, Academic Documentation

Alzheimer's Disease Detection Using MRI Images with Deep Learning Techniques

Aburvaa M, Jeya Sree S, Mr. G. David Raj
Department of Computer Applications,
B.S Abdur Rahman Crescent Institute of Science and Technology, Chennai.

ABSTRACT

Neurodegenerative diseases such as Alzheimer's present significant challenges in detection and diagnosis due to overlapping symptoms and progressive cognitive and motor decline. This study proposes a unified, intelligent diagnostic platform powered by Deep Learning (DL) and Machine Learning (ML) techniques for the efficient classification and detection of Alzheimer's disease. The system integrates advanced Convolutional Neural Networks (CNNs) for imaging data analysis alongside methods for clinical and behavioral data interpretation, enabling multi-modal disease prediction. By leveraging patient datasets comprising MRI scan images, combined with clinical records and symptom scores, the proposed framework aims to enhance diagnostic accuracy and support timely medical intervention. Experimental results demonstrate the system is potential to achieve high precision and recall in differentiating between neurodegenerative conditions, facilitating personalized treatment pathways and improving patient outcomes. The proposed intelligent diagnostic platform combines deep learning and machine learning techniques for accurate detection and classification of Alzheimer's disease using multimodal patient data. By integrating CNN-based MRI image analysis with clinical and behavioral data interpretation, the system improves diagnostic precision and enables early medical intervention. Experimental results indicate high accuracy, recall, and reliability, supporting personalized treatment planning and improved healthcare outcomes for neurodegenerative diseases.

Keywords: Neurodegenerative Diseases, Alzheimer's, , Machine Learning, Deep Learning, CNN, MRI, Clinical Data, Disease Prediction, Medical Imaging.

Cloud Analytics: Warehouse Cost & Pipeline Optimization

R. Pushpanathan, Jagadeesh R, Department of Master of Computer Applications, Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

Cloud computing enables scalable data processing but also introduces significant challenges in managing operational cost and pipeline efficiency. This paper presents Cloud Analytics: Warehouse Cost and Pipeline Optimization**,** a data-driven framework developed to analyze cloud resource usage and improve system performance. The proposed framework utilizes MySQL for data storage, Python for data processing and analytical operations, and Power BI for visualization and reporting. The system evaluates critical parameters including compute hours, storage utilization, data transfer volume, and pipeline execution time to identify high-cost services and operational inefficiencies. An efficiency model is incorporated to measure resource utilization and detect cost-intensive usage patterns. Based on the analytical outcomes, the framework provides optimization insights for query execution and pipeline performance enhancement. Experimental evaluation indicates that the proposed system improves resource utilization and achieves cost reduction of approximately 20–30%. The framework supports better decision-making, enhanced scalability, improved reliability, and efficient cloud resource management.

Keywords: Cloud Analytics, Cost Optimization, Pipeline Optimization, Database, Resource Utilization, Cloud Computing.

Cardiovascular Disease Detection

Dr.R.Selvi, Vijayalakshmi S, Rohitha M, Information Technology,
Dr. M.G.R. Educational and Research Institute, Chennai.

ABSTRACT

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, demanding timely diagnosis and continuous monitoring to prevent severe complications. Traditional cloud-based healthcare systems often suffer from latency, bandwidth limitations, and privacy concerns, making them unsuitable for real-time applications. To overcome these challenges, this project proposes a Real-Time Cardiovascular Disease Detection System using Fog Computing, which leverages the power of edge and fog layers to process patient data locally and provide instant health insights. In the proposed system, biomedical sensors continuously monitor critical physiological parameters such as heart rate, blood pressure, ECG, oxygen saturation, and temperature. A machine learning model is deployed at the fog layer to detect possible cardiovascular abnormalities in real time. This localized computation reduces latency, enhances data security, and ensures that urgent alerts are generated instantly for patients and healthcare professionals. Fog computing acts as an intermediary between the IoT- enabled sensing devices and the centralized cloud, enabling efficient data management and decision-making closer to the data source. The system architecture integrates real-time analytics, anomaly detection, and secure communication protocols to guarantee reliable performance even in low-bandwidth or remote environments. The fog layer not only reduces dependency on the. cloud but also optimizes bandwidth consumption and energy usage, making the entire framework cost- effective and scalable. This research aims to revolutionize remote healthcare monitoring by combining IoT, Fog Computing, and Machine Learning to deliver an intelligent, low-latency, and privacy-preserving cardiovascular disease detection system. Experimental results and simulations validate that the proposed framework achieves faster response times, improved accuracy, and enhanced reliability compared to conventional cloud-centric approaches. The system holds great promise for deployment in hospitals, elderly care centers, and smart health environments to facilitate early detection and prevention of life-threatening cardiac events.

Keywords: IoT, Artificial Intelligence, Health Monitoring, Duty Management, Stress Detection, Fatigue Monitoring, Biomedical Sensors.

A Quantum-Inspired Model for Community Detection in Networks

Dr. A. Jensila Smile, Vani deve Jayavijayan, Ifrath Fathima S, Subbulakshmi V
Department of Computer Science, Sri Sarada College for Women, Salem.

ABSTRACT

Identifying communities in weighted graphs is a fundamental problem in network analysis, with significant applications in social network analysis and the study of complex systems. But finding the best community structures is hard for computers because graph partitioning problems are NP-hard. We suggest a hybrid quantum-classical framework for community detection in this paper. It is based on the Quantum Approximate Optimization Algorithm (QAOA). The method treats the job of finding communities as a Max-Cut optimization problem on weighted graphs made from transaction-based co-occurrence data. In these graphs, nodes stand for items and edge weights show how often they co-occur. By using a classical optimizer to improve parameterized quantum circuits, QAOA finds an approximate solution to the Max-Cut problem. We suggest a recursive splitting technique that uses QAOA on subgraphs over and over again to make binary partitioning work for multi-community discovery. This lets you break down networks in a hierarchical way. The proposed method is evaluated on multiple datasets of differing sizes, showcasing its ability to identify significant and interpretable clusters. The experimental results demonstrate that the recursive QAOA-based method effectively identifies structural linkages within the data while providing a scalable framework for community detection. This research illustrates the efficacy of quantum-inspired optimization techniques in addressing complex graph-related problems.

Keywords: Quantum Approximation Optimization Algorithm, Max-Cut, Community detection, Networks.

AI-Driven Power Transformer Inefficiency Detection And Energy Optimization

Shafiullah Shariff S, Sriramkesaven M, Dr..K.M.N.Syed ali fathima,
Department of Computer Applications
BS Abdur Rahman Crescent Institue of Science and Technology, Chennai.

ABSTRACT

Power transformers play a critical role in electrical power systems by ensuring efficient transmission and distribution of electricity. However, inefficiencies in transformers due to factors such as overloading, temperature rise, voltage fluctuations, and aging components can lead to significant energy losses, reduced performance, and increased operational costs. Traditional monitoring methods rely on manual inspection and basic threshold-based systems, which are not capable of detecting early-stage inefficiencies or handling la rge-scale real-time data effectively. This project proposes an AI-Driven Power Transformer Inefficiency Detection and Energy Optimization System to address these challenges. The system utilizes machine learning algorithms to continuously monitor key transformer parameters such as load, voltage, temperature, power factor, and energy consumption. By analyzing these parameters, the system can identify abnormal patterns, detect inefficiencies at an early stage, and predict potential faults before they lead to failures. The proposed system follows a structured approach that includes data collection from sensors or datasets, preprocessing, feature engineering, model training, and real-time prediction. Machine learning algorithms such as Random Forest, Support Vector Machine (SVM), Decision Tree, and Artificial Neural Networks are used to improve detection accuracy and reliability. Technologies such as Python, Pandas, NumPy, Scikit-learn, and TensorFlow/Keras are used for implementation.

Keywords: Artificial Intelligence, Machine Learning, Artificial Neural Network, Internet of Things, Application Programming Interface, Root Mean Square Error.

NG-IDS: A Hybrid Real-Time Intrusion Detection System using Machine Learning

Golden Jeffrey J , Kanish M , Krishna R, Dr. S. Jagadeesh Kumar,
Computer Science and Engineering, Kathir College of Engineering, Coimbatore.

ABSTRACT

Intrusion Detection Systems (IDS) are essential for identifying cyber threats in modern network environments. However, traditional IDS approaches face challenges in detecting unknown attacks and maintaining real-time performance in resource constrained systems. This paper proposes NG-IDS, a hybrid real-time intrusion detection system that integrates signature-based detection with machine learning-based anomaly detection. The system employs lightweight feature extraction, Random Forest classification, and a streaming-based processing mechanism to ensure efficient real time analysis. By combining rapid detection of known threats with adaptive identification of unknown attacks, the proposed model enhances overall detection capability. Experimental results demonstrate that NG-IDS achieves an accuracy of up to 98%, with low false positive rates and millisecond-level response times. The system is specifically designed for small-scale environments such as laboratories and offices, offering a practical and efficient solution for real-time cybersecurity. The proposed NG-IDS system combines signature-based detection and machine learning-driven anomaly detection to provide efficient real-time cybersecurity monitoring. Using lightweight feature extraction, Random Forest classification, and streaming-based processing, the system accurately detects both known and unknown cyber threats with low latency. Experimental results demonstrate high detection accuracy, low false positive rates, and suitability for resource-constrained environments such as laboratories and office networks.

Keywords: Intrusion Detection System, Machine Learning, Anomaly Detection, Real-Time Detection, Cybersecurity.

CERTVERIFY - AI Certificate Verification System

Ms. Subashini R C, Kishore B, Makesh P, Hariharan V,
Artificial Intelligence and Data Science
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

The proliferation of digitally forged academic and professional certificates has emerged as a critical challenge for educational institutions, employers, and credentialing authorities worldwide. Traditional manual verification methods are time-consuming, error-prone, and insufficient against sophisticated forgery techniques enabled by modern image editing tools. This paper presents an AI-driven framework for automated fake certificate detection that integrates Convolutional Neural Networks (CNNs), Natural Language Processing (NLP), and metadata forensic analysis to identify tampered, counterfeit, or fraudulent certificates with high accuracy. The proposed system extracts multi-modal features from certificate images, including font inconsistencies, layout anomalies, signature irregularities, and textual semantic patterns, and classifies them using an ensemble learning approach. Experimental evaluation on a curated dataset of genuine and forged certificates demonstrates that the proposed model achieves a detection accuracy of 97.4%, with a precision of 96.8% and a recall of 96.1%, significantly outperforming existing rule-based and single-modal detection methods. The system provides a scalable, explainable, and deployable solution for real-time certificate verification across academic, corporate, and government sectors.

Keywords: Fake Certificate Detection, Deep Learning, Convolutional Neural Network (CNN), Document Forensics, Image Tampering, Natural Language Processing (NLP), Ensemble Learning.

Vision Transformer Based Framework for Detection and Classification of Real and AI-Generated Visual Media

Mr.Raja R, Nandhitha B, Priya Mahalakshmi S, Sandhya B, Tejasri K
Computer Science and Engineering
Velammal Institute of Technology, Panchetti

ABSTRACT

Artificial Intelligence has enabled the creation of highly realistic deepfake images and videos, raising concerns about misinformation, identity misuse, and cybersecurity. This project proposes a Vision Transformer (ViT)-based framework for detecting and classifying real and AI-generated visual media. Unlike traditional Convolutional Neural Networks (CNNs), the Vision Transformer divides images into patches and uses self-attention mechanisms to capture both local and global features effectively. The system follows a structured pipeline including data collection, preprocessing, patch embedding, feature extraction, and classification. By leveraging transformer architecture, the model achieves improved accuracy, robustness, and generalization across diverse datasets and emerging deepfake techniques. The proposed approach offers a scalable and efficient solution for deepfake detection, with applications in digital forensics, media verification, and cybersecurity to ensure the authenticity and reliability of visual information. The proposed Vision Transformer (ViT)-based framework provides an effective solution for detecting and classifying AI-generated deepfake images and videos. By utilizing patch embedding and self-attention mechanisms, the model captures both local and global visual features, improving accuracy and robustness compared to traditional CNN-based methods. The system supports scalable deepfake detection for applications in digital forensics, media authentication, and cybersecurity, helping to combat misinformation and identity misuse.

Keywords: Deepfake Detection, Vision Transformer (ViT), Artificial Intelligence, Image Classification, Cybersecurity.

Reverse Engineering on Fileless Malware using Memory Forensics

Ajay.A, Harish.S, Sanjay.S
Computer Science and Engineering(Cyber Security),
Gojan School of Business and Technology, Chennai

ABSTRACT

Traditional Veterinary Information Management Systems (VIMS) face major challenges related to data security, reliability, and interoperability due to their centralized architecture, which creates a single point of failure. This paper proposes a Reliable and Intelligent Veterinary Information Management System (RIVIMS) that integrates blockchain technology with predictive analytics to improve data management and healthcare forecasting. The system uses a permissioned blockchain framework, Hyperledger Fabric, along with the SHA-256 algorithm to ensure data integrity and prevent unauthorized tampering. In addition, machine learning models such as Deep Neural Networks (DNN) and Multi-directional Long Short-Term Memory (MD-LSTM) are utilized to analyze historical appointment data and predict future healthcare trends and resource requirements. Experimental results show that the MD-LSTM model achieves high predictive accuracy with an R^2 score of 0.979. The blockchain component also demonstrates efficient performance with high transaction throughput and low latency. The proposed system provides a secure, reliable, and intelligent framework for modern veterinary information management and healthcare planning.

Keywords: Blockchain Technology, Veterinary Information System, Predictive Analytics, MD-LSTM, Hyperledger Fabric

Cardio Vascular Risk Prediction

Mr. R. Selvamanikandan, Mr. R. Nithish Sriram, Mr. M. Sivasakthi,
Artificial Intelligence and Data Science,
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

Cardiovascular diseases are one of the major health problems worldwide, especially among people with diabetes. Early prediction of cardiovascular risk can help reduce serious health complications and improve patient care. This project proposes a cardio-diabetic risk prediction system that uses machine learning and a mobile application to estimate the risk of cardiovascular disease. The system is developed using a Flutter mobile application where users can enter health details such as blood pressure, heart rate, blood sugar level, and body mass index. These values are sent to a backend server through REST APIs, where a trained machine learning model analyzes the data and predicts the cardiovascular risk score. The predicted result is displayed in the mobile application as a percentage along with risk levels such as low, medium, or high. The application also allows users to track their risk history and monitor their health parameters through a dashboard. This system helps patients and healthcare providers monitor health conditions and take preventive measures at an early stage.

Keywords: Cardiovascular Disease Prediction, Diabetes, Machine Learning, Flutter Mobile Application, Health Monitoring, Risk Score Prediction.

CROPDEFENDER AI: An Automated Deep Learning Architecture for Securing Agricultural Perimeters Against Wildlife Intrusion

Aishwarya K M, Hari Nivetha Mary, Mohana Sankari, Roshan Sabeha M, Ms. J. Vasugi,
Artificial Intelligence and Data Science,
M.A.M. College of Engineering and Technology, Tiruchirappalli.

ABSTRACT

Agricultural crop damage caused by animal intrusion is a significant challenge, particularly during nighttime and low-visibility conditions, leading to economic losses and human-wildlife conflicts. Traditional protection methods are often costly and prone to false alarms. This paper proposes "CropDefender AI," an intelligent, real-time animal intrusion detection and alert system leveraging Convolutional Neural Networks (CNN) and YOLO algorithms. By continuously monitoring the farm environment using high-resolution cameras, the system accurately detects and localizes animals. Upon detection, it instantly triggers automated alert mechanisms, such as sirens and mobile notifications, enabling timely intervention. Evaluated through a dedicated web interface, this scalable framework minimizes manual effort, enhances farm security, and supports sustainable smart farming practices. The proposed "CropDefender AI" system utilizes CNN and YOLO-based deep learning models for accurate real-time animal intrusion detection in agricultural fields. By integrating automated alerts such as sirens and mobile notifications, the system enables quick response and reduces crop damage during low-visibility conditions. The scalable and intelligent framework improves farm security, minimizes manual monitoring, and promotes sustainable smart farming practices.

Keywords: Deep Learning, YOLO, Animal Intrusion, Smart Farming, Computer Vision.

Geo-Aware Driver Vitality Surveillance System using Multi-Modal Health and Drowsiness Analytics

Asarudeen R, Eljin Shiney E J , Jerry George Joshi,
Artificial Intelligence and Data Science
Vel Tech Muti Tech Dr.Rangarajan Dr.Sakunthala Engineering college, Chennai.

ABSTRACT

Driver fatigue remains one of the leading causes of road accidents worldwide, necessitating robust real-time intervention systems. This paper proposes a hybrid, IoT-enabled detection framework that combines computer vision with physiological monitoring to enhance detection accuracy. The system utilizes Python and OpenCV to analyze facial landmarks in real time, monitoring head position and eye closure. Simultaneously, an Arduino-based heart rate sensor measures pulse variability to identify physiological signs of fatigue. When drowsiness is detected, the system immediately triggers an alarm and updates an onboard LCD display. In parallel, data is transmitted to an IoT platform for remote monitoring, fleet management, and logging. This multi-modal approach overcomes the limitations of single-sensor systems and provides a cost-effective, scalable solution suitable for both personal and commercial vehicles, thereby contributing to improved road safety. The proposed IoT-enabled driver fatigue detection system combines computer vision and physiological monitoring to accurately identify drowsiness in real time. Using OpenCV-based facial analysis and Arduino-supported heart rate sensing, the framework detects fatigue symptoms and instantly triggers alerts for driver safety. Integrated IoT connectivity enables remote monitoring, data logging, and fleet management, making the system a scalable and cost-effective solution for improving road safety in personal and commercial vehicles.

Keywords: OpenCV, Physiological Monitoring, Arduino UNO, Real-Time Safety Systems, Driver Fatigue Detection, IoT.

Quantum Error Correction in Topological Quantum Computing: A Novel Approach Using Surface Codes

P. Ramu, Electronics and Communication Engineering, Jaya Institute of Technology,
S P Vimal, Sri Ramakrishna Engineering College

ABSTRACT

Quantum computing represents a paradigm shift in computational capability; however, quantum error correction remains a critical challenge for practical implementation. This study presents a novel approach to quantum error correction using surface codes in topological quantum computing architectures. A hybrid error correction algorithm is developed that combines surface code topology with machine learning-based error prediction. Experimental results demonstrate a 34% improvement in logical qubit fidelity compared to conventional surface code implementations. The system is evaluated on a simulated 49-qubit quantum processor, achieving error rates below the fault-tolerance threshold. This research provides a scalable framework for building fault-tolerant quantum computers and advances the field toward achieving practical quantum advantage in complex computational problems. The proposed hybrid quantum error correction approach integrates surface code topology with machine learning techniques to enhance logical qubit reliability in quantum computing systems. Experimental evaluation on a simulated 49-qubit processor demonstrates improved qubit fidelity and reduced error rates below the fault-tolerance threshold. This scalable framework contributes to the development of fault-tolerant quantum computers and supports future advancements toward practical quantum advantage.

Keywords: Quantum Error Correction, Surface Codes, Topological Quantum Computing, Fault Tolerance, Quantum Algorithms, Machine Learning.

IoT-Based Automated Vehicle Health Monitoring System

Jai Prakash K, Aadhithiya T, Mohamed Rizwan S, Mrs. Anitha M,
Artificial Intelligence and Data Science,
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

The rapid growth of the Internet of Things (IoT) has enabled the development of intelligent systems capable of collecting and processing data from connected devices through network communication. This project presents the design and implementation of an IoT-based system that detects the approximate geographical location of a laptop using network connectivity. The proposed system utilizes the device's public IP address to determine its location through online geolocation services. It is implemented using Python and leverages external geolocation APIs along with mapping libraries to retrieve and visualize location data. Experimental results confirm that the system provides reliable location detection with reasonable accuracy across varied network conditions, making it a practical and cost-effective solution without requiring additional GPS hardware. The proposed IoT-based laptop location detection system uses public IP addresses and geolocation APIs to identify and visualize the approximate geographical position of a device. Implemented using Python and mapping libraries, the system offers a cost-effective and hardware-free solution for location tracking. Experimental results demonstrate reliable performance and practical applicability for device monitoring and network-based tracking applications.

Keywords: IoT, Vehicle Health Monitoring, Python, Real-Time Monitoring, Network Communication, Edge Computing, Fleet Management, Fault Detection.

SEP-GCN: Leveraging Similar Edge Pairs with Temporal and Spatial Contexts for Location-Based Recommender Systems

Rajalakshmi. I, Swetha G, Master of Computer Applications,
Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

SEP-GCN (Similar Edge Pair Graph Convolutional Network) is a novel framework designed to enhance location-based recommender systems by effectively capturing both spatial and temporal dynamics of user behavior. The model introduces the concept of similar edge pairs, enabling the system to learn relationships between user–location interactions with similar contextual patterns. By integrating spatial proximity and temporal sequences, SEP-GCN constructs a more expressive graph representation that improves recommendation accuracy. The framework leverages graph convolutional networks to propagate contextual information across nodes while preserving structural dependencies. Additionally, it addresses data sparsity and dynamic user preferences by modeling evolving interaction patterns over time. Experimental evaluations demonstrate that SEP-GCN outperforms traditional recommendation models in terms of precision and recall. This approach provides a robust and scalable solution for real-world applications such as tourism, navigation, and personalized location suggestions in mobile and web platforms.

Keywords: Location-Based Recommendation, Graph Convolutional Network (GCN), Similar Edge Pairs, Spatial-Temporal Context, Recommender Systems, Deep Learning, User Behavior Modeling, Data Sparsity.

Blockchain-Driven Electronic Health Record on Cloud

Ms. Rajalakshmi, Thirisha D,
Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

Block chain-based Electronic Health Records (HER) integrated with cloud computing provide a secure, scalable, and efficient healthcare data management system. Block chain ensures data security, transparency, integrity, and privacy by storing records in a decentralized and tamper-proof manner. It enables patient-centric control, where patients grant permission to healthcare providers through smart contracts and consent management. Cloud infrastructure enhances scalability, accessibility, and interoperability, allowing seamless data sharing among authorized users. This integration reduces risks of unauthorized access, data breaches, and misuse while ensuring compliance with standards like HIPAA. A unified data exchange layer eliminates compatibility issues across different healthcare systems, enabling better collaboration and decision-making. Additionally, cloud platforms support large-scale data storage and dynamic system performance. Overall, this system improves patient care, data security, stakeholder collaboration, and healthcare efficiency, making it a reliable solution for modern digital healthcare challenges.

Keywords: Blockchain Technology, Electronic Health Records, Cloud Computing, Data Security, Smart Contracts

Smart Waste Classification and Monitoring System Using AI and IoT

Dr. B. Shanthini, Karthik K , Mythreya Srinivasaraj, Narayanan B,
Computer Science and Engineering
St. Peter's Institute of Higher Education and Research, Chennai

ABSTRACT

Rapid urbanization has resulted in a significant increase in solid waste generation, leading to serious environmental and public health challenges. This paper presents a Smart Waste Classification and Monitoring System that integrates Artificial Intelligence (AI) and Internet of Things (IoT) technologies to enable efficient waste management. A convolutional neural network based on MobileNetV2 is trained for waste classification and deployed using TensorFlow Lite with INT8 quantization on a Raspberry Pi for edge-based inference. The system performs real-time classification of waste materials while simultaneously monitoring bin levels using IoT sensors. Experimental results demonstrate high classification accuracy of approximately 90% along with efficient real-time performance. The proposed system offers a scalable, cost-effective, and sustainable solution for smart city waste management applications. The proposed Smart Waste Classification and Monitoring System combines AI and IoT technologies to improve waste segregation and real-time bin monitoring in urban environments. By deploying a MobileNetV2-based model on Raspberry Pi with TensorFlow Lite optimization, the system achieves efficient edge-based waste classification with high accuracy and low computational cost. The solution supports sustainable smart city initiatives by reducing manual effort, improving waste management efficiency, and promoting environmental cleanliness.

Keywords: Edge AI, Waste Classification, Smart Waste Management, Internet of Things (IoT), Convolutional Neural Network (CNN), TensorFlow Lite, Raspberry Pi, Deep Learning.

AI Asset System: A Generative AI-Agentic RAG Powered Context-Aware System for Intelligent Data Querying

Gayathri S J, Jeya Renukaa S , Mr. T. R. Chenthil,
Artificial Intelligence and Data Science, Jeppiaar Engineering College, Chennai.

ABSTRACT

This paper presents a Generative AI-based Agentic Retrieval-Augmented Generation (RAG) system developed for a land records portal. The proposed system enables government officials and citizens to query land records using natural language, eliminating the need for Structured Query Language (SQL) expertise. To improve accessibility across diverse user groups, the system supports multilingual interaction in both native languages and English. The architecture follows a multi-agent pipeline consisting of an Orchestrator, SQL Agent, RAG Agent, and Synthesizer. The Orchestrator classifies user queries into database, document, or hybrid categories. The SQL Agent translates natural language queries into PostgreSQL statements for structured database retrieval, while the RAG Agent performs semantic document search using vector embeddings. The Synthesizer integrates outputs from both agents to generate coherent and context-aware responses. To ensure secure access, the system incorporates OTP-based authentication and role-based data access control according to jurisdictional roles such as Collector, Tahsildar, and Citizen. It supports ingestion of PDF, DOCX, text, and image documents using multilingual embedding models based on transformer architectures. The complete system is deployed online using locally hosted large language models through Ollama, including SQLCoder and Mistral. Experimental evaluation demonstrates a query accuracy of 72%–82%, while maintaining strong data privacy and operational reliability.

Keywords: Agentic RAG, Generative AI, LLM, NLP, Vector Embeddings

Predicting Employee Burnout using Multidimensional Psychometric Scoring and Data Twin Modeling

Priyanka Diwan,Swasti HS, Subhra Sulagana Debata, Jayashree M Kudari,
Computer Science and Information Technology, Jain University, Bengaluru.

ABSTRACT

Employee burnout is a psychological state of emotional, physical, and mental exhaustion caused by prolonged work-related stress. It not only affects employee well-being but also significantly influences organizational productivity and workforce sustainability. This paper presents a predictive framework for employee burnout detection using multidimensional psychometric scoring and data twin modeling. A validated 48-item psychometric questionnaire was used to collect responses from 303 employees. Following data preprocessing, strict feature selection and class balancing techniques were applied to improve model robustness. Five machine learning classifiers were implemented and evaluated to compare predictive performance. Among the tested models, the Random Forest classifier achieved the highest testing accuracy of 91.30%, demonstrating superior classification capability for burnout prediction. Confounder analysis using Chi-square testing and Cramér's V identified recognition, motivation, and stress as the most influential factors affecting employee burnout, with Cramér's $V = 0.5013$ and $p < 0.001$. The proposed predictive framework is deployed through a backend developed using FastAPI with PostgreSQL support to enable data twin-based employee burnout monitoring and analysis. The system offers a scalable and practical solution for proactive workforce mental health assessment in organizational environments.

Keywords: Employee Burnout, Machine Learning, Psychometric Scoring, Confounder Analysis, Data Twin, Random Forest, Workforce Analytics.

HAND 2 VOICE : IoT-Based Smart Glove for Speech-Impaired People for Real-Time Communication

Dr. B. Shanthini, Bolagani Sri Charan Durga, Mohan Kumar N, Prashanth A. S.
Computer Science and Engineering
St. Peter's Institute of Higher Education and Research, Chennai.

ABSTRACT

Communication is an essential part of human interaction; however, speech-impaired individuals face significant challenges in expressing their thoughts to others. This paper presents an Internet of Things (IoT)-based smart glove system designed to translate hand gestures into speech and text in real time. The proposed system utilizes flex sensors embedded in a wearable glove to detect finger movements corresponding to predefined sign language gestures. These signals are processed using a microcontroller and transmitted through an IoT communication module to a mobile or web interface. The interpreted gesture is then converted into readable text and audible speech, enabling effective communication between speech-impaired individuals and people unfamiliar with sign language. The system is designed to be portable, cost-effective, and user-friendly. Experimental results demonstrate that the smart glove can accurately recognize common gestures and convert them into meaningful communication outputs. This technology can significantly enhance the independence and social interaction of speech-impaired individuals. The proposed IoT-based smart glove enables real-time conversion of hand gestures into speech and text, improving communication for speech-impaired individuals. By integrating flex sensors, microcontrollers, and IoT connectivity, the system provides accurate, portable, and user-friendly gesture recognition. The technology enhances accessibility, independence, and social interaction while offering a cost-effective assistive communication solution.

Keywords: Internet of Things, Smart Glove, Sign Language Recognition, Assistive Technology, Gesture Recognition, Speech Conversion.

Oreste Methodology within a Circular Intuitionistic Fuzzy Framework for Preferential Outranking in Hybrid Cloud Service Selection

Selvaraj.A, Narmadha. D, Department of Master of Computer Applications,
Thirumalai Engineering College, Kilambi, Kanchipuram

ABSTRACT

This paper advances the ORESTE (Organisation, Rangement Et Synthèse de Données Relarionnelles) methodology within the Circular Intuitionistic Fuzzy (CIF) framework, highlighting its potential in practical decision analytics. The study first enhances CIF aggregation by employing the generalized mean technique, offering a flexible way to combine evaluative ratings and significance weights. Through modulation of the averaging parameter, decision-makers are able to accentuate either lower or higher values, thereby overcoming the constraints associated with conventional arithmetic means. The framework further improves decision precision through CIF similarity-driven appraisal indices, which utilize refined similarity metrics grounded in axiomatic properties such as symmetry, boundedness, identity, and monotonicity. These indices quantify the similarity between evaluative ratings and anchor references, while also revealing indifference and incomparability thus equipping decision-makers with a comprehensive toolset for handling uncertainty. The CIF ORESTE framework comprises two methodologies. CIF ORESTE I delivers a global weak ranking using similarity-driven indices and generalized projection-related distances. CIF ORESTE II addresses the limitations of weak rankings by incorporating an Indifference-Preference-Incomparability (I-P-R) structure, which uses mean and net preference intensities to establish thresholds and clarify outranking relations. Applied to the evaluation of hybrid cloud services for a technology corporation, the CIF ORESTE framework demonstrates its effectiveness in resolving group decisions, managing uncertainty, and structuring preferences. Comparative analyses further underscore its robustness in handling CIF-based data and delivering reliable results.

Keywords: Multi-Criteria Decision Making, ORESTE Method, Circular Intuitionistic Fuzzy Sets; Outranking Relations, Generalized Mean Aggregation.

An Integrated Digital Portal for Tracking Academic Progression in Higher Education Institutions

Dr. B. Shanthini, Daniya Sulthana Begam Shaik, Sai Chandanaa N, Tarunika Sri H
Computer Science and Engineering,
St. Peter's Institute of Higher Education and Research, Chennai.

ABSTRACT

The campus placement process plays a crucial role in shaping students' career opportunities and institutional reputation. However, traditional placement systems are often manual, fragmented, and inefficient, resulting in communication gaps, data inconsistencies, a lack of transparency, and increased administrative workload. These limitations highlight the need for a centralized and automated solution. This paper presents the design and development of the SPIHER Progression Portal, a web-based system that automates and streamlines placement and student progression activities. The system is implemented using HTML, CSS, and JavaScript for the frontend, and PHP with MySQL for backend processing and data management. It provides a centralized platform for students, Placement Officers (TPOs), Heads of Departments (HoDs), and administrators with role-based access control. The portal includes multiple dashboards, such as Student Dashboard, TPO Dashboard, HoD Dashboard, and Admin Dashboard, each designed to support specific functionalities. Students can manage profiles, upload resumes, apply for placements, and track application status in real time. The TPO dashboard enables efficient management of companies, applications, and recruitment processes. The HoD dashboard provides analytical insights into student eligibility and placement performance, while the admin dashboard ensures overall system control and data management. The proposed system improves efficiency, enhances transparency, ensures real-time updates, and reduces manual workload. It offers a scalable, secure, and user-friendly solution for modern placement management, and can be extended in the future with advanced features such as mobile support and intelligent analytics.

Keywords: Placement Management System, Student Progression Portal, Web-Based Application, Role-Based Access Control, Real-Time Analytics

AI-Powered Smart Tourism Companion with CNN-Based Destination Recognition

Dr. M. Buvana,Kamalnath L, Rubesh Dev S, Samuvel John S,
Computer Science and Engineering,
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College,Chennai

ABSTRACT

The ongoing advancements in digital technology have had a significant impact on how people plan and enjoy their travels, making intelligent and customized tourism systems increasingly necessary. This article presents the Smart Tourism Companion, a smartphone app that integrates image identification, real-time travel assistance, and booking features into a single system. The system gives users an intuitive interface to access vital information via a cloud-based system. Instant recognition and site knowledge are made possible by the use of a Convolutional Neural Network (CNN) to identify photos that users upload of tourist destinations. Additionally, the system offers comprehensive location data, personalized travel recommendations, and simple methods for booking trips, activities, and accommodations. The proposed approach combines artificial intelligence techniques with mobile technologies to enhance user experience and journey planning efficiency. The Smart Tourism Companion is a crucial step in the development of secure, data-driven, and user-focused smart tourism systems.

Keywords: Convolutional Neural Network (CNN), Image Recognition, Cloud-Based System, Artificial Intelligence, Smart Tourism, Personalized Recommendations, Intelligent Travel Assistant, and Destination Identification.

Toward an Automated Cross-Multimodal Verification of Mobile App Bug Fixes Integrating User Feedback, Developer Responses, Changelogs, and UI Visual Analysis

Harini A, Selvaraj A
Thirumalai Engineering College, Kilambi, Kanchipuram.

ABSTRACT

Verifying claimed bug fixes in mobile applications is crucial, yet the "fixed but not resolved" phenomenon remains a persistent challenge. Existing bug analysis tools focus on pre-fix tasks like detection and reproduction, but lack mechanisms to holistically verify a fix post-deployment by cross-referencing developer claims, visual UI changes, and subsequent user feedback. This gap leads to persistent bugs, wasted developer effort, and user dissatisfaction. This paper introduces BUGFixChecker, the first framework for automated, multimodal cross-verification of mobile app bug fixes. Our primary goal is to determine if a claimed fix has truly resolved a user-reported issue. BUGFixChecker integrates five data sources: the original user bug report, the developer's fix claim, "before" and "after" UI screenshots, and post-fix user reviews. The core methodology employs a Multimodal Large Language Model (MLLM) guided by a Chain-of-Thought prompt to perform a comparative reasoning task. We evaluated the framework on a curated dataset of 53 real-world bug fix cases from Android applications.

Keywords: Bug Fix Verification, Multimodal Large Language Model, Mobile Application Testing, User Feedback Analysis, Automated Bug Validation

Prediction of used Car Prices using Machine Learning

Thirumalai E, Dr. S. Tamilarasi, Mrs. K. Aishwarya, Information Technology
St. Peter's Institute of Higher Education and Research, Chennai.

ABSTRACT

The rapid growth of the automobile market has significantly increased the demand for accurate price prediction of used cars. Estimating the resale value of vehicles is a complex task due to the influence of multiple factors such as brand, model, fuel type, year of manufacture, and mileage. This paper presents a machine learning-based system designed to predict the resale value of used vehicles using historical car sales data. Various machine learning algorithms, including Linear Regression, Decision Tree, and Random Forest, are applied to analyze the data and generate accurate price predictions. The system also incorporates a web-based interface that enables users to input vehicle details and obtain estimated market prices in real time. The proposed model enhances pricing transparency, assists buyers and sellers in making informed decisions, and reduces uncertainty in the used car market. The proposed system leverages machine learning techniques to accurately predict used car prices based on multiple vehicle attributes and historical sales data. By integrating algorithms such as Linear Regression, Decision Tree, and Random Forest, the model provides reliable and real-time price estimation through a user-friendly web interface. The system improves transparency, supports informed decision-making, and enhances efficiency in the used automobile market.

Keywords: Machine Learning, Price Prediction, Used Cars, Regression Models, Data Analysis.

An Enhanced Method for Detecting Smart Traffic Violation Using Predictive Analytics System

Dr.M.Sukanya, Mythili.A, Sinega.P, VinothKumar.V,
Computer Science and Engineering, Kathir College of Engineering , Coimbatore

ABSTRACT

This paper presents an automated Smart Traffic Violation Detection and Predictive Analytics System that leverages deep learning and computer vision techniques for intelligent traffic monitoring. The proposed system is designed to identify common traffic violations, including red-light jumping, over-speeding, helmet non-compliance, and illegal lane changes in real time. The framework utilizes advanced object detection models such as YOLOv8 and YOLOv9 in combination with Optical Character Recognition for automated vehicle number plate recognition. Once a violation is detected, the corresponding vehicle details are extracted and securely stored in cloud-based databases for record maintenance and analysis. To support enforcement, the system generates real-time alerts that are transmitted to traffic authorities through SMS notifications and web-based dashboards. Experimental evaluation shows that the proposed framework achieves detection accuracy ranging from 85% to 95%, significantly outperforming conventional CCTV-based traffic monitoring systems. The proposed solution offers a scalable, efficient, and cost-effective framework for improving urban traffic governance, enhancing road safety, and reducing manual surveillance requirements in smart city environments.

Keywords: Traffic Violation Detection, YOLOv8, YOLOv9, Predictive Analytics, Computer Vision, Deep Learning, OCR, Smart City.

Traceable and Revocable Large Universe Multi-Authority Attribute-Based Access Control with Resisting Key Abuse

Mr.A.Selvaraj , V.Abinayam, Department of Master of Computer Applications, Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

The proposed traceable, revocable, and collusion-resistant multi-authority ABE scheme (TRKAD-ABE) provides secure, fine-grained, cross-domain access control. It handles dynamic changes in attributes, users, and authorities, supporting large universes and efficient, authorized outsourcing of decryption. The scheme resists key abuse by preventing authorities and cloud service providers (CSPs) from generating full keys or accessing data. Additionally, it features high efficiency, user traceability, and strong forward/backward security, making it suitable for secure, dynamic, large-scale cloud data sharing, ensuring only authorized, non-revoked users access sensitive information. The TRKAD-ABE scheme offers secure and fine-grained access control for large-scale cloud data sharing environments. It ensures user traceability, revocation support, and resistance against collusion attacks while maintaining strong forward and backward security. The proposed approach improves efficiency and privacy by enabling secure outsourced decryption and preventing unauthorized access by malicious users, authorities, or cloud service providers.

Keywords: Attribute-based encryption; Access control; Large universe; Multi-authority; Key abuse.

Web-Based Intelligent Farmer Support System of Precision Agriculture and Market Awareness

Mrs. Tamilselvi.C, Meghavarshiny.S R, Perinban.S, Information Technology,
Dr.M.G.R. Educational and Research Institute, Chennai

ABSTRACT

The agricultural industry is still one of the pillars of the economy in the developing countries but farmers still grapple with a series of problems like farm illnesses, weather insecurity, low level of market transparency and poor knowledge on government welfare programs. To address such problems, the given work suggests the AI-Powered Farmer Advisory and Crop Price Forecasting Web Application created within the Flask framework and incorporating machine learning, deep learning, and real-time data services that can be used to make informed decisions in agriculture. The system integrates various smart units, such as the deep learning-based crop and fruit leaf disease detection unit that examines pictures of leaves and labels diseases and prescribes appropriate chemical and organic treatment interventions. An OpenWeather API is used in a real time weather monitoring and forecasting module to access localized real time weather conditions and five-day weather predictions. There is also a crop information and guidance module, which provides detailed information about soil suitability, climatic conditions, nutritional advantages, and general pests using an interactive card-based system. The platform also incorporates machine learning-based crop suggestions depending on environmental conditions, immediate market price updates so that it can provide connectivity between farmers and buyers and incorporate individual suggestions of applicable government initiatives. Through AI-based prediction synchronization with real-time data and a friendly web interface, the proposed system is expected to provide farmers with the ability to achieve higher agricultural productivity, minimize crop losses, and get a readily available, dependable, and timely agricultural intelligence.

Keywords: Smart Agriculture, Artificial Intelligence, Machine Learning, Deep learning, Crop Disease Detection, Weather Forecasting, Crop Recommendation, Market Price Forecasting, Decision support system.

AI-Driven Hospital Management System with Intelligent Scheduling and Encrypted Access Control

Anitha T, Tamil Kumaran K, Venkat Raj B L, Visakh S
Artificial Intelligence and Data Science,
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai.

ABSTRACT

The work of a hospital is highly complicated and requires constant, perfect control over patient information, employee shifts, and access control. This paper introduces the AI-based Integrated Hospital Management System (IHMS) based on the methods of machine learning and deep learning to automatize and optimize the most important hospital operations. The system is designed based on microservices architecture, which combines React.js frontend, a backend based on Spring Boot and a Python-based Flask machine learning inference service. Intelligent appointment booking, staff shift assignment, and patient review prioritization are done using supervised learning algorithms like Gradient Boosting and Random Forest model on the basis of the parameters like patient severity, doctor availability, and workload distribution. An automated contactless staff attendance is implemented with facial recognition module implemented by DeepFace through a deep learning model. Role-Based Access Control (RBAC), dynamic credential masking, hashing passwords with SHA-512/Argon2 and token authentication with JWT are used to provide security. The system allows two interface access to clinical staff and patients, and real-time data matching and audit trails. The experimental outcomes reveal that scheduling efficiency, less manual intervention and better data security would be more effective than traditional hospital management.

Keywords: Hospital Management System, machine learning, intelligent scheduling, facial recognition, role-based access control, deep learning, microservices, spring boot, flask, react.js.

Cloud -Fog Enabled and Privacy-Preserving IoT Data Market Platform based Block Chain

Ms. I. Rajalakshmi, S. Tabusum,
Master of Computer Applications,
Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

The rapid growth of the Internet of Things (IoT) is transforming how devices communicate, enabling advanced applications such as Smart Grids (SG). However, challenges such as high data-sharing overhead and the lack of trusted centralized systems affect both efficiency and security. To address these issues, this paper proposes a hierarchical and secure IoT data-sharing platform using cloud–fog integration. The system introduces a non-interactive zero-knowledge proof-based group authentication and key agreement protocol, ensuring secure and privacy-preserving communication. It supports efficient one-to-many data sharing while maintaining data integrity and confidentiality. Security analysis through formal verification confirms mutual authentication and resistance to various attacks. Compared to existing approaches, the proposed method significantly reduces computational and communication overhead while offering improved scalability, making it suitable for large-scale IoT environments with increasing users and data volume. The proposed cloud–fog integrated platform enhances secure and efficient data sharing in large-scale IoT environments such as Smart Grids. By utilizing zero-knowledge proof-based authentication and key agreement mechanisms, the system ensures privacy, data integrity, and resistance against cyberattacks. Experimental and security analyses demonstrate reduced computational overhead, improved scalability, and reliable performance for real-time IoT communication applications.

Keywords: Internet of Things (IoT), Smart Grid, Cloud-Fog Computing, Security, Authentication, Scalability.

Detection of DDOS Attacks in Vehicular ADHOC Networks using Ensemble Deep Learning Model and Optimization Technique

Ms. I. Rajalakshmi,Vigneshwari.A, Department of Master of Computer Applications, Thirumalai Engineering College, Kilambi, Kancheepuram

ABSTRACT

Vehicular Ad-Hoc Networks (VANET) stands vital in integrating intelligent transport services but equally remains vulnerable to DDoS attacks because of the inherent dynamic and wireless routing. This paper proposes a secure VANET architecture with SDN and an ensemble deep learning model-AlexInception-BiLSTM-AttentionNet-to increase detection level as well as ensure an immediate security response. This architecture supports layered operation that decouples the control plane from the data plane of VANETs through the help of SDN controllers. Further, it brings in a novel SDN controller framework design for edge base stations. To this end, a two-layer classification and clustering scheme is piled on top of metaheuristic nature of Dung Beetle Optimization (DBO) to speed up convergence and fine-tune model parameters. Two benchmark datasets, UNSW_NB15 and Litnet20, were used in comparison with the ton of literature-based state-of-the-art models. This method recorded promising results: 99.31 % Accuracy, 99.28 % Specificity on UNSW_NB15 and 98.94 % Accuracy and 99.13 Specificity on Litnet20. The results illustrate the ability of the proposed model to robustly identify DDoS attacks with a high precision. Having a serious ability of adaptation, scaling, and executing, therefore, our SDN-driven VANET security model is suggested to be an insightful and efficient platform to monitor the traffic in real-time and work for its mitigation against some attacks via the next-generation intelligent transportation system.

Keywords: Vehicular Ad-Hoc Networks, DDoS Attack Detection, Software-Defined Networking, Deep Learning Ensemble Model, Intelligent Transportation Systems

Schnorr Approval-based Secure and Privacy-Preserving Data Aggregation with Dynamic Clustering and Fault Tolerance for Wireless Body Area

A. Selvaraj, A. Srilekha, Department of Master of Computer Applications
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

Wireless Body Area Networks play a pivotal role in enabling real-time health monitoring within the Internet of Medical Things, yet their resource-constrained sensors, sensitive health data, and dynamic operational conditions introduce significant challenges, such as limited battery life, network disruptions due to patient mobility, and vulnerabilities to sensor failures or cyberattacks that jeopardize data integrity and privacy. This paper presents a novel Schnorr Approval-Based Secure and Privacy Preserving Data Aggregation system tailored for Wireless Body Area Networks, integrating a lightweight Modified MuSig2 Schnorr signature scheme for robust, rogue key-resistant data authentication, a Dynamic Clustering Algorithm to seamlessly adapt to mobility-induced topology changes, a Recoverable Masking Technique leveraging SipHash-2–4 to ensure data confidentiality without sacrificing accuracy, and a Fault-Tolerant Data Approval mechanism employing Reed-Solomon coding to mitigate failures and malicious attacks. In contrast to computationally intensive methods like homomorphic encryption, this proposed system is meticulously designed for low-power sensors, achieving a balance of security, efficiency, and scalability. Comprehensive simulations across diverse scenarios demonstrate that our system delivers minimal latency, low energy consumption, high aggregation accuracy, and exceptional resilience against data injection attacks and node failures, positioning it as an innovative and reliable solution for next generation of Wireless Body Area Networks for health monitoring applications.

Keywords: Wireless Body Area Networks, Internet of Medical Things, Privacy-Preserving Data Aggregation, Schnorr Signature Scheme, Fault-Tolerant Health Monitoring.

AI Secure Plugin Framework against Prompt Injection in LLM

Dr. C. Vimalarani, Saranya M, Merlinpriya M, Sibin Joseph CJ
Computer Science and Engineering, Kathir College of Engineering, Coimbatore.

ABSTRACT

Large Language Models (LLMs) are widely used in modern applications for tasks such as content generation, automation, and decision support. When integrated with external tools and plugins, these models become more powerful but also more vulnerable to security threats. One of the major risks is prompt injection, where malicious inputs manipulate the model to perform unintended actions or expose sensitive information. This paper proposes a Secure AI Plugin Framework that adopts a layered security approach to protect LLM-based systems. The framework consists of multiple security layers, including input validation, sanitization, context isolation, access control, source verification, data protection, and logging. Each layer is designed to detect and mitigate different types of attacks such as prompt injection, data leakage, and unauthorized access. A prototype system was implemented using a JavaScript-based web application to demonstrate the effectiveness of the proposed framework. The results indicate that the layered approach successfully blocks malicious inputs while allowing legitimate user queries to pass through. This work emphasizes the importance of integrating robust security mechanisms into AI systems to ensure safe and reliable operation.

Keywords: Large Language Models, Prompt Injection, AI Security, Data Protection, Access Control.

Attendance Monitoring System of Students Based on Biometric and GPS Tracking System

Mr. Yuvaraj K, Rajalakshmi I
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

This paper is a study of a fingerprint recognition system based on minutiae-based fingerprint algorithms used in various techniques. This line of track mainly involves extraction of minutiae points from the model fingerprint images and fingerprint matching based on the number of minutiae pairings among to fingerprints. This paper also provides the design method of fingerprint-based student attendance with help of GSM. This system ignores the requirement for stationary materials and personnel for keeping of records. The main objective of this project is to develop an embedded system, which is used for security applications. The biometrics technology is rapidly progressing and offers attractive opportunities. In recent years, biometric authentication has grown in popularity as a means of personal identification in college administration systems. The prominent biometric methods that may be used for authentication include fingerprint, palmprint, and handprint, face recognition, speech recognition, dental and eye biometrics. In this paper, a microcontroller-based prototype of attendance system using fingerprint sensor and face recognition module is implemented. The tracking module is used here to identify the location of the missing person.

Keywords: Fingerprint Recognition System, Biometric Authentication, Student Attendance System, Face Recognition, GSM Tracking Module.

AI-Powered Smart Navigation System for Visually Impaired Persons

V.C Nathiya, Moumiya.S, Gopika.P, Maayavathi.S, Computer Science and Engineering, Kathir College of Engineering, Coimbatore

ABSTRACT

Visually impaired individuals face significant challenges in outdoor navigation due to limited situational awareness and absence of real-time environmental intelligence. This paper presents an AI-powered Smart Navigation System developed as a software-only platform comprising twelve integrated intelligent modules. The system combines Whisper ASR for multilingual voice recognition, YOLOv8 for real-time obstacle detection, K-Means clustering for personalized route memory, a crowdsourced hazard map with Geohash-based proximity queries, a React.js caregiver live dashboard with WebSocket synchronization, CNN-based Indian currency detection, Tesseract OCR for medicine label reading, and context-aware warnings using weather and Open Street Map data. The system operates on standard smartphones without specialized hardware, making it accessible and deployable. The proposed system enhances independent mobility for visually impaired users by providing real-time navigation assistance, obstacle detection, and voice-guided interaction. By integrating artificial intelligence, computer vision, and location-based services, the platform improves environmental awareness and user safety in outdoor environments. Its software-only architecture ensures cost-effectiveness, scalability, and easy deployment on standard smartphones without the need for additional hardware.

Keywords: visually impaired navigation; YOLOv8; Whisper ASR; crowd sourced hazard map; K-Means clustering.

Real-Time Spatial-Temporal Anomaly Detection in Surveillance Videos using YOLOv8 and BiLSTM

Ms.Meenatchi M, Vikas M V, Vijay M, Aruna R,
Computer Science and Engineering, Kathir College of Engineering, Coimbatore

ABSTRACT

This paper presents a real-time system for detecting abnormal activities in surveillance videos using deep learning techniques. The proposed approach combines YOLOv8 for object detection and Long Short-Term Memory with Bidirectional processing (BiLSTM) for analyzing movement patterns over time. YOLOv8 identifies objects such as people and vehicles in each frame, while BiLSTM analyzes their motion across consecutive frames to detect unusual behavior. By integrating spatial and temporal analysis, the system improves anomaly recognition accuracy and enables more effective surveillance monitoring. The proposed method reduces the need for continuous human supervision and enhances surveillance efficiency. It is particularly suitable for applications such as public safety, traffic monitoring, and smart city environments. The system provides real-time monitoring and rapid detection of suspicious activities, helping authorities respond quickly to potential threats. By combining YOLOv8 and BiLSTM, the framework achieves improved accuracy in identifying abnormal motion patterns in dynamic environments. The proposed solution is scalable, efficient, and suitable for intelligent surveillance applications in public spaces, transportation systems, and smart city infrastructures.

Keywords: YOLOv8, Long Short-Term Memory, Surveillance Video Analysis, Anomaly Detection, Spatial-Temporal Learning, Deep Learning, Real-Time Monitoring, Public Safety, Smart Surveillance Systems.

Multimodal Assistive Technology for Disabled Users

S. Usha, Harini S, Subasree R, Iniya M, Artificial Intelligence and Data Science,
Kathir College of Engineering, Coimbatore

ABSTRACT

This paper presents an artificial intelligence-driven multimodal assistive framework developed to support individuals with visual and hearing impairments. The proposed system integrates real-time object detection and sign language recognition to improve accessibility and independent interaction. For visually impaired users, object detection is performed using the YOLOv8 architecture, which identifies surrounding objects and provides audio-based environmental descriptions through text-to-speech (TTS) output. This enables enhanced situational awareness in daily activities. For hearing-impaired users, the system incorporates a hand gesture recognition pipeline based on MediaPipe landmark tracking and a Convolutional Neural Network trained on static Indian Sign Language (ISL) alphabets. Recognized gestures are converted into text or synthesized speech, enabling effective communication support. Multilingual text-to-speech functionality, available in both online and offline modes, improves accessibility across diverse language communities. The system also operates without continuous internet connectivity and includes a user-driven training module for personalized functionality. Experimental results demonstrate 85% object detection precision at 30 FPS and 92% ISL gesture classification accuracy, indicating strong practical feasibility for real-world deployment.

Keywords: Assistive Technology, Object Detection, YOLOv8, Sign Language Recognition, MediaPipe, CNN, Text-to-Speech, Multimodal AI, Accessibility, Deep Learning.

Automated Vehicle Service System with Online Booking and Budget Approval

Santhosh Kumar N, Pushpanathan R
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

The increasing complexity of urban mobility and the growing number of privately owned vehicles have amplified the demand for efficient and streamlined vehicle servicing solutions. Traditionally, vehicle maintenance processes involve manual appointment scheduling, service delays due to communication gaps, and uncertainty in service costs, leading to user dissatisfaction. The project titled "Automated Vehicle Service System with Online Booking and Budget Approval" addresses these challenges by leveraging web technologies to automate and optimize the entire vehicle service lifecycle. The system is designed and implemented using Java as the core programming language**,** with JSP, CSS, and JavaScript for the frontend interface**,** and MySQL as the backend database**.** It allows service centers to register on the platform, which are then reviewed and approved by an administrator before gaining access. Registered users can log in, book vehicle services online by entering car and fault details, and choose a preferred service center. Once a booking is made, the respective service center can view the request and respond with an estimated service cost. The user then has the option to approve or reject the budget. Only after the budget approval by the user, the service is confirmed. After the service is completed, the status is updated, and users can provide feedback on their experience. This automated system ensures improved communication, cost transparency, and convenience for both customers and service centers. By replacing manual processes with a digitized workflow**,** the system not only enhances user satisfaction but also contributes to efficient service center operations.

Keywords: Automated Vehicle Service System, Online Service Booking, Budget Approval System, Vehicle Maintenance Management, Web-Based Service Platform.

Smart Contract-based Public Integrity Auditing for Cloud Storage Against Malicious Auditors

Lokesh E, A. Selvaraj
Thirumalai Engineering College, Kilambi, Kancheepuram.

ABSTRACT

Cloud storage, a vital component of cloud computing, faces significant challenges in ensuring data integrity, which hinders its widespread adoption. Public auditing models, which rely on third-party auditors (TPAs), have been developed to address these issues by offloading computation from users. However, maintaining the consistent trustworthiness of TPAs remains a major challenge, especially in preventing dishonest behaviors, such as collusion, procrastination, and forgery. In this paper, we propose a novel smart contract-based public integrity auditing scheme for cloud storage, introducing a transparent, non-black-box auditing process. This scheme adopts certificate less authentication, significantly reducing the overhead associated with traditional key management and certificate handling. To mitigate TPA dishonesty, we introduce a blockchain-based challenge generation algorithm and an auditing process preservation mechanism. The challenge algorithm ensures fair random sampling by leveraging block chain's immutability, reducing the risk of collusion between TPAs and Cloud Service Providers. The auditing process preservation mechanism prevents procrastination by recording task completion times and preserving metadata, ensuring full traceability and accountability. We also present a post-auditing validation mechanism that enhances the verifiability of auditing results, comprising two components: auditing computation proof, which verifies the correctness of computationally intensive steps, and auditing process replay, which replays the entire auditing using preserved metadata. Finally, we formally prove the security of our scheme and conduct a comprehensive performance comparison with existing solutions. The results demonstrate that our approach offers strong security, reduces computational overhead, and maintains comparable communication overhead to other schemes.

Keywords: Cloud Storage, Data Integrity, Public Auditing, Smart Contracts, Blockchain, Certificateless Authentication, Third-Party Auditor (TPA), Data Security.

Iris Recognition for Secure and Automated Attendance Management

Subramani V, Kishore, Bharath P
St.Peter's Institute of Higher Education and Research, Chennai.

ABSTRACT

Attendance management is an important task in educational institutions and organizations, but traditional methods such as manual registers and roll calls are time-consuming, error-prone, and inefficient. To overcome these limitations, this project proposes an automated iris-based attendance system that ensures high accuracy, security, and efficiency. The system uses the ESP32-CAM module to capture real-time images of the user's eye and transmits them to a backend server through a wireless network. The backend, developed using Python and Flask, processes the images using OpenCV techniques. During the enrollment phase, multiple eye images of each student are captured and stored in a dataset to train the recognition model. Once trained, the system can identify individuals in real- time and automatically mark attendance. The attendance data, along with date and time, is stored in a MySQL database for efficient management and retrieval. The system also includes a user-friendly web interface developed using ReactJS, which allows administrators to manage student records, monitor attendance, and generate reports. Additional hardware components such as LCD display, buzzer, and LED indicators provide real-time feedback to users. The system integrates a MySQL database to store student information and attendance records in a structured manner. Real-time feedback is provided through hardware components such as LCD display, buzzer, and LED indicators, enhancing user interaction. The proposed system offers several advantages, including high accuracy, prevention of proxy attendance, reduced manual effort, and improved efficiency. It is scalable and can be deployed in various environments such as schools, colleges, offices, and secure facilities. By combining IoT devices, biometric recognition, and modern web technologies, the system provides a reliable and advanced solution for automated attendance management.

Keywords: Iris-Based Attendance System, Biometric Recognition, ESP32-CAM Module, Automated Attendance Management, Internet of Things (IoT).

Blood Supply Chain Management: A Centralized System for Efficient Blood Donation and Distribution

Mrs. S. Nagalakshimi, Mr. S.Dhivagar, Mr. K.S.Mithiles Siva, Ms. S.Pooja Shri,
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

Blood is a critical resource in healthcare systems and is essential for saving lives during surgeries, accidents, and medical emergencies. However, hospitals often face difficulties in quickly identifying available blood units and contacting eligible donors. This paper presents a dashboard-based Blood Supply Chain Management System developed using Python that connects blood banks, hospitals, and donors through a centralized digital platform. The system provides separate dashboards for administrators, hospitals, and donors to manage blood inventory, submit blood requests, and track donation activities. Features such as donor eligibility tracking, blood camp scheduling, and emergency alerts help improve coordination between stakeholders. The proposed system enables real-time information sharing and reduces delays in blood availability, thereby supporting faster response in critical healthcare situations. The system also enhances transparency and accuracy in blood stock management through automated updates and monitoring features. Secure data management and user authentication ensure reliable access to sensitive healthcare information. Overall, the proposed platform improves efficiency, minimizes blood wastage, and supports effective healthcare service delivery during emergencies.

Keywords: Blood Supply Chain Management, Blood Bank Management System, Donor Management, Hospital Blood Request System, Healthcare Information Systems.

Smart Solar Dryer with IoT-Based Intelligent Temperature Control

Ms. Nagavani.C, Ragavan.N, Kishore Kumar.R, Vadivel.M,
Electronics and Communication Engineering,
Kamaraj College of Engineering and Technology, Virudhunagar.

ABSTRACT

Post-harvest losses remain a critical hurdle for farmers, as moisture and microbial growth often ruin crops before they reach the market. While traditional sun drying is the standard, it is frequently plagued by unpredictable weather, dust contamination, and significant nutrient loss. To solve this, our project introduces a Smart Solar Dryer that combines renewable energy with IoT-based intelligence for a more reliable, hygienic result. At its core, an Arduino-based system uses sensors to track internal conditions, automatically activating fans or backup heaters to keep the environment stable. We have also integrated image processing via a Convolutional Neural Network (CNN) so the dryer can "see" what type of food is inside and automatically adjust its settings for the best results. Farmers can monitor the entire process from their smartphones, drastically reducing the need for constant manual labor. This portable and eco-friendly design is specifically built for rural communities, ensuring that high-quality food preservation is both affordable and sustainable. By bridging the gap between traditional farming and modern technology, this system empowers small-scale growers to protect their harvests and boost their income regardless of the weather.

Keywords: Smart Solar Dryer, Internet of Things, Convolutional Neural Network, Food Preservation, Renewable Energy System.

AI Business Location Analyzer: AI System for Smart Business Recommendation and Location Analysis

Aathiswaran S, Vikram K, Karthikraja N, Dr. N. Gajalakshmi
Artificial Intelligence and Data Science
Kamaraj College of Engineering and Technology, Virudhunagar.

ABSTRACT

Starting a business without proper location analysis often leads to failure due to high competition, poor demand, and lack of strategic planning, especially in rapidly growing urban and semi-urban areas, and existing tools such as map services and business directories are limited as they only provide basic information like nearby shops and locations but do not offer intelligent recommendations or decision- making support; to address this, an AI Business Location Analyzer is proposed that provides smart business recommendations, market analysis, and risk prediction using advanced data analytics and machine learning techniques, where the system analyzes multiple factors such as population density, nearby competitors, income levels, and consumer demand patterns, with the collected data processed through an AI-based recommendation engine that evaluates market gaps and suggests the most suitable business opportunities for a given location, while also providing additional features such as investment planning, success probability estimation, and comparative analysis of nearby businesses, making the system scalable, efficient, and practical for real-world applications as an intelligent decision-support tool for entrepreneurs and startups.

Keywords: AI Business Location Analyzer, Machine Learning Recommendation System, Market Analysis, Business Risk Prediction, Decision Support System

Packet Level Analyst for Zero Day Intrusion Detection

Bhadhrunnisha R, Kanishka S, Kesiya S
Computer Science and Engineering (Cyber Security)
Gojan School of Business and Technology, Chennai.

ABSTRACT

The proposed anomaly-driven packet-level intrusion detection system addresses a critical limitation of traditional security mechanisms by focusing on real-time packet behavior analysis rather than predefined signature matching. This enables the detection of zero-day attacks that may remain unnoticed by conventional intrusion detection systems. The system employs ensemble anomaly scoring techniques to accurately identify suspicious activity while reducing false positives and ensuring that genuine threats are effectively captured. By analyzing packet-level characteristics, the framework provides granular visibility into network traffic and supports early identification of malicious behavior. Suspicious sessions are automatically flagged for logging and forensic investigation, offering valuable insights for strengthening network defense strategies. The proposed approach enhances real-time incident response capability, reduces potential damage, and efficiently handles large volumes of network traffic. Ensemble-based detection improves robustness by combining multiple analytical models, while continuous learning enables adaptation to evolving threat landscapes. This proactive framework remains effective against unknown and emerging cyber threats, significantly improving modern network security.

Keywords: Anomaly Detection, Zero-Day Attacks, Packet-Level Analysis, Intrusion Detection, Network Security, Machine Learning.

Secure Cluster-Based MANET Communication System with Shortest Path Routing

Dr.J.Suresh, P.Dharsaa, N.Karthick Raja, T.Kaviya,
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

Communication in Mobile Ad-hoc Networks (MANETs) for tactical environments is often hindered by dynamic topology changes and security vulnerabilities. Solution: This paper presents a cluster-based communication framework designed for soldier-to-base-station coordination. The system utilizes a dynamic clustering architecture to organize mobile nodes, automatically calculating the shortest routing path to the base station through optimal cluster heads. To ensure mission-critical security, all data including text, files, images, and real-time GPS coordinates are encrypted using the Advanced Encryption Standard (AES). Implementation: The system is developed with a Python-based Graphical User Interface (GUI) that provides real-time visualization of the network's topology, node status, and active routing paths. Significance: By combining automated path selection with end-to-end encryption, the proposed system enhances data throughput and security in infrastructure-less environments. Experimental results indicate that this clustered approach reduces routing overhead and latency compared to standard flat-routing protocols, making it a viable solution for modern digital battlefield operations.

Keywords: Mobile Ad-hoc Networks, Dynamic Clustering, Advanced Encryption Standard, Secure Tactical Communication, Shortest Path Routing

Encrypted Threads: Analyzing C2 Patterns to Detect Malware

Amudesh S, Hemalatha S, Pandiyan N
Computer Science and Engineering (Cyber Security)
Gojan School of Business and Technology, Chennai

ABSTRACT

With the rapid adoption of encryption protocols such as TLS and HTTPS, traditional network security mechanisms have become less effective in detecting malicious activities. Malware authors increasingly exploit encrypted channels to conceal Command-and-Control (C2) communications, making detection significantly more challenging. This paper focuses on analyzing encrypted traffic patterns without decrypting the payload, leveraging metadata, statistical features, and behavioral characteristics. The proposed approach emphasizes machine learning-based traffic analysis to identify anomalies associated with malware C2 communication. By extracting flow-level features such as packet size, timing intervals, and session duration, the system distinguishes benign from malicious encrypted traffic. This ensures privacy preservation while maintaining strong detection capabilities. The study also reviews existing techniques, including deep packet inspection limitations, signature-based detection weaknesses, and the evolution of behavioral analysis methods. A hybrid detection model is proposed to improve accuracy and scalability in modern network environments. Experimental evaluation demonstrates that the proposed method effectively detects encrypted C2 channels with high accuracy and low false-positive rates. The results highlight the potential of intelligent traffic analysis in strengthening cybersecurity defenses against advanced persistent threats.

Keywords: Encrypted traffic analysis, Malware detection, Command and control (C2) pattern, Machine learning.

A Systematic Evaluation of DBI Frameworks for JOP Detection in Stripped Binaries

Mohandass.B, Tamilselvi.G, Computer Science and Engineering (Cyber Security),
Gojan School of Business and Technology, Chennai.

ABSTRACT

Code-reuse attacks, such as Return-Oriented Programming (ROP) and Jump-Oriented Programming (JOP), remain the dominant post-Data Execution Prevention (DEP) exploitation techniques on Linux systems. Detecting active gadget chains at runtime requires efficient instrumentation tools; however, a systematic comparison of native Linux tracing facilities for this purpose is lacking. This paper presents StripJOP, a five-phase benchmark framework designed to evaluate the effectiveness of five Linux analysis tools objdump, strace, bpftrace, radare2 (r2), and a custom ptrace-based chain detector. The evaluation is conducted on three real-world binaries, including a synthetic vulnerable binary, */bin/ls*, and */usr/bin/curl*, using a novel metric called Recall@Overhead**.** This metric is defined as Recall% divided by (1 + Overhead%/100), producing a unified deployability score ranging from 0 to 1. Experimental results indicate that static analysis using objdump achieves the highest score of 1.0, while bpftrace provides the best runtime performance with scores up to 0.20 on *curl*. In contrast, ptrace-based single-step analysis incurs extremely high overhead (22,000%–107,000%), making it impractical for real-time use. Additionally, bpftrace requires root privileges for accurate probe data, and radare2 exhibits excessive overhead (40,000%–125,000%), limiting its suitability for online detection. The complete benchmark suite, JOPBench v10**,** is open-sourced to ensure reproducibility and further research. The findings provide valuable insights into selecting efficient tools for runtime detection of code-reuse attacks in Linux environments.

Keywords: Jump-Oriented Programming (JOP), Return-Oriented Programming (ROP), Code-Reuse Attacks, eBPF, bpftrace, strace, ptrace, radare2, Recall@Overhead, Gadget Detection, Linux Kernel Tracing, JOPBench

Obesmart – Unravelling the role of lifestyle factors in the rise of Obesity

Anwesha Pal, Bhavya Jivani, Jain University, Bengaluru

ABSTRACT

Recently, the effects of globalization, the rise in the number of people living in urban areas, and the fast pace of change in technology have significantly impacted daily life. These changes have affected the foods people eat, the ways in which they exercise and move, the ways in which they sleep, and the ways in which they cope with stress. These changes have resulted in the increase of non-communicable diseases in daily life. This paper will discuss the role of lifestyle in the increase of obesity in society and how these factors impact people at various stages of their lives, with the impact being more significant in the lives of the younger generation of people. In addition, the impact of lifestyle will be discussed in relation to the challenges people face in society and the challenges people face in trying to live a healthier lifestyle. The role of the environment in the preservation of energy, the consumption of plant-based foods produced locally, and the use of environmentally friendly transportation will be discussed.

Keywords: Non-communicable diseases, Cardiovascular disease, Obesity.

Blockchain-Based Secure Key Exchange Framework using Diffie-Hellman Protocol for Real Time Data Transmission Long-Range Monitoring

Dhanush V, Vignesh R, Vinith A ,
Computer Science and Engineering(Cyber security),
Gojan School of Business and Technology, Chennai.

ABSTRACT

Secure data transmission in modern networks is highly vulnerable to key interception and man-in-the-middle attacks. This work proposes a real-time secure key exchange system that integrates the Diffie-Hellman protocol with a lightweight blockchain framework. The system enables communicating parties to generate a shared secret key without transmitting it, while public keys are securely stored and verified on the blockchain to ensure integrity and prevent tampering. The derived key is used with AES encryption for protecting messages and files. Implemented using Python and Flask with RESTful APIs, the system supports real-time secure communication. Experimental results demonstrate improved confidentiality, integrity, and strong resistance to man-in-the-middle attacks, making it suitable for applications such as secure messaging, file sharing, IoT, and cloud security. The proposed system combines Diffie-Hellman key exchange, blockchain technology, and AES encryption to ensure secure and tamper-resistant data transmission. By storing and verifying public keys on the blockchain, the system effectively prevents key manipulation and man-in-the-middle attacks. The lightweight and real-time architecture makes it suitable for secure communication applications including messaging, cloud services, and IoT environments.

Keywords: Globalization, Obesity, Non-Communicable Diseases, Healthy Lifestyle, Sustainable Living

Energy-Efficient WSN Optimised Duty Cycling for Long-Range Monitoring

Arasa Kumaran T , Saimathan V, Srikanth A S, Manimuthu M, M. Mehaboob Subahani
Information Technology, Velammal Institute of Technology, Chennai.

ABSTRACT

This paper proposes an energy-aware environmental monitoring system based on Wireless Sensor Network technology to detect harmful chemical gases. The system operates using Internet of Things (IoT)-based sensor nodes integrated with long-range communication and power management mechanisms, thereby increasing battery life and improving overall operational efficiency. The system incorporates duty-cycling techniques, in which sensor nodes are periodically switched on and off to minimize total power consumption. This approach significantly reduces unnecessary energy usage while maintaining reliable environmental monitoring performance. The primary objective is to develop an affordable, low-power, and scalable Wireless Sensor Network for environmental surveillance, capable of supporting long-term deployment in remote and hazardous areas.

Keywords: Wireless Sensor Networks, Internet of Things, Environmental Monitoring, Gas Detection, Energy Efficiency, Duty Cycling, Battery Optimization.

Artificial Intelligence Driven Real Time Interface for Sign Language Communication

Dr. M. Rajesh Khanna, Meenakshi. R, R. Bawyah, K. Bharathi,
Information Technology,
Vel Tech Multi Tech Dr.Rangarajan Dr. Sakunthala Engineering College, Chennai.

ABSTRACT

Sign language recognition plays a key role in reducing the communication gap between individuals with hearing and speech impairments and the general public. This paper presents a real-time, vision-based sign language recognition system developed using computer vision and machine learning techniques. The proposed system utilizes a webcam to capture hand gestures and employs the MediaPipe framework to detect hand landmarks from each video frame. The extracted landmark coordinates are used as features for gesture classification using a Random Forest-based model. The system is designed to operate efficiently without requiring specialized hardware, making it suitable for real-time applications. Experimental results demonstrate that the proposed model achieves high recognition accuracy while maintaining consistent performance across varying hand positions and lighting conditions. The system supports both alphabet-based gestures and commonly used sign language words, enabling more practical and effective communication. The results indicate that the proposed approach provides an efficient and reliable solution for real-time sign language recognition using lightweight resources.

Keywords: Sign Language Recognition, MediaPipe, Machine Learning, Gesture Recognition, Computer Vision.

SafeLink: Advanced Phishing URL Detection using Cybersecurity Intelligence

Pooranavishwa M, Sonu M, Vishal Dev P
Computer Science and Engineering(Cyber security),
Gojan School of Business and Technology, Chennai.

ABSTRACT

Phishing attacks remain one of the most prevalent cybersecurity threats, exploiting user trust through deceptive messages and fraudulent digital content. Traditional phishing detection approaches rely on static feature extraction and single-stage classification models, which often fail to adapt to evolving attack strategies and contextual variations. To address these limitations, this paper proposes TACS-PD (Trust-Adaptive Contextual Scoring for Phishing Detection)**,** a machine learning–based framework that integrates Gene Regulatory Networks (GRN) and Long Short-Term Memory (LSTM) models. The GRN component captures structural and relational dependencies among phishing features, while the LSTM model learns sequential and temporal behavioral patterns. Additionally, a trust-adaptive scoring mechanism is introduced to dynamically adjust feature importance based on historical confidence and feedback. Experimental results demonstrate that the proposed system achieves improved detection accuracy and reduced false positives, making it highly effective against zero-day phishing attacks.

Keywords: Phishing Detection, Machine Learning, Gene Regulatory Networks, Long Short-Term Memory (LSTM), Trust-Adaptive Scoring, Cybersecurity, Zero-Day Attacks.

ChurnSense AI: An AI-Driven Customer Lifetime Value Prediction System

Mr. R. Selvamanikandan, Ms. S. Pooja Lakshmi, Ms. G. Mahalakshmi, Mr. C. Jeyachandran,
Artificial Intelligence and Data Science
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

Customer retention is a critical challenge for modern businesses, particularly in industries such as e-commerce, telecommunications, and subscription-based services. Many organizations primarily focus on customer acquisition while lacking the analytical tools required to effectively understand and retain existing customers. Customer churn, which occurs when customers discontinue their interaction with a product or service, leads to significant revenue loss over time. This paper proposes ChurnSense AI, an AI-driven customer lifecycle intelligence platform designed to predict customer churn and analyze engagement patterns using machine learning techniques. The system processes customer behavioral data, engagement metrics, and transaction records to identify customers at risk of leaving. It integrates engagement prediction, churn prediction, and Customer Lifetime Value (CLV) estimation to provide a comprehensive view of customer behavior. The combined outputs are used to generate a Customer Health Score, enabling businesses to assess customer loyalty and potential value effectively. The results are visualized through an interactive Streamlit-based dashboard, allowing organizations to monitor customer trends and implement proactive retention strategies. The proposed system enhances decision-making, improves customer retention, and maximizes long-term business profitability.

Keywords: Customer Churn Prediction, Customer Lifetime Value, Engagement Analysis, Machine Learning, Customer Retention, Data Analytics.

JobGenie: A Skill-Based Job Recommendation Platform for Fairer and Smarter Recruitment

Aadhi Shangar M, Balaji A, Sabarish B, Dr.E. Mercy Beulah,
Computer Science and Engineering,
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai.

ABSTRACT

Most online job platforms primarily rely on resume keyword matching and formal educational credentials, which often results in highly capable candidates being overlooked due to poor resume formatting or non-traditional academic backgrounds. In addition, hiring decisions may be unintentionally influenced by demographic factors, leading to bias in recruitment processes. This paper presents JobGenie**,** a skill-based job recommendation platform designed to support fairer and smarter recruitment by evaluating practical candidate capabilities rather than conventional resume attributes. The system assesses real-world technical performance through coding contributions, problem-solving records, and continuous learning indicators obtained from platforms such as GitHub and LeetCode. The proposed platform incorporates blind screening by masking personal identifiers during the initial evaluation stage, thereby reducing demographic bias. Automated skill-based scoring is performed using semantic similarity techniques, while personalized career roadmaps are generated to guide candidate development. Experimental evaluation across multiple candidate profiles and recruiter workflows demonstrates that JobGenie provides more relevant job recommendations, achieving a match relevance score of 4.2/5 compared to 3.1/5 in conventional keyword-based systems. The platform also contributes to significantly reduced hiring bias and enables approximately 30% faster recruitment decisions. The proposed framework aims to create a recruitment ecosystem in which candidates are selected based on demonstrated skills, regardless of educational pedigree or demographic background.

Keywords: Skill-Based Hiring, AI Recruitment, Semantic Similarity, Blind Screening, ATS Scoring, Candidate Ranking, Career Roadmap.

An Efficient Approach for Drone Detection in Aerial Images Using Deep Learning

Carinelia W Corthis, Deepika J, Abinaya S,
Computer Science and Engineering, Panimalar Engineering College. Chennai.

ABSTRACT

The increasing use of drones across applications such as surveillance, delivery services, agriculture, and defense has raised significant concerns related to airspace safety and security monitoring. Identifying drones in real-time is a challenging task due to their small size, rapid movement, and visual similarity to other flying objects such as birds, especially under varying environmental and lighting conditions. This paper presents a simplified deep learning-based approach for drone detection using aerial image datasets. The proposed work focuses on utilizing modern object detection techniques to accurately identify drones while maintaining efficient processing speed suitable for real-time applications. The system includes basic preprocessing steps, model training, and evaluation under different scenarios to ensure robustness. Performance is assessed using common evaluation metrics such as accuracy, precision, and detection speed, providing a clear understanding of the model's effectiveness. Experimental results indicate that deep learning methods can reliably detect drones with satisfactory accuracy while maintaining low computational complexity. The study emphasizes the suitability of lightweight detection models for deployment in real-world surveillance systems where quick response and resource efficiency are critical. Overall, this work demonstrates the potential of deep learning-based solutions in addressing emerging challenges in drone monitoring and airspace management.

Keywords: Drone Detection, Deep Learning, Object Detection, Aerial Surveillance, Computer Vision, Real-Time Monitoring.

Smart Bioremediation: Integrating Machine Learning and Multi-Omics for the Remediation of Recalcitrant Pollutants

Kalaivani B, Yashica, Dr S. Shantkriti
SRM University of Science and Technology, Tiruchirappalli

ABSTRACT

The intersection of Artificial Intelligence (AI) and synthetic biology has triggered the creation of so-called Cognitive Bioremediation, which is characterized by autonomous, data-driven recovery of degraded ecosystems. This article offers high-fidelity technical specifications of the union between the multi-omics information streams and the progressive machine learning frameworks in the 2026 biotechnological environment. We consider the application of so-called Physics-Informed Neural Networks (PINNs) in particular, which combine thermodynamic constraints and Monod kinetics into the loss functions of deep learning models, in order to bridge the so-called Laboratory-to-Field Gap. The review also discusses how Generative Protein Diffusion Models are used to evolve high-stability enzymes in vivo to be able to break down complex anthropogenic contaminants such as microplastics and PFAS. We show that with the synthesis of existing case studies on Environmental Digital Twins, we can be certain of a 3545 percent increase in pollutant mineralization rates of traditional bioaugmentation when using Smart Bioremediation 4.0 through real-time Bayesian optimization of nutrient flux. Lastly, the most important is the critical transition to the concept of black-box algorithms to the concept of Explainable AI (XAI) frameworks to align with the emerging global regulatory requirements. This paper sets up an advanced technical pathway on implementation of self-healing ecological infrastructures, which places algorithmic intelligence as the key driver to global environmental sustainability and precision-guided microbial engineering.

Keywords: Smart Bioremediation 4.0, Machine Learning, Multi-Omics Integration, Precision Environmental Biotechnology, Recalcitrant Pollutants.

Hospital Management System – Advanced Web-Based Healthcare Management Platform

D. Jackulin Sharmi, P. L. Rubeshkanna, S. Santhosh, M. Vignesh
Computer Science and Engineering, Thirumalai Engineering College

ABSTRACT

This paper presents an advanced Hospital Management System (HMS) designed to automate and optimize healthcare administrative operations. The system integrates patient management, doctor scheduling, appointments, medical records, and billing services into a centralized web-based platform. Built using Django, Python, and SQLite database, the system ensures scalability, security, and efficiency in handling hospital workflows. Modern healthcare institutions require reliable digital infrastructure to manage large volumes of patient data and operational processes. Traditional paper-based or semi-digital systems often lead to inefficiencies, delays, and data inconsistencies. The proposed system addresses these issues by providing a centralized, role-based, and secure platform where administrators, doctors, and staff can access and manage information in real time. The system provides modules such as user authentication, patient registration, appointment scheduling, and medical record management. Security features include authentication mechanisms and role-based access control to ensure that sensitive medical data is protected. The proposed solution improves hospital productivity, reduces manual workload, and enhances patient care quality.

Keywords: Hospital Management System, Healthcare IT, Django Framework, Patient Management, Web Application.

An Enhanced Method for Detecting Aerial Object and Laser Target Indication System using Artificial Intelligence

Dr.M.Sukanya, Velan S, Samith C, Vishnu M
Computer Science and Engineering, Kathir College of Engineering, Coimbatore.

ABSTRACT

This paper presents an AI-based aerial object detection and laser target indication system designed for real-time surveillance applications. The system utilizes a YOLOv8 deep learning model to detect aerial objects such as drones and aircraft from live video input. The detected object's position is calculated and transmitted to a Arduino Uno, which controls a servo motor to align a laser module towards the target. This integration of artificial intelligence with embedded systems enables both detection and physical target indication. The proposed system provides a low-cost, efficient, and real-time solution for intelligent monitoring and tracking applications. The system combines deep learning and embedded hardware to achieve accurate aerial object detection and automatic laser target indication in real time. By using the YOLOv8 model with Arduino-controlled servo mechanisms, the system can efficiently track and point toward detected aerial targets. The proposed solution is cost-effective, scalable, and suitable for surveillance, security, and intelligent monitoring applications.

Keywords: Aerial Object Detection; YOLOv8; Computer Vision; Real-Time Monitoring; Arduino Uno; Servo Motor Control; Laser Target Indication; Artificial Intelligence; Embedded Systems

AI-Driven Digital Twin Framework for Real-Time Smart Farm Decision Support using Data Engineering Pipelines

Dr. Sivakumar. K, Ms. Hemadharshini V, Ms. Maria Mirza. J, Ms. Priyanga. G
Agricultural Engineering, Nandha Engineering College, Tamil Nadu.

ABSTRACT

Agriculture is increasingly challenged by climate variability, water scarcity, and inefficient decision-making practices rooted in experience rather than data-driven insight. While recent advancements in smart farming have introduced predictive analytics using Internet of Things (IoT) sensor networks and machine learning models, these systems remain fundamentally reactive, offering forecasts without prescriptive guidance. This paper proposes a comprehensive AI-driven Digital Twin (DT) framework that constructs a dynamic virtual replica of a physical farm environment, enabling real-time simulation, multi-scenario evaluation, and prescriptive decision support. The framework integrates five architectural layers: physical IoT sensing, data engineering pipelines, digital twin modeling, a scenario-based simulation engine, and an intelligent decision support interface. Machine learning models including Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks are embedded within the twin to predict soil behavior, crop growth trajectories, and meteorological trends. A what-if simulation engine evaluates multiple intervention strategies simultaneously, ranking outcomes by expected yield, water consumption, and resource efficiency. Experimental validation on benchmark agricultural datasets demonstrates a prediction accuracy of 90.3%, a 20% reduction in irrigation water usage, and a 15.4% improvement in crop yield over baseline methods. This work advances the state of precision agriculture by transitioning farm management systems from passive predictors to active, prescriptive decision-support platforms.

Keywords: Digital Twin, Smart Agriculture, Precision Farming, Machine Learning, IoT, Data Engineering, Decision Support System, LSTM, XGBoost, Scenario Simulation.

Sustainable Agriculture Through AI: Leveraging Scalable Technologies

Ramya, Priya Darshini K, Shantkriti S
Biotechnology, School of Biosciences and Technology,
SRM Institute of Science and Technology Tiruchirappalli

ABSTRACT

Artificial Intelligence (AI) is transforming environmental sustainability by improving bioremediation and agricultural practices. This review highlights the integration of AI techniques such as machine learning, deep learning, and low-cost intelligent systems in environmental biotechnology and smart agriculture. Traditional bioremediation methods often face inefficiencies due to complex environmental conditions, which AI addresses through predictive modelling, optimization of microbial processes, and real-time monitoring of soil, water, and air quality. In agriculture, AI-driven tools like mobile applications, IoT sensors, and basic algorithms enable precision farming by optimizing irrigation, fertilizers, and pest control, thereby enhancing productivity while reducing environmental impact. Applications include pollutant degradation prediction, crop disease detection, and yield forecasting. The democratization of AI makes these technologies accessible to smaller stakeholders, though challenges such as limited data, infrastructure gaps, and low digital literacy persist. Overall, AI offers a promising, scalable solution for sustainable development.

Keywords: Artificial Intelligence, Bioremediation, Smart Agriculture, Sustainable Farming, Environmental Biotechnology.

Smart Vision: AI-Guided Navigation for the Visually Impaired

Ganesh K, Pranav AV, Mohamed ImthiyasA, Liyandas D
Computer Science and Engineering
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai.

ABSTRACT

The smart assistive device combines AI-based navigation, real-time object detection, and voice feedback, utilizing IoT and sensor technology to enhance mobility for visually impaired users. The global assistive technology market is projected to grow significantly by 2030, showing the increasing reliance on intelligent systems in daily life. The device uses ESP32 CAM and ESP32 microcontrollers for connectivity and data transfer, enabling users to detect objects, recognize faces, and identify currency in real time. The system also includes vision-based features such as image recognition, object detection, facial recognition, and optical character recognition to support intuitive interaction and environmental awareness. With IoT functionality, low power consumption, and cloud integration, the device provides real-time assistance and supports remote monitoring. Studies indicate that real-time assistive devices can improve navigation efficiency by 30% and enable safer independent movement in 20% of users, contributing to enhanced accessibility and timely decision-making. The smart assistive device uses AI, IoT, and sensor technologies to support visually impaired users through real-time object detection, facial recognition, currency identification, and voice feedback. Using ESP32 CAM and cloud integration, the system enables safe navigation, environmental awareness, and remote monitoring with low power consumption. The proposed solution improves independent mobility, accessibility, and decision-making, offering a cost-effective and intelligent assistive technology.

Keywords: Assistive Technology, AI Navigation, Real-Time Object Detection, ESP32 Microcontroller, Vision AI, Visually Impaired Support.

An Enhanced Machine Learning Approach for Customer Churn Prediction using Classification and Clustering Techniques

Akshaya Vishal S V, Sakthivel A, Dinesh Pandiyan S
Artificial Intelligence and Data Science,
Kamaraj College of Engineering and Technology, Virudhunagar

ABSTRACT

Customer churn prediction is a critical challenge for businesses in the telecommunications and service sectors, directly impacting revenue retention and customer relationship management. This paper presents a comprehensive machine learning framework for predicting customer churn by integrating multiple classification algorithms with unsupervised clustering. Logistic Regression, Decision Tree, Random Forest, and AdaBoost classifiers are evaluated alongside K-Means clustering for customer segmentation. The Synthetic Minority Oversampling Technique (SMOTE) is employed to address class imbalance, while GridSearchCV enables systematic hyperparameter optimization. Models are rigorously evaluated on a telecom dataset of 7,043 records using accuracy, precision, recall, F1-score, and AUC-ROC. Random Forest achieves the highest accuracy of 88%, F1-score of 0.84, and AUC of 0.93. SMOTE improves churn recall by up to 53% across models. K-Means clustering identifies three actionable customer risk segments. Pseudocode for all algorithms is presented alongside experimental results to facilitate reproducibility.

Keywords: Customer Churn, Random Forest, AdaBoost, SMOTE, K-Means Clustering, Hyperparameter Tuning, ROC-AUC, Telecom Analytics

FinServe AI Assistant: A Three-Tier Intelligent Response System for BFSI Domain

Mrs. Nisha R., Ms. K. Lawanya, Ms. J. Abinaya, Ms. R. Lejancy
Artificial Intelligence and Data Science
CARE College of Engineering, Tiruchirappalli, Tamil Nadu

ABSTRACT

The Banking, Financial Services, and Insurance (BFSI) sector manages a large volume of customer queries related to loan eligibility, EMI calculations, interest rates, and account services. Traditional support systems often suffer from inconsistent responses, limited availability, and difficulty maintaining financial compliance. This project presents the FinServe AI Assistant, a domain-specific BFSI AI Assistant designed for secure and reliable query handling. The system is built on a structured three-tier pipeline that ensures both accuracy and efficiency. In Tier 1, cosine similarity matching is performed using the sentence-transformers all-MiniLM-L6-v2 model on a curated dataset of 178 Alpaca-format BFSI query-response pairs. When the similarity threshold (0.82) is met, a verified response is returned instantly, eliminating hallucination risks. Tier 2 uses a Small Language Model (SLM), specifically TinyLlama enhanced with LoRA fine-tuning, to handle domain-related queries that do not meet the similarity threshold. For more complex queries, Tier 3 implements Retrieval-Augmented Generation (RAG) using a FAISS vector database, ensuring responses are grounded in verified financial documents. The backend is implemented using FastAPI with a modular architecture, while the frontend uses React with TypeScript to deliver a responsive and user-friendly interface. The system operates entirely on local infrastructure, ensuring data privacy and regulatory compliance. Experimental results demonstrate accurate query routing, fast response times, and consistent performance across all tiers.

Keywords: Three-Tier Pipeline, BFSI Domain, Cosine Similarity, Small Language Model, Retrieval-Augmented Generation, TinyLlama, LoRA Fine-Tuning, FastAPI, FAISS

Automated Detection of Security Vulnerabilities in Static Websites using AI Agents

Aravind Pandi, Vijaya Ramanan, Priya Dharsan
Artificial Intelligence and Data Science,
Kamaraj College of Engineering and Technology, Virudhunagar, Tamil Nadu

ABSTRACT

With the rapid growth of static websites across diverse use cases including developer portfolios, product landing pages, documentation hubs, open-source project sites, and startup MVPs many developers unknowingly expose sensitive information such as API keys, credentials, and configuration data in client-side code. This paper presents an AI-driven multi-agent system for automated detection of security vulnerabilities in static websites. The proposed system employs autonomous agents to analyze HTML and JavaScript files, identify patterns of sensitive data exposure, and generate structured security reports accompanied by remediation recommendations. By integrating rule-based detection with intelligent contextual analysis, the system improves detection accuracy and markedly reduces false positives. Experimental results on 45 real-world static websites demonstrate effective identification of common vulnerabilities including hardcoded API keys, Firebase configuration leaks, and insecure authentication tokens, achieving a mean detection accuracy of 90.6%. This work highlights the potential of AI agent frameworks in enhancing web security and empowering developers to build safer applications from the ground up.

Keywords: AI Agents, Web Security, Static Analysis, Static Websites, API Key Leakage, CrewAI, Vulnerability Detection, Multi-Agent Systems

Fake News Detection Using Natural Language Processing and Deep Learning: A Comparative Study

Arunjunai Rajan M, Ponkumar N, Mohan Kumar T
Kamaraj College of Engineering and Technology

ABSTRACT

The rapid proliferation of misinformation across digital platforms has made automated fake news detection a critical priority. This paper presents a comparative study of Natural Language Processing (NLP) techniques and deep learning models for detecting fake news articles. We evaluate traditional machine learning classifiers Naive Bayes, Support Vector Machine (SVM), and Logistic Regression against modern deep learning architectures including Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model. Experiments are conducted on the LIAR benchmark dataset comprising 12,836 labeled statements across six veracity classes, simplified to binary classification (fake vs. real). Text preprocessing involves tokenization, stop-word removal, lemmatization, and TF-IDF vectorization for traditional models and contextual embeddings for deep models. Results indicate that fine-tuned BERT achieves the highest accuracy of 91.3% and F1-score of 0.90, substantially outperforming classical approaches. BiLSTM achieves competitive performance at 87.6% accuracy. The study demonstrates the superiority of transformer-based models for fake news detection and offers a practical framework applicable to social media monitoring systems.

Keywords: Fake News Detection, BERT, LSTM, NLP, Misinformation, Text Classification, Deep Learning

A Review of Computer Vision-based AI Techniques for Solid Waste Segregation

Jeyavarshini R P, Parineetha Viswanaath, Dr Shantkriti S
Biotechnology, School of Biosciences and Technology,
SRM Institute of Science and Technology (SRMIST) Tiruchirappalli Campus,

ABSTRACT

The current study presents the most recent developments in the field of artificial intelligence-base automatic waste segregation, in particular, the progress of waste sorting systems based on computer vision technology and image recognition capabilities. Traditional approaches to waste classification have several weaknesses errors, limiting the effectiveness of recycling procedures. However, the modern developments in artificial intelligence involve the usage of complex deep learning algorithms and architectures, including convolutional neural networks (CNNs), vision transformers (ViTs), and hybrid solutions for automatic classification and sorting of different waste materials, such as plastic products, metal objects, papers, and organic items. Such smart waste processing solutions benefit from connection to the Internet of Things technologies, robotics, as well as edge computing. In addition, other innovations in this area include using transfer learning, data augmentation, and multimodal sensing that improve the quality of recognition under various circumstances. Finally, another important development worth noting is cloud computing analytics and smart waste bins equipped with AI technologies. The potential challenges in this area are associated with relatively expensive deployment costs and small-size datasets but still, all these innovations significantly facilitate waste classification and contribute to environmental sustainability.

Keywords: Artificial Intelligence (AI), Computer Vision, Waste Segregation, Deep Learning, Smart Waste Management

Smart Iot Based Home Automation with Real Time Alerts

Dr. B.S. Charulatha , Elam Parithi. P, Ravichanthiran. R, Govindarasu. G
Computer Science and Engineering,
St. Peter's Institute of Higher Education and Research, Chennai, India

ABSTRACT

This paper presents the design and implementation of a Smart IoT-Based Home Automation and Security System using the ESP32 microcontroller. The system integrates multiple sensors, including an ultrasonic sensor for human detection, an MQ2 gas sensor for leakage detection, a flame sensor for fire monitoring, and a DHT11 sensor for temperature measurement. A servo motor enables automatic door control. The ESP32-CAM module enhances the system by capturing images of visitors and sending them to the user through Telegram for real-time remote monitoring. In addition, the system continuously analyzes environmental conditions and triggers alerts in case of abnormal situations such as gas leaks, fire hazards, or high temperature. The use of WiFi-based communication allows seamless interaction without the need for cloud infrastructure. This system offers a cost-effective, scalable, and efficient solution for modern smart home applications.

Keywords: IoT, ESP32, Smart Home Security, Gas Detection, Fire Detection, Temperature Monitoring, Ultrasonic Sensor, Telegram Alerts, Real-Time Monitoring

AI-Enhanced Cybersecurity Frameworks: Integrating Machine Learning, Blockchain-IoT, and Privacy-Preserving Techniques for Ethical Threat Management

Dr.C.Seelammal , S.Vallimayil
Solamalai College of Engineering, Madurai.

ABSTRACT

This paper explores the convergence of AI-driven security, block chain-integrated IoT, and privacy-preserving AI to address modern cybersecurity challenges ethically. We propose a hybrid framework that leverages machine learning for automated threat detection, block chain for secure IoT data management, and federated learning for model training without exposing sensitive data. Through methodology involving simulation and descriptive analysis, we demonstrate a 35% improvement in threat detection accuracy while preserving user privacy. Findings underscore the ethical imperative of balancing innovation with data protection, offering actionable insights for scalable deployment. The proposed framework enhances cybersecurity by integrating intelligent threat analysis with decentralized and secure data handling mechanisms. Machine learning algorithms continuously monitor network activities to identify anomalies and potential attacks in real time. Blockchain technology ensures data integrity, transparency, and tamper-proof communication among IoT devices. Federated learning enables collaborative model training while keeping sensitive user data localized and protected from external exposure. Experimental results indicate improved scalability, reduced security risks, and stronger privacy preservation, making the framework suitable for next-generation secure IoT ecosystems.

Keywords: Artificial Intelligence Security, Blockchain-Integrated IoT, Privacy-Preserving AI, Federated Learning, Cybersecurity Threat Detection

Interactive Tutor -Student Collaboration Platform Using React

R.Nithish Sunil Raj, M.Kawin K.Yoganarasimman S.Logeshwaran
Computer Science and Engineering
Vel Tech Multi Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai.

ABSTRACT

The Interactive Tutor-Student Collaboration Platform is a web-based application developed using React that aims to improve communication and learning between tutors and students. The platform provides a user-friendly interface where students can easily connect with tutors, schedule sessions, share study materials, and ask questions in real time. which help create an engaging and collaborative learning environment. Tutors can manage multiple students, provide personalized guidance, and monitor their performance effectively. The platform also includes features such as notifications, progress tracking, and secure login authentication to enhance usability and security. Students can access recorded sessions and uploaded resources anytime for better learning flexibility. Tutors can create assignments, evaluate student performance, and provide feedback through the system. The application is designed with responsive web technology, ensuring smooth access across desktops and mobile devices. Overall, the platform promotes interactive, accessible, and personalized online learning experiences for both tutors and students.

Keywords: Tutor-Student Collaboration Platform, Web-Based Learning System, Real-Time Communication, Personalized Learning, React Application

A Review on Industrial Drying Technologies and AI-Based Energy Efficiency Optimization

Juhitha D S, Pramila Murugesan
Biotechnology, School of Biosciences and Technology,
SRM Institute of Science and Technology, Tiruchirappalli

ABSTRACT

Industrial drying is a crucial unit operation in industries such as food processing, pharmaceuticals, textiles, and chemicals. It plays an important role in material preservation by removing moisture and improving shelf life. However, drying is one of the most energy-intensive industrial processes, accounting for a significant portion of industrial energy consumption. This paper presents a comprehensive study of both traditional and modern drying technologies, including their operating mechanisms, advantages, and limitations. In addition, the integration of Artificial Intelligence (AI) techniques such as Artificial Neural Networks (ANN), Machine Learning (ML), Genetic Algorithms (GA), and Fuzzy Logic is explored to improve energy efficiency and optimize the drying process. AI-based systems support predictive modelling, real-time monitoring, and intelligent process control, thereby enhancing system performance and reducing energy consumption. The paper also discusses the challenges and future scope associated with the implementation of AI-driven drying systems in industrial applications. Overall, this study highlights the potential of AI in promoting sustainable, cost-effective, and energy-efficient industrial drying processes.

Keywords: Industrial Drying, AI-Based Optimization, Energy Consumption, Drying Kinetics, Smart Manufacturing

Medimart: Online Medicine Ordering Application

Ramesh R, Dhasvinth S, Sobaan Areeb K S
Information Technology, Dr.M.G.R. Educational and Research Institute, Chennai

ABSTRACT

The process of procuring medicines from physical pharmacies can often be time-consuming and inconvenient, especially for elderly individuals, differently-abled persons, and patients suffering from chronic illnesses who require frequent refills. Although online medicine delivery platforms have emerged as a convenient alternative, many existing solutions still depend heavily on manual prescription verification by pharmacists. This manual process not only slows down the order fulfillment cycle but also increases the chances of human error, leading to potential delays and inefficiencies in medicine delivery.

Keywords: Convolutional Neural Networks(CNN), Optical Character Recognition (OCR), AI-powered prescription recognition system, Pharmacist Module, digital healthcare ecosystem , digital healthcare ecosystem,

External Parasites Detection Using Thermal Imaging on Pets

Drishya Darwin, Akash. S, Priyaharshika. R
Information Technology
Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

The early detection of external parasites in pets is crucial for their health and for stopping the spread of diseases. However, traditional diagnosis relies heavily on manual observation and invasive methods, which are not very effective in the early stages. This paper presents a non- invasive and smart system designed for the early detection of external parasites in pets. It uses thermal imaging techniques combined with the Internet of Things (IoT) and artificial intelligence. The system employs low-cost thermal sensors to identify temperature changes on the pets' surfaces, as inflammation caused by parasites leads to thermal differences. The thermal images are preprocessed and analyzed using a lightweight Convolutional Neural Network (CNN) that is optimized for low-resolution images to determine if parasites are present. The IoT and cloud features of the system support real-time data transmission, secure storage, and automated notifications to pet owners. This setup makes it easier for them to take prompt action. Additionally, a learning mechanism is included to enhance detection accuracy. Overall, this system offers a scalable, affordable, and ethical solution for smart pet healthcare monitoring and shows the potential of AI-enabled IoT systems to change modern veterinary diagnosis.

Keywords: Thermal Imaging, IoT, Parasite Detection, Pet Healthcare, Machine Learning, Cloud Computing, CNN, Non-Invasive Diagnosis.

Real-Time Big Data Visualization of Experiments

Mrs.S.Agalya, Y.Nithesh Raj,
Information Technology, Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

The Modern experimental environments generate massive volumes of heterogeneous data from sensors, simulations, laboratory instruments, and user-driven processes. Traditional batch-oriented analysis methods are insufficient for extracting timely insights from such continuously evolving datasets. This project presents a scalable framework for real-time big data visualization of experiments, enabling continuous data ingestion, processing, and interactive visualization. The proposed system integrates distributed data streaming, real-time processing engines, and dynamic visualization dashboards to transform raw experimental data into actionable insights. By employing stream-processing techniques and efficient data pipelines, the framework minimizes latency while ensuring reliability and scalability. Interactive visualization components allow researchers to monitor experimental parameters, detect anomalies, and analyze trends as data is generated. The architecture is designed to support high-throughput data streams and adaptable visualization modules suitable for scientific research, industrial monitoring, and simulation-based experiments. Performance evaluation demonstrates improved responsiveness, reduced processing delay, and enhanced decision-making capability compared to conventional post-processing approaches. The system provides a flexible and extensible solution for real-time experimental data analysis in data-intensive environments.

Keywords: Real-time Real-Time Data Processing, Big Data Analytics, Stream Processing, Data Visualization, Experimental Data Monitoring, Distributed Systems, Interactive Dashboard.

NeuroScent: A Scent-Inspired Multi-Layer Encryption System for Secure Communtication

Mrs.D.Madhura , Karan.K, Praveen Kumar.T
Information Technology Dr.M.G.R.Educational and Research Institute Chennai,

ABSTRACT

A new perspective on secure data transfers across digital platforms demonstrates the shortcomings of previous approaches. Something new replaces fixed keys and endless web addresses connected to large hubs. Imagine short-lived connections between devices that are only long enough to safely transfer bits of data. Each step is concealed by a combination of timed codes that take inspiration from the layers of scents. The handshake is guided by temporary tags that are cleaned after each use. No enduring trails. Every round adds to the jumbled pieces, cycling quickly before disappearing. Short windows avoid leaks that others leave open and prevent recurring breaches. Each session ends after one minute of inactivity, but complete activity continues for up to twenty minutes after logging in; careful timing ensures that access is secure. Data begins as letters, changes to numbers, goes through layered mappings, splits into pieces, inserts random fillers, wraps under AES-GCM encryption, and is finally locked in a custom manner.nsf form. Firebase manages who gets in when, while a Flutter interface communicates with Python to handle crypto tasks. Each ticket for NeuroScent only permits one entry. stops copied tokens cold and automatically reuses blocks. Session IDs only exist as SHA-256 shadows, which make tracking more difficult and leave less trace. Combining strict control over active sessions with encryption that verifies authenticity is one method it uses to protect data. Its ability to protect secrets, stop changes, function flawlessly, and run effectively during testing is what makes it stand out. This configuration works well for short- term teamwork, secret messaging, and safe file transfers without causing lag. It works well in situations where privacy is crucial, according to real-world use cases.

Keywords: NeuroScent, Secure Data Exchange, Online Session Key, One-Time Token, AES-GCM, Scent Mapping, Multi-Round Encryption, .nsf File Format, Firebase Session Management, Replay Attack Prevention, Confidentiality, Integrity

SPIN-X: Solar Power Invisible Transmission Using Radio Frequency Technology

Mithun L., Mohamed Harish.A,
Information Technology, Dr. M.G.R Educational and Research Institute, Chennai, India

ABSTRACT

The increasing global demand for sustainable and renewable energy sources has encouraged researchers to explore innovative technologies for efficient energy generation and distribution. Solar energy has emerged as one of the most promising renewable energy resources due to its abundance and environmental friendliness. However, conventional solar power transmission systems rely heavily on wired infrastructure, which results in significant energy losses, higher installation costs, and limitations in remote or inaccessible locations. To address these challenges, this research proposes SPIN-X (Solar Power Invisible Transmission eXperiment), a conceptual system that enables wireless transmission of solar-generated power using Radio Frequency (RF) technology. On the receiving end, the rectenna converts the RF signals back into usable electrical energy, which can be utilized to power small electronic devices. The main objective of this project is to explore the potential of wireless power transmission technologies as a complementary solution to traditional power distribution systems. By removing the need for wired connections, the SPIN-X system reduces transmission losses caused by resistance in electrical wires and minimizes infrastructure requirements. The proposed SPIN-X model highlights the future potential of integrating wireless power transmission with renewable energy technologies. Although the current prototype is developed for low-power applications, the concept can be further expanded for larger-scale implementations such as smart cities, wireless charging infrastructure, and remote energy distribution networks. In conclusion, SPIN-X represents an innovative step toward improving the efficiency and flexibility of solar energy transmission while contributing to the advancement of sustainable and wireless energy solutions.

Keywords: Solar Energy, Wireless Power Transmission, Radio Frequency (RF), Renewable Energy, Rectenna Technology.

HireMate Software: Resume Builder, Optimizer, and ATS-Ready Enhancement Solution.

Mr.V.Raghavan, Kabilan.S, Mukesh Kumar.M
Information Technology Dr.M.G.R.Educational and Research Institute
Chennai,TamilNadu

ABSTRACT

Recruiters widely use Applicant Tracking Systems (ATS) as an essential tool for screening and selecting job applicants. This paper presents an advanced AI-enabled resume building and optimization platform called *HireMate*, designed to generate ATS-friendly resumes tailored for specific job roles. The system requires users to upload a single resume, which can then be dynamically optimized for multiple job descriptions. Using advanced Natural Language Processing (NLP) and semantic search techniques, HireMate enhances resume quality by improving keyword relevance, skill matching, and overall compatibility with ATS systems. The platform incorporates several NLP components, including transformer-based BERT and SBERT models, keyword extraction, skill relevance scoring, and heuristic-based ATS validation techniques to generate optimized resumes along with meaningful ATS scores. Experimental evaluation of the optimized resumes demonstrated improved semantic matching, better keyword relevance, and higher ATS compatibility compared to conventional resume systems. The proposed Resume Optimizer provides a complete, scalable, and intelligent end-to-end framework for resume enhancement, making it highly suitable for real-world recruitment and career development applications.

Keywords: HireMate, Intelligent, Natural Language Processing (NLP), Resume Parsing, Skill Extraction, Job Candidate Matching, Semantic Similarity Analysis, TF-IDF Vectorization, Cosine Similarity, Automated Resume Screening. Semantic Similarity, AI Resume Building, ATS Scoring.

Banana Pseudostem-based Biocomposites from Agro-waste with Machine Learning-driven Optimization for Sustainable Packaging

Vanshitha R, Nivedha N, Shantkriti S
Biotechnology, School of Bioscience and Technology
SRM Institute of Science and Technology Tiruchirappalli

ABSTRACT

Plastic pollution is one of the major environmental problems that individuals must deal with. Thus, there is a need to develop sustainable alternatives based on agricultural biodegradable polymers and waste. One of the biopolymers which show potential to be used as a substitute for plastic are biocomposites made of banana pseudostems, since they contain a high level of cellulose, have strong mechanical properties and are available as agro-waste. This review aims to summarize the latest finds in natural fibre-reinforced biocomposites, with particular focus on banana fibres, starch-based and lipid-based additives like waste cooking oil. Combining these materials in a synergetic manner increases the mechanical, barrier, as well as biodegradation properties of the material that can be used in packaging. Also, some environmental advantages, scalability issues, and future perspectives such as machine learning-based optimization are touched upon. Such materials may provide effective avenues of developing sustainable circular economy-centred packaging solutions.

Keywords: Banana fibre, Biocomposites, Sustainable packaging, Agro-waste valorisation, Biodegradable polymers.

Explainable AI–Integrated Congestion Control Mechanism for Mission-Critical Wireless Applications

Karthika K, Computer Science and Engineering,
Sri Venkateswaraa College of Technology
Sundarambal B, Computer Science and Engineering, Chennai Institute of Technology

ABSTRACT

Healthcare monitoring systems, autonomous vehicles, military communications, industrial automation, and other mission-critical wireless applications require a high level of reliability, low latency, and adaptability in the network. In these types of environments, the slightest delays or packet losses can result in serious failures and safety hazards. Although conventional congestion control techniques work well in a stable environment, they are usually unable to cope with highly dynamic wireless environments and are not transparent in decision-making. This constraint decreases their applicability to mission-critical applications, where performance and trust are crucial to overcome these issues. This study presents an explainable artificial intelligence (XAI)-based congestion control system that can improve the efficiency and interpretability of a network. The suggested framework uses machine learning algorithms to forecast congestion using real-time network parameters, including the packet arrival rate, queue length, and bandwidth usage. Moreover, methods of explainability, namely Shapley additive explanations (SHAP), were also implemented to provide straightforward and understandable information about the choices of the model. The outcomes of the simulations prove that the model under consideration performs significantly better than traditional and standard AI-based methods because it provides higher throughput, lower latency, and less packet loss overall. This study can be beneficial in the development of intelligent and transparent congestion management solutions, which will facilitate the adoption of reliable AI-based solutions within mission-critical wireless communication settings.

Keywords: Explainable AI, Congestion Control Wireless Networks, Machine Learning, Network Optimization

Dynamic Capsule Routing Model of Behavioral Analysis and Prevention of Keylogging Intrusions

Jeyshankar, Avinash G, Information Technology
Dr. M.G.R. Educational and Research Institute Chennai

ABSTRACT

Keylogging software is one of the most secretive and devastating types of cyber threat, as it allows to intercept user input and supply the authentication, financial, and secretive communications indefinitely. The traditional security measures which are based on signature matching and rule analysis which is done statically proves to be less and less effective in face of advanced obfuscation, polymorphing and behavioral evasion techniques. The current study paper offers partial alleviation with the help of the Artificial Neural Network models, but the performance reduction becomes common in a high-dimensional feature setting because of the hierarchical attenuation to the information. The current research paper proposes a Capsule Network–based detection and prevention framework aimed at increasing structural feature conservation and behavioral understanding. Spatial correlations, instantiation parameters and activity relationships are coded in capsules, which allow robust detection of disguised malicious processes. Systematic extraction of behavioral indicators such as abnormal frequency of keystroke logs, pattern of unauthorized API invocation, abnormal system calls and outbound network communication are processed by dynamically routed to undergo refined classification with a detection accuracy of 98.72 compared to traditional ANN-based methods. A decrease in the false positive rate to 1.84% is a sign of a better reliability in the case of complex intrusion. The deployed deployment model has a response latency of less than 120 milliseconds in real time, which means the interception of malicious activity. High-quality generalization aids flexibility to zero-day and polymorphic variants of a keylogger to create an effective and scalable cybersecurity defense system.

Keywords: Capsule Network (CapsNet), Keylogger Malware Detection, Cybersecurity - Threat Analysis, Behavioral Feature Extraction, Dynamic Routing Mechanism, Malware Classification Accuracy, Intrusion Prevention Framework, Network Security Analytics.

Smart Bag for Women Safety

Dr.N.Kanya , R.Aswin , A.Bagavath Raj
Dr MGR Educational and Reserarch Institute, Chennai,India.

ABSTRACT

The world is becoming a more hazardous place for women in every manner. Crime committed by women is increasing. Working women are feeling more insecure as a result of rising crime. This study recommends a quick- response strategy for helping women in need. If a lady feels that someone is harassing her, she can click the button to reveal where she is. Based on latitude and longitude, information is sent as an SMS alert to the pre-defined numbers. The Arduino Uno is the controller in use. It has an LCD (16x2) display, a GPS module, a GSM modem, and a push button interface. The controller transfers the GPS module's current position data to the designated phone number when the switch is pressed via a GSM modem. The The "C" language is used to construct the software. The purpose of this endeavour is to provide women a sense of security. The system is designed to provide immediate assistance during emergency situations by sending real-time location details to trusted contacts. It ensures faster response and enhances personal safety through continuous location tracking. The compact and portable design makes it convenient for everyday use by women, students, and elderly individuals. The integration of GPS and GSM technologies improves reliability and communication efficiency during distress situations. This smart safety device offers a low-cost and effective solution for enhancing women's security in modern society.

Keywords: GSM, LCD, SMS, Embedded C, Arduino UNO, Safety bag.

BlueChat: A Bluetooth-Based Offline Messaging System for Emergency Communication

Mr. A. Kameshwaran , Dr. N. Kanya , John paul Mathews k , Manikandan B
Information Technology Dr. M.G.R. Educational and Research Institute
Chennai, India

ABSTRACT

Mobile phones and internet-based applications are now the primary means of communication for most people. However, this dependence creates serious difficulties when internet access is unavailable. During situations such as floods, earthquakes, power failures, or sudden network disruptions, communication becomes extremely limited, even when people are located nearby. This issue is frequently observed in rural and remote regions where network connectivity is already unreliable. To address this challenge, this paper proposes BlueChat, an offline messaging application that uses Bluetooth for communication. The purpose of BlueChat is to allow nearby users to connect directly and exchange messages without the need for mobile data or Wi-Fi. The system is built to be straightforward, battery-efficient, and capable of maintaining stable connections between devices. Testing results show that BlueChat provides dependable message delivery and maintains consistent performance in offline environments. These findings indicate that the system can be an effective, low-cost communication alternative for emergency situations, local coordination, and areas with limited network availability.

Keywords: Bluetooth Communication, Offline Messaging, Peer- to-Peer Networking, Emergency Communication, Device-to- Device Communication, Mobile Applications

Human-Centric Smart Gas Leak Detection and Public Alert System

Mr. J. Praveen Kumar, Mr. J. Jayan, Mrs.Drishya Darwin

ABSTRACT

In public places, commercial establishments and residential facilities, it has been proven that the failure to detect unsafe gases in a timely fashion is a leading cause of fatal accidents and other health risks. To maintain an inclusive and proactive approach to safety via real time monitoring and early warning systems; this study proposes a Human-Centric Intelligent Gas Leak Detection and Public Alert System. The proposed gas detection system will detect unsafe levels of gases including carbon monoxide, methane and LPG by integrating MQ series gas sensors with either ESP32 or Arduino microcontrollers. Sensor data will be sent to the cloud for the purposes of identifying anomalies, performing historical analysis and providing real time display. To ensure timely notification of users, building managers or authorities, a multi-channel alert system will be implemented via local alarms, SMS, mobile device notifications and email alerts. The human-centered design of the proposed method is different from traditional reactive systems because it promotes community-based safety, and accessibility for people with disabilities, while at the same time collecting and providing gas consumption and emission data that can be used to promote sustainability, enforce government regulations, and promote environmental monitoring. Overall, the proposed method will allow for proactive, scalable, and cost-effective methods to improve gas safety in modern IoT-enabled environments.

Keywords: ESP32, MQ sensors, Internet of Things, Human-centered, Predictive analysis, Smart Safety Systems, and Gas Leak Detection.

AI-Driven Personalized Career Path Prediction and Skill Gap Analysis System Integrated with an Intelligent Chatbot

Mr.Muthukrishnan, S.Arulselvam, R.Vengadesh, R.Robert Reshil
Artificial Intelligence and Data Science
Dhanalakshmi Srinivasan College of Engineering and Technology.
Chennai, Tamilnadu

ABSTRACT

In the rapidly evolving digital economy, the demand for skilled professionals across diverse domains has increased significantly, creating a widening gap between academic learning and industry requirements. Students and job seekers often face difficulty in identifying suitable career paths due to the lack of personalized guidance systems that align their existing skills, interests, and learning capabilities with current job market demands. Traditional career counseling methods are largely generic, static, and fail to provide actionable insights such as skill gap identification and structured learning pathways. This limitation leads to inefficient career decisions, skill mismatches, and reduced employability. To address these challenges, this paper proposes an AI-Driven Personalized Career Path Prediction and Skill Gap Analysis System integrated with an intelligent chatbot, designed to deliver dynamic, data-driven, and user-centric career guidance. The proposed system leverages advanced machine learning techniques, including cosine similarity-based vector matching and content-based recommendation algorithms, to analyze user skill profiles and map them to the most relevant career roles. By representing both user skills and job requirements as high-dimensional vectors, the system calculates similarity scores to rank potential career paths with high precision. Furthermore, the system incorporates an automated skill gap analysis module that compares the user's current skill set with the required competencies for selected job roles, identifying missing or weak areas that need improvement. Based on this analysis, a heuristic-driven learning plan generator produces a structured, phase-wise roadmap that includes recommended courses, timelines, and milestones, enabling users to systematically acquire the necessary skills.

Keywords: AI-Driven Career Prediction, Skill Gap Analysis, Machine Learning Recommendation System, Personalized Learning Path, Intelligent Career Chatbot

Remote Monitoring of Hazardous Environment at Mining Sites using LoRa Network

Dr.R.Selvi, Kiran Kishore.R, Kannadasan.R
Information Technology Dr.M.G.R.Educational and Research Institute
Chennai,TamilNadu

ABSTRACT

The mining industry remains one of the most high-risk industrial sectors, characterized by volatile environments where the accumulation of toxic gases, such as Methane and Carbon Monoxide (CO), poses a constant threat to personnel safety. Traditional wired monitoring systems are difficult to maintain in shifting underground terrains, while high-frequency wireless solutions like Wi-Fi and ZigBee suffer from severe signal attenuation and limited range in non-line- of-sight (NLOS) conditions.This paper proposes a robust, remote monitoring framework leveraging LoRa (Long Range) technology to facilitate long-distance data transmission in complex mining topologies. By utilizing the Chirp Spread Spectrum (CSS) modulation technique at sub- GHz frequencies, the system achieves superior penetration through rock strata and metallic obstructions. Experimental simulations and field-testing demonstrate that the proposed system maintains a Packet Delivery Ratio (PDR) exceeding 95% at distances of up to 800 meters within confined underground galleries, significantly outperforming 2.4 GHz-based alternatives. Furthermore, the implementation of an adaptive sleep-cycle algorithm extends the battery life of end-nodes to over 18 months, ensuring a maintenance-free safety net. The results indicate that the LoRa-based approach provides a scalable, cost-effective, and highly reliable solution for real-time telemetry in hazardous mining environments, potentially reducing emergency response times and improving overall occupational safety

Keywords: Internet of Things (IoT), LoRaWAN, Mining Safety, Hazardous Gas Detection, LPWAN, Wireless Sensor Networks (WSN), Remote Monitoring.

Clinical Decision Support System for Brain Tumour Diagnosis

Ms. Kanchana K, Sham Sam Press Sj, Manikandan A, Gowtham M
Computer Science and Engineering, Kathir College of Engineering, Coimbatore

ABSTRACT

This paper presents a Clinical Decision Support System (CDSS) designed to assist in the diagnosis of brain tumors using artificial intelligence. The system utilizes deep learning techniques, specifically Convolutional Neural Networks (CNN), to analyze MRI brain images and classify tumors accurately. The model is trained on labeled datasets to detect tumor presence and type. The proposed system aims to support medical professionals by providing faster and more accurate diagnostic insights. It reduces human error and improves early detection, thereby enhancing patient treatment outcomes. The system is cost-effective and suitable for real-time clinical applications. The proposed CDSS integrates image preprocessing, feature extraction, and classification modules to improve the accuracy and reliability of tumor detection. The system can classify different categories of brain tumors and assist radiologists in making informed clinical decisions. Experimental results demonstrate high accuracy, sensitivity, and specificity in tumor identification when compared with traditional diagnostic approaches. A user-friendly interface is also incorporated to allow seamless interaction between healthcare professionals and the system. This intelligent framework has the potential to enhance healthcare efficiency and can be further extended with cloud integration and advanced predictive analytics for remote medical support.

Keywords: Brain Tumor; Clinical Decision Support System; Deep Learning; MRI Imaging; CNN; Artificial Intelligence

Smart Toll Flow: A Vision-Based Intelligent Toll Collection System for Seamless Highway Operations.

Mrs G. Deepa, Aseel Ahamed.K, Dhanasekar.R,
Information Technology Dr. M.G.R. Educational and Research Institute
Chennai, India

ABSTRACT

Traffic congestion and the operational inefficiencies of toll plazas continue to be important issues facing modern highway systems. Although toll collection has been completed for decades through manual collection systems, there are also several semi-automated methods currently in use, such as the RFID-based FASTag system. These methods, however, have limitations, including their hardware dependencies, the possibility of scanner malfunctions and issues related to human error. This paper describes the Smart Toll Flow, an intelligent and fully autonomous toll collection mechanism, which can facilitate seamless and contactless transactions using deep learning/network-based approaches in conjunction with computer vision technology. This innovative system employs high-definition (HD) video feeds for real-time vehicle detection and for identifying number plates using advanced algorithms, such as YOLOv5 and/or Faster R-CNN. After detection, the corresponding Optical Character Recognition (OCR) module, based on the Tesseract engine, uses OCR to detect alphanumeric character data from number plates and to convert it to digital text. The resulting digital text data is cross-referenced with the data stored in a centralised CSV file/dataset for retrieval of the vehicle specifications and/or owner information, allowing for the Toll Calculation Engine to log the toll fee, without any need for human interaction. Experimental considerations focus on achieving maximum accuracy in varying environmental conditions, including variable lighting levels and/or vehicle orientations. This research shows that the system decreases both the amount of time spent waiting and using fuel. The beginning of this line supports the United Nations' SDGs #9, #11, and #13 through the development of resilient infrastructure and climate action. In addition, this modular design offers a scalable and economical way to upgrade national highway networks or smart cities' transportation systems.

Keywords: Deep Learning, YOLOv5, Optical Character Recognition (OCR), Automated Toll Collection, Intelligent Transportation Systems (ITS), Computer Vision.

A Standalone Cyber Forensics Toolkit For Digital Evidence Analysis

Tamil Selvi C, Abdul Hameed T, Aadheesh
Information Technology Dr. M.G.R. Educational and Research Institute Chennai, India

ABSTRACT

The current evolution of digital forensics has changed how cybercrime investigations operate. Digital evidence must now be secured, reliable, and tamperproof when handling digital evidence throughout the forensic lifecycle. A growing list of forensic analysis tools is available to investigators today. However; existing products tend to be excessively resource-intensive as well as highly complicated so as to require online environments; making them inappropriate for use in lightweight, offline, or air-gapped forensic environments. To address these limitations this paper presents a Standalone Offline Cyber Forensics Toolkit (SOCFT); which integrates key forensic analysis and security mechanisms into a common standalone toolkit. The SOCFT provides investigators with the means to acquire digital evidence, perform binary-level inspections, extract metadata and securely encrypt digital evidence, as well as recover digital evidence from a controlled environment. A case-study evaluation will demonstrate how well the SOCFT provides investigators with an efficient and simplified approach to performing forensic analysis while still providing investigators with robust cryptographic protection of digital evidence. The modular and offline-based design of the SOCFT will facilitate future enhancements such as improved support for advanced forensic types of activity, automated evidence-chain management, and enhanced analytical capability; thereby, providing adequate use for both academic research and secure forensic environments.

Keywords: Cyber Forensics, Digital Evidence, Offline Forensics, AES Encryption, Metadata Analysis, Evidence Integrity

Marine Debris Detection using Artificial Intelligence

Mrs.S.Agalya, E.Ragavan, K.Rahul
Information Technology
Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

Marine pollution, especially underwater waste, has emerged as one of the most severe environmental challenges of the modern era. The accumulation of marine debris including plastics, metals, rubber, and discarded electronic items poses critical threats to marine biodiversity, ecological balance, and human livelihood. Traditional monitoring and cleanup methods are labor-intensive, time-consuming, and limited in scope, making automated detection a crucial necessity. This project addresses the issue of marine debris detection using Deep Learning techniques, particularly the YOLOv8 (You Only Look Once, Version 8) object detection algorithm. YOLOv8's advanced architecture, featuring an anchor-free detection mechanism, decoupled head, and C2f modules, significantly enhances detection accuracy and inference speed while maintaining computational efficiency.A dataset containing over 5,000 images representing 15 categories of underwater debris was collected and annotated using the Roboflow Annotation Tool. Model training and evaluation were conducted in the PyCharm IDE within the Anaconda environment, and experiment tracking was managed using Weights and Biases (WandB) for transparency and reproducibility. The proposed system provides an automated, scalable, and efficient solution for monitoring marine ecosystems. By enabling realtime detection of underwater debris, it contributes to the protection of marine biodiversity and supports global sustainability efforts. The integration of deep learning with environmental monitoring technologies marks a significant step forward in addressing the growing crisis of marine pollution.

Keywords: Real-time Marine Debris Monitoring, Ocean Pollution Tracking, Plastic Waste Detection, Coastal Surveillance Systems, Environmental Data Analytics, Sensor-Based Monitoring, Remote Sensing

Comparative Performance Analysis of EfficientNet-B0 and Custom CNN for Automated Pneumonia Detection from Chest X-Ray Images

Jagath M , Kamalesh Prabhu K , Kishore P, Artificial Intelligence and Data Science
Dr. M.G.R. Educational and Research Institute, Chennai

ABSTRACT

Pneumonia is a serious lung infection that continues to affect millions of individuals globally each year. Early diagnosis is essential to prevent complications and reduce mortality. Chest X-ray imaging is widely used for detection; however, accurate interpretation requires clinical expertise and may be affected by human error or delays. In recent years, deep learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated strong potential in medical image analysis. This study presents a comparative evaluation of a custom-built CNN model and EfficientNet-B0, a pre-trained architecture utilizing transfer learning, for the classification of chest X-ray images into normal and pneumonia categories. Both models were trained using the Kaggle Chest X-ray dataset with identical preprocessing and training conditions. Their performance was evaluated using accuracy, precision, recall, F1-score, and loss analysis. The results show that EfficientNet-B0 achieves better generalization and faster convergence compared to the custom CNN model. These findings highlight the advantages of transfer learning for medical image classification, especially when working with limited datasets.

Keywords: Pneumonia Detection, Chest X-Ray, Deep Learning, CNN, EfficientNet-B0, Transfer Learning

Dynamic Pricing for Online Retail Using Artificial Intelligence and Machine Learning

Ashin Bosco Raj. S , Manoj Kumar . M, Ramya V
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

Dynamic pricing has become an essential strategy in modern online retail, where market conditions and customer preferences change rapidly. This project focuses on developing an intelligent dynamic pricing system for e-commerce platforms using Artificial Intelligence (AI) and Machine Learning (ML) techniques. The proposed system analyzes multiple factors such as historical sales data, customer behavior, competitor pricing, demand fluctuations, inventory levels, and market trends to determine optimal product prices in real time. Machine learning algorithms including regression analysis and demand forecasting are used to predict customer demand and price elasticity. The system can automatically increase or decrease prices based on market conditions to maximize profit, improve customer satisfaction, and maintain competitiveness. In addition, the model supports personalized pricing and promotional strategies based on customer preferences and purchasing patterns. The system also provides analytical dashboards for monitoring sales performance, revenue growth, and inventory management. By integrating AI-driven analytics with automated pricing mechanisms, the proposed solution enhances business efficiency, supports better decision-making, and enables scalable real-time pricing for online retail environments.

Keywords: Artificial Intelligence (AI), Dynamic Pricing, Online Retail, E-commerce, Machine Learning, Demand Forecasting, Price Optimization

AI-Based Bird Call Classification and Migration Pattern Intelligence System

Kathikeyan G, Dhashketh Eliyacip T, V. Ramya
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

Bird monitoring and migration analysis are important for maintaining ecological balance and protecting biodiversity. Traditional methods of identifying bird species and tracking migration patterns require significant time, manual effort, and expert knowledge. This project proposes an AI-Based Bird Call Classification and Migration Pattern Intelligence System that combines audio signal processing, machine learning, and geospatial visualization into a unified intelligent platform. The system captures real-time bird sounds using microphones and processes the audio signals through Fast Fourier Transform (FFT) techniques to generate Mel-spectrograms representing frequency and time-domain characteristics. These spectrograms are then analyzed using a Convolutional Neural Network (CNN) trained to classify bird species based on their unique vocal patterns. This approach enables accurate and automated bird identification, even in noisy environments. After identifying the bird species, the system predicts migration patterns using a Random Forest algorithm. The model analyzes spatio-temporal features such as geographic location, seasonal variations, and environmental conditions to estimate migration routes and stopover locations. The predicted migration paths are visualized through an interactive map interface, improving understanding of bird movement behavior. The system also includes a Habitat Explorer module that provides real-time habitat insights using geolocation data and AI-based search techniques. In addition, a Creative Studio module allows users to generate bird images and animations for educational and awareness purposes. The performance of the proposed system is evaluated using Precision, Recall, F1-score, and ROC-AUC metrics. Overall, the system offers a scalable, intelligent, and efficient solution for bird classification, migration prediction, and ecological research.

Keywords: Bird Call Classification, Convolutional Neural Network, Random Forest, Migration Prediction, Spectrogram, Machine Learning, Geospatial Analysis, Artificial Intelligence, Biodiversity Conservation

Online Exam Cheating Behavior Detection

Udhaya Prakasam R, Sriram B, V.Ramya
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of an AI-powered cheating detection system for online examinations. With the rapid growth of digital education and online assessments, maintaining academic integrity has become a major challenge. Students may use unauthorized resources, collaborate with others, or display suspicious behaviors during exams, making manual monitoring difficult. The proposed system addresses these challenges by using Artificial Intelligence (AI) and Machine Learning (ML) techniques to automatically detect cheating activities in real time. The system collects and analyzes data from multiple sources, including webcam video, screen activity, browser usage, keystroke patterns, and examination logs. Advanced ML and Natural Language Processing (NLP) techniques are applied to identify abnormal behaviors such as frequent tab switching, unusual eye movement, multiple face detection, and inconsistent answering patterns. Based on these analyses, the system classifies suspicious activities and generates alerts and detailed reports for instructors. A major feature of the system is its adaptive learning capability, which improves detection accuracy over time by continuously learning from historical behavior patterns. The solution also provides analytics dashboards for monitoring exam integrity, student behavior, and risk assessment summaries. Automated reporting and behavior tracking reduce manual invigilation effort and support large-scale online examinations efficiently. The proposed system enhances the security, fairness, and reliability of online assessments. By automating monitoring and behavior analysis, it acts as an intelligent invigilation assistant that supports educators in maintaining academic integrity and strengthening trust in digital education environments.

Keywords: Artificial Intelligence (AI), Online Exam Monitoring, Cheating Detection, Natural Language Processing (NLP), Machine Learning, Behavior Analysis, Anomaly Detection, Academic Integrity, Remote Proctoring, Automated Surveillance

Automated Assessment System for Placement Test Preparation

Ganga Priya E, Wisely Joe J
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the development of a web-based Multiple Choice Question (MCQ) system designed to enhance placement exam preparation for students. The system follows a three-tier architecture consisting of Presentation, Application, and Data layers to ensure usability, functionality, and data integrity. It provides a user-friendly interface with secure login and personalized test settings for an improved user experience. The platform introduces two key modes: Test Mode, which simulates real exam conditions with time constraints, and Practice Mode, which allows students to attempt questions without time limits while receiving instant feedback. The system dynamically generates randomized questions, ensuring a diverse problem set, and supports auto-evaluation with immediate results. Additionally, it enables certificate generation after test completion, motivating students to track their progress. The inclusion of industry-relevant (high-tech) questions further strengthens students' readiness for placement exams. The Data Layer securely manages question banks, user data, and performance records. By integrating features such as real-time feedback, auto-grading, and performance analysis, the system addresses limitations of traditional assessment methods. Overall, the platform offers a flexible, scalable, and efficient solution that improves student confidence and promotes effective self-evaluation for placement success

Keywords: Automated Assessment System, Placement Test Preparation, Multiple Choice Questions, Web-Based Learning Platform, Auto Evaluation, Performance Analysis

AI Based Finance Tracker with Expense Insights

Grace Prasana M , Arputha K , Dharani K
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of an AI-based personal finance tracking system that helps users efficiently manage their daily financial activities through intelligent analysis and real-time insights. In today's digital environment, individuals often face difficulties in tracking expenses, maintaining budgets, and making informed financial decisions due to the absence of effective financial management tools. The proposed system transforms traditional expense tracking into a smart, data-driven financial management solution. The system collects and manages financial data such as daily expenses, income details, and categorized spending records including food, transportation, utilities, entertainment, and savings. Using Machine Learning (ML) and data analytics techniques, the system analyzes user spending patterns, identifies unnecessary expenses, and provides meaningful financial insights. This enables users to better understand their financial behavior and improve budgeting practices. A major feature of the system is real-time financial calculation, which provides instant updates on total expenditure, remaining balance, and category-wise spending distribution. The system also incorporates predictive analytics to forecast future expenses based on historical data, helping users plan their finances effectively. In addition, it generates intelligent recommendations such as reducing overspending and optimizing budget allocation. The application includes personalized alerts, reminders, visual dashboards, monthly summaries, and financial reports to support better financial planning. By automating expense tracking and analysis, the system acts as a virtual financial assistant that promotes financial awareness, responsible spending, and long-term financial stability.

Keywords: Artificial Intelligence (AI), Personal Finance Tracker, Machine Learning, Expense Analysis, Predictive Analytics, Budgeting, Financial Insights, Real-Time Calculation, Data Visualization, Financial Management

AI-Based Student Course Recommendation System with Grade Prediction

Aathithyan K, Sathesh Kumar S, Grace Prasana M
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the development of an AI-Based Student Course Recommendation System with Grade Prediction designed to assist students in selecting suitable courses based on their academic performance, interests, and learning patterns. In higher education, students often face challenges in choosing appropriate subjects that match their abilities and career goals. The proposed system addresses this issue by providing intelligent course recommendations along with predicted academic performance. The system collects and analyzes student data such as previous semester grades, subject preferences, attendance, and academic history. Using Machine Learning (ML) algorithms, the system predicts students' future grades and recommends courses that align with their strengths and interests. The recommendation engine helps students make informed academic decisions and improves their chances of achieving better performance. A major feature of the system is its ability to provide personalized suggestions by continuously analyzing student learning behavior and performance trends. The system also generates performance reports and visual analytics to help students and faculty monitor academic progress. Additionally, the platform supports data-driven academic planning and reduces the risk of students selecting unsuitable courses. By integrating Artificial Intelligence and predictive analytics, the proposed solution enhances academic guidance, improves student confidence, and supports better educational outcomes. Overall, the system acts as an intelligent academic advisor that promotes effective course selection and improved academic success.

Keywords: Machine Learning, Course Recommendation, Grade Prediction, Student Performance Analysis, Artificial Intelligence

Remedy Sense – AI Natural Beauty Remedies

Aliana Begum K, Jayasree G, Ramya V
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT
This project presents the design and development of "Remedy Sense – AI Natural Beauty Remedies," an intelligent system that provides personalized natural beauty and skincare recommendations using Artificial Intelligence (AI). In today's fast-paced lifestyle, many individuals experience skin and hair-related issues due to pollution, stress, unhealthy food habits, and the excessive use of chemical-based cosmetic products. Traditional methods of finding suitable remedies often require expert consultation and may not provide personalized solutions. The proposed system addresses these challenges by offering AI-driven natural remedy suggestions based on user-specific conditions and preferences. The system collects user inputs such as skin type, hair condition, common beauty concerns, allergies, and environmental factors. Using Machine Learning (ML) techniques and data analysis, the application identifies suitable herbal and natural remedies for skincare and haircare problems such as acne, dryness, dandruff, pigmentation, and hair fall. The platform also provides ingredient-based recommendations using commonly available natural products like aloe vera, turmeric, neem, honey, and coconut oil. A key feature of the system is its ability to generate personalized beauty care routines and provide preventive suggestions for maintaining healthy skin and hair. The application also includes educational content, remedy preparation steps, and precautionary measures for safe usage. By integrating Artificial Intelligence with natural healthcare knowledge, the proposed system promotes safe, affordable, and chemical-free beauty solutions while improving user awareness and self-care practices.

Keywords: Artificial Intelligence, Natural Remedies, Beauty Care, Skincare Recommendation, Machine Learning

AI-Based Smart Infant Cry Analyzer

Brindaa Tamilarasi K, Mawheba T, Anitha V.B.
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of an AI-Based Smart Infant Cry Analyzer, an intelligent system developed to identify and analyze infant crying patterns using Artificial Intelligence (AI) and Machine Learning (ML) techniques. Infants are unable to communicate their needs verbally, making it difficult for parents and caregivers to understand the reason behind their crying. The proposed system aims to assist caregivers by automatically detecting and classifying different types of infant cries such as hunger, pain, discomfort, tiredness, and illness. The system captures infant cry audio signals through microphones and processes them using audio signal processing techniques such as Fast Fourier Transform (FFT) and Mel-frequency cepstral coefficients (MFCC). The extracted features are analyzed using Machine Learning and Deep Learning models to classify the cry patterns accurately. By identifying the emotional and physical needs of infants in real time, the system helps caregivers respond more effectively and quickly. A major feature of the system is its real-time monitoring and alert mechanism, which provides immediate notifications and recommendations to parents through a user-friendly interface. The application also stores historical cry data and generates analysis reports to help track infant behavior and health patterns over time. By integrating Artificial Intelligence with audio analytics, the proposed solution improves infant care, reduces caregiver stress, and supports timely attention to infant needs. Overall, the system acts as an intelligent assistant for smart childcare and infant well-being.

Keywords: Artificial Intelligence, Infant Cry Analysis, Machine Learning, Audio Signal Processing, Smart Healthcare

AI-Powered Browser Virtual Assistant

David Freeman D N, Tharunesh P, Grace Prasana M
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of a Virtual Assistant for Browser using JavaScript, aimed at improving user interaction and productivity while browsing the internet. In today's digital environment, users frequently perform repetitive browser tasks such as searching information, opening websites, managing tabs, and accessing online services. The proposed system provides an intelligent browser-based virtual assistant that simplifies these activities through voice and text-based commands. The system is developed using JavaScript along with web technologies such as HTML, CSS, and browser APIs to create an interactive and responsive assistant. It supports functionalities including voice recognition, speech synthesis, web searching, opening and closing tabs, fetching online information, setting reminders, and providing quick navigation support. By using Natural Language Processing (NLP) techniques, the assistant can understand user commands and respond intelligently in real time. A key feature of the system is its ability to automate common browser tasks and provide hands-free interaction, improving accessibility and user convenience. The assistant also supports personalized responses and customizable commands based on user preferences. Additionally, the system is lightweight, platform-independent, and can be integrated easily into modern web browsers. The proposed solution enhances browsing efficiency, reduces manual effort, and provides a smart interactive experience for users. Overall, the project demonstrates how JavaScript and AI-based technologies can be combined to create an effective virtual assistant for modern web applications.

Keywords: Virtual Assistant, JavaScript, Browser Automation, Natural Language Processing, Voice Recognition

Food Calorie Estimation System Using CNN

Hariharan A, Gokul M, Lavanya R
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of a Food Calorie Estimation System using Convolutional Neural Networks (CNN) to identify food items and estimate their calorie content automatically from images. Maintaining a healthy diet and monitoring calorie intake are essential for fitness and disease prevention. However, manually calculating calories is often time-consuming and inaccurate. The proposed system addresses this challenge by providing an intelligent image-based calorie estimation solution. The system captures or uploads food images and processes them using image preprocessing and deep learning techniques. A Convolutional Neural Network (CNN) model is trained on a food image dataset to recognize different food categories accurately. After identifying the food item, the system estimates the approximate calorie value using nutritional databases and portion-based analysis. A key feature of the system is its ability to provide real-time calorie estimation along with nutritional information such as carbohydrates, proteins, and fats. The application also supports multiple food item detection within a single image, improving usability for everyday meal tracking. Additionally, the system offers personalized dietary suggestions and maintains food consumption history for monitoring health goals.The proposed solution enhances health awareness, simplifies calorie tracking, and supports healthier lifestyle choices. By integrating Artificial Intelligence, image processing, and deep learning technologies, the system acts as a smart dietary assistant for fitness enthusiasts, patients, and general users.

Keywords: Convolutional Neural Network, Food Image Recognition, Calorie Estimation, Deep Learning, Image Processing

Intelligent Student Complaint and Maintenance Management System Using NLP

Harini S, Husvanthini R, Anitha V.B
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of an Intelligent Student Complaint and Maintenance Management System using Natural Language Processing (NLP). In educational institutions, students often face difficulties in reporting complaints related to infrastructure, maintenance, academic issues, and hostel facilities through traditional manual methods. These methods are time-consuming, lack proper tracking, and may delay issue resolution. The proposed system provides an intelligent and automated platform for managing student complaints efficiently. The system allows students to submit complaints in natural language through a web-based interface. Using NLP techniques, the system analyzes the complaint text, identifies keywords, classifies the complaint category, and automatically forwards it to the appropriate department or authority. This automation reduces manual effort and improves the speed of complaint handling and maintenance management. A major feature of the system is its real-time tracking and notification mechanism, which enables students to monitor complaint status updates and receive alerts regarding resolution progress. The platform also maintains complaint history, generates analytical reports, and provides dashboards for administrators to monitor pending and resolved issues effectively. Additionally, the system uses Machine Learning techniques to improve classification accuracy and identify frequently occurring problems within the institution. By integrating Artificial Intelligence and NLP technologies, the proposed solution enhances communication between students and administration, increases transparency, and improves overall campus maintenance efficiency.

Keywords: Natural Language Processing, Complaint Management System, Machine Learning, Student Support System, Artificial Intelligence

AI-Based Duckworth-Lewis-Stern Score Predictor

Janakiraman S, Ramkumar R
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of an AI-Based Duckworth-Lewis-Stern (DLS) Score Predictor for cricket matches affected by interruptions such as rain or bad weather. In limited-overs cricket, interruptions can significantly impact match outcomes, making accurate score prediction essential for fair gameplay. The proposed system aims to provide intelligent and accurate target score predictions using historical match data and machine learning techniques based on the Duckworth-Lewis-Stern method. The system collects and analyzes cricket match data including overs played, wickets lost, current run rate, team performance, pitch conditions, and historical scoring patterns. Using Machine Learning (ML) algorithms and statistical analysis, the system predicts revised target scores for interrupted matches while considering match situations and resource availability. A key feature of the system is its ability to generate real-time score predictions during ongoing matches. The application also provides visual analytics, match summaries, and performance insights to improve understanding of game progression and prediction accuracy. Additionally, the platform supports comparative analysis between predicted and actual match outcomes for better evaluation. The proposed solution enhances decision-making in interrupted cricket matches by providing reliable and automated score calculations. By integrating Artificial Intelligence, data analytics, and cricket statistics, the system improves fairness, accuracy, and efficiency in match target prediction. Overall, the project acts as an intelligent sports analytics tool for modern cricket match management.

Keywords: Duckworth-Lewis-Stern Method, Machine Learning, Cricket Analytics, Score Prediction, Artificial Intelligence

Customer Relationship Optimization and Lead Conversion Automation Using Zoho CRM

Madhumitha M, Vishnu Priya B
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of a Customer Relationship Optimization and Lead Conversion Automation System using Zoho CRM. In modern business environments, managing customer relationships and converting leads efficiently are critical for improving sales performance and business growth. Traditional customer management processes often involve manual tracking, delayed follow-ups, and inefficient lead handling, resulting in reduced productivity and missed business opportunities. The proposed system addresses these challenges through automation and intelligent customer relationship management techniques. The system utilizes Zoho CRM to manage customer data, track interactions, automate lead generation workflows, and improve communication between businesses and customers. It integrates features such as lead scoring, automated email notifications, task reminders, customer segmentation, and sales pipeline management. By analyzing customer behavior and engagement patterns, the system helps identify high-potential leads and supports faster lead conversion. By integrating CRM automation with intelligent data analysis, the proposed solution improves customer satisfaction, increases sales conversion rates, and enhances business productivity. Overall, the system acts as a smart customer relationship management solution for efficient lead handling and business growth.

Keywords: Zoho CRM, Lead Conversion, Customer Relationship Management, Sales Automation, Business Analytics

AI-Driven Smart Urban Water Demand Forecasting and Decision Support System Using Machine Learning

Ravi Krishna A R, Vikram L, Wisely Joe J
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of an AI-Driven Smart Urban Water Demand Forecasting and Decision Support System using Machine Learning. Rapid urbanization and population growth have increased the demand for efficient water resource management in cities. Traditional water management methods often fail to accurately predict consumption patterns, leading to water shortages, wastage, and inefficient distribution. The proposed system addresses these challenges by using Artificial Intelligence (AI) and Machine Learning (ML) techniques to forecast urban water demand and support effective decision-making. The system collects and analyzes historical water consumption data, weather conditions, population statistics, seasonal variations, and environmental factors. Machine Learning algorithms are applied to identify consumption trends and predict future water demand with improved accuracy. These predictions help authorities and utility providers optimize water distribution, reduce wastage, and plan resource allocation efficiently. By integrating AI, predictive analytics, and smart monitoring technologies, the proposed solution enhances sustainable water management and supports smarter urban planning. Overall, the system acts as an intelligent decision support tool that improves water resource utilization, operational efficiency, and environmental sustainability.

Keywords: Artificial Intelligence, Water Demand Forecasting, Machine Learning, Smart City, Decision Support System

Real-Time Pothole Detection System Using Object Detection

Sivanesh S, Vignesh R
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of a Real-Time Pothole Detection System using Object Detection techniques to improve road safety and support smart transportation management. Poor road conditions and potholes are major causes of vehicle damage and road accidents, especially in urban and rural transportation networks. Traditional road inspection methods are time-consuming, costly, and inefficient. The proposed system addresses these challenges by providing an automated and intelligent pothole detection solution using Artificial Intelligence (AI) and computer vision technologies. The system captures real-time road images and video streams using cameras mounted on vehicles or mobile devices. Using deep learning-based object detection algorithms such as YOLO (You Only Look Once) and Convolutional Neural Networks (CNN), the system identifies and classifies potholes accurately under different environmental conditions. The detected potholes are highlighted and their locations are recorded using GPS integration for effective road monitoring and maintenance planning. By integrating Artificial Intelligence, object detection, and geolocation technologies, the proposed solution improves road safety, reduces vehicle damage, and supports efficient infrastructure maintenance. Overall, the system acts as an intelligent road monitoring and management tool for safer transportation systems.

Keywords: Object Detection, Pothole Detection, Computer Vision, Deep Learning, Smart Transportation

Real-Time Language Translation System Using Natural Language Processing

Thirumalaiselvam T, Yunesh T
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This project presents the design and development of a Real-Time Language Translation System using Natural Language Processing (NLP) to enable effective communication between users speaking different languages. In today's globally connected world, language barriers often create challenges in communication, education, business, and travel. Traditional translation methods may not provide fast and accurate results in real-time interactions. The proposed system addresses these challenges by providing intelligent and instant language translation using Artificial Intelligence (AI) and NLP techniques. The system accepts text or speech input from users and processes it using NLP algorithms for language identification, text analysis, and translation. Machine Learning and deep learning models are used to improve translation accuracy and contextual understanding between multiple languages. The translated output is generated instantly in text or speech format, enabling smooth and natural communication. The proposed solution enhances communication efficiency, reduces language barriers, and supports educational, professional, and social interactions across different languages. By integrating Artificial Intelligence, Natural Language Processing, and speech technologies, the system acts as an intelligent multilingual communication assistant for real-time translation environments.

Keywords: Natural Language Processing, Real-Time Translation, Machine Learning, Speech Recognition, Artificial Intelligence

Optimized Transfer Learning with Federated Face Embeddings for Privacy-Aware Attendance Tracking System

Mr. Issac Moses D , Mr. Thirumalai Nambi S , Mrs. Lavanya R
Artificial Intelligence and Data Science
St. Peter's College of Engineering and Technology

ABSTRACT

This paper presents the design and implementation of a real-time face recognition system on the ESP32 microcontroller, addressing the need for low-power, privacy-preserving biometric authentication in IoT environments. Unlike cloud-based systems, the proposed approach performs on-device inference, reducing latency, bandwidth usage, and security risks. The system integrates an OV2640 camera with an ESP32-CAM and evaluates three models: LBPH, a lightweight CNN, and an INT8-quantized MobileNetV2, trained offline using TensorFlow and OpenCV and deployed via TensorFlow Lite. Experimental results show a peak accuracy of 97.1% with MobileNetV2 at 389 ms latency, while LBPH achieves 91.3% accuracy at 112 ms latency, making it suitable for real-time applications. The system also incorporates hardware components such as servo motors, relay modules, OLED display, and MQTT-based alerts, supporting features like live enrollment, multi-face detection, anti-spoofing, and attendance logging, with an overall cost of approximately USD 18.

Keywords: Face Recognition, ESP32-CAM, Internet of Things (IoT), TensorFlow Lite, Edge Computing, Biometric Authentication, MobileNetV2, Real-Time Monitoring, Embedded Systems, Anti-Spoofing

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